

Single-Stage PFC LLC Converter Design

with the STMicroelectronics L6790A

Worked design

90 to 264 Vac, 47 to 63 Hz

25 V / 26.3 A = 657.5 W

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Before you start

The L6790A behaviour described here comes from a **draft** datasheet dated 30 April 2026. That draft has items marked TBD and at least one place where it contradicts itself. Check every controller constant against the released datasheet before the design goes to production.

The worked design has not been built. Only the numbers marked as measured are measured; the rest are calculated.

All figures were drawn for this note. Where a figure follows the argument of a published application note, the caption says so.

Revision history

Version	Date	Change
V1.0	August 2026	First issue. Worked design: 25 V / 26.3 A, 657.5 W.
V1.1	September 2026	Background section on LLC and power factor correction added, with circuit figures. Transformer design, the closed form for $f_{sw}(\theta)$, the compensator design method, the gain-margin check and the input capacitor added. Worked design moved to the $N_p:N_s = 15:2$ transformer.
V1.2	September 2026	Renamed single-stage PFC LLC throughout. Verification margins defined. Worked values moved out of the design procedure and into the design example, where each one names the equation it comes from and shows the numbers put into it.
V1.3	September 2026	Design example on one transformer (ETD 49/25/16DG) with the winding and pin assignment drawn. Compensator chapter rewritten with the loop equations, the op-amp equivalent of the TL431 network and a worked loop design. Full read-through; Korean edition brought to the same content.

Conventions

- Frequencies are switching frequencies unless written f_l , which is the line frequency.
- $f_n = f_{sw}/f_r$ is the switching frequency normalised to the series resonance. Gain curves are always drawn against it, so that one curve serves every tank. Figures reproduced from other documents may write f_s for f_{sw} ; that is the same quantity.
- M is the voltage gain of the resonant network: the output voltage referred to the primary, divided by the fundamental of the square wave driving it. **$M = 1$ at the series resonance**, whatever the load.
- $Z_0 = \sqrt{L_r/C_r}$ is the characteristic impedance of the tank, and $Q = Z_0/R_{ac}$ is its quality factor — how heavily the tank is loaded. $Q = 0$ is no load; larger Q is more power and a flatter, lower gain curve.
- $V_{o,eff} = V_{out} + N_{rect} V_f$ is the output voltage the primary actually sees, with the rectifier drop added. N_{rect} is the number of rectifier drops in the conduction path: 1 for a centre tap, 2 for a full bridge.
- n is the equivalent-model turns ratio used in the gain equation; n_T is the physical turns ratio of a transformer whose leakage is integrated into L_r .
- $\lambda = L_r/L_m$, and $m = 1 + 1/\lambda$ is the ratio of total primary to resonant inductance used in older literature.
- θ is the line phase angle, zero at the mains zero crossing.

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- “Equivalent input” is what the resonant tank sees after topology morphing, which is not the mains voltage.

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Single-Stage PFC LLC Converter Design

1 Introduction

A conventional universal-input LLC power supply has two stages: a boost power-factor corrector that holds a 400 V bus, and an LLC converter that steps that bus down. The capacitor between them does two jobs. It stores the energy that a unity-power-factor input cannot deliver evenly, and it gives the LLC an almost constant input, so the resonant tank only has to cover a narrow range.

A **single-stage PFC LLC** removes both the boost stage and that capacitor. The rectified mains feeds the resonant tank directly. The result is a converter whose input is a 100/120 Hz half sine, whose gain is no longer something the controller can set, and whose output capacitor must absorb everything the bus capacitor used to. It is a different converter, not an LLC with one stage removed, and most of the design habits from the two-stage case do not apply.

This note explains how such a converter works and how to design one with the STMicroelectronics **L6790A** controller. One worked design runs through it, from specification to component values: **90 to 264 Vac in, 25 V / 26.3 A = 657.5 W out**, a low-voltage, high-current dc rail.

Scope. Covered: the power stage, the resonant tank, the output bank, the controller network and the voltage loop.
Not covered: EMI filtering, magnetics construction, layout and safety approval.

2 The two converters this one is made of

What comes later is easier to follow if the two halves are understood separately first. This section covers an ordinary LLC and ordinary power factor correction. A reader who has designed both can skip to Section 3.

2.1 What an LLC converter is

An LLC converter is a square-wave generator that drives a resonant network; the network feeds a transformer and a rectifier. The switches only chop the input into a square wave whose frequency can be varied. There is no duty cycle to set and no inductor current to program.

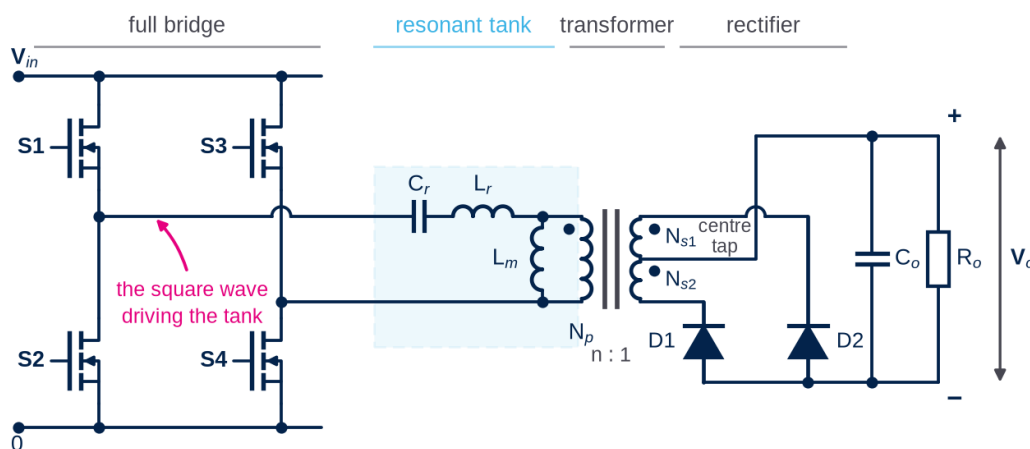


Figure 1: An LLC stage: a square-wave generator, the resonant network, a transformer and a rectifier. The switches only set the frequency — the tank decides how much power flows and the transformer sets the voltage.

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The three tank elements are named in the order they appear in “LLC”: the series inductance L_r , the magnetising inductance L_m of the transformer, and the series capacitance C_r . The middle one is not a separate part: it is the transformer’s own magnetising inductance. So only C_r and L_r are added components, and L_r is often the leakage inductance of the transformer itself.

- **The square wave sets the frequency, and nothing else.** The output voltage is controlled by moving f_{sw} , because the impedance of the tank, and so its voltage division, depends on frequency.
- **The tank current is almost a sine wave.** That is the purpose: the switches turn on and off while the current is small, or while it flows through the body diode, so most of the switching loss disappears.
- **The transformer steps the voltage and provides isolation**, and its leakage inductance can be used as L_r instead of being a problem. That is the usual construction at these power levels.

2.2 Why an LLC, and not something simpler

An LLC is more work than a forward or a flyback converter, and it does not regulate by duty cycle, which is what most designers are used to. At a few hundred watts and above, it is worth the extra work for these reasons:

- **Soft switching over the whole load range, on both sides.** The primary switches turn on at zero voltage, and below resonance the secondary rectifiers turn off at zero current. A phase-shifted full bridge loses ZVS at light load; an LLC does not, because the magnetising current that does the work is present at any load.
- **The frequency can be raised.** In a hard-switched converter, switching loss limits the frequency, and with it the size of the magnetics and the filter. Remove that loss and hundreds of kHz become practical.
- **The transformer leakage is used, not fought.** In a forward converter the leakage inductance is a problem that needs a snubber. Here it can be L_r , so the resonant inductor costs nothing and the snubber is not needed.
- **No output inductor.** The rectifier feeds the output capacitor directly. That removes a wound component, and its loss, from the high-current side, where it hurts most.
- **Low noise.** The currents are sine waves rather than trapezoids, and the switching edges are slow and damped, so the conducted and radiated emissions are much lower than in a hard-switched converter of the same power.

The costs are real too. Regulation is by frequency, so the operating frequency moves with line and load, and the magnetics must work over that whole range. The magnetising current circulates whether or not the load takes power, so light-load efficiency is not a strong point. And the design does not come from a duty-cycle expression: it comes from the gain curve, which is what the rest of this section is about.

2.3 What resonance is for

In a hard-switched converter a switch turns on with the full input voltage across it, so the energy stored in its output capacitance is lost in the channel every cycle. That loss grows with frequency. It is what limits the frequency, and with it the size of the magnetics.

An LLC that runs **in its inductive region** presents an inductive load to the bridge. The tank current then lags the drive voltage. In the dead time between the two switches of a leg, that lagging current discharges the mid-point capacitance by itself, and the next switch turns on at zero volts. **That is zero-voltage switching, and it is the whole reason for the topology.** Section 2.10 shows the mechanism and what it requires of the design.

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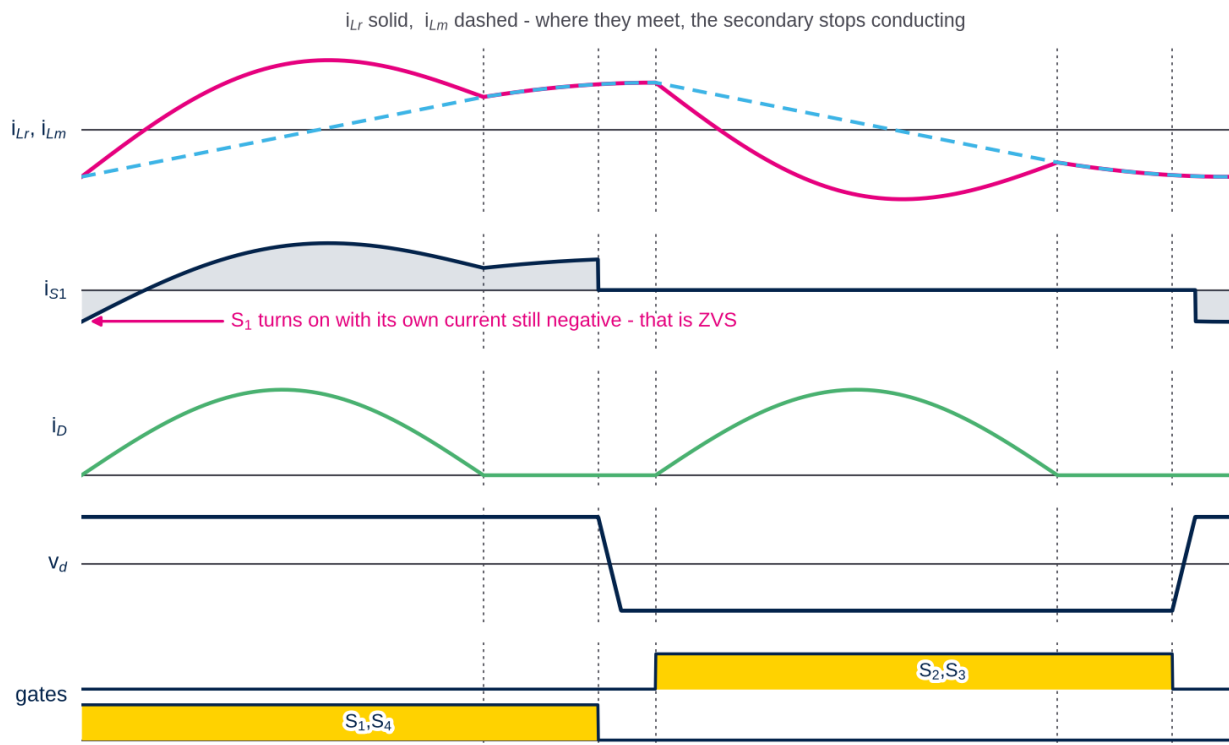


Figure 2: One switching period below resonance. i_{Lr} is the tank current and i_{Lm} the magnetising current; the difference between them is what crosses to the secondary, and it is gone once the two meet. The switch turns on while its own current is still negative, and that is what discharges the mid-point.

The two frequencies that bound this are the subject of Section 2.5, and the operating regions they divide are in Section 2.6.

The condition is *inductive operation*, not a particular frequency. Go low enough in frequency and the tank becomes capacitive: the current leads, the dead time works against the transition instead of for it, and the bridge hard-switches into a low impedance. It is the one failure mode that destroys parts rather than just heating them. **Where that happens is not a fixed frequency:** the boundary moves with load, and Section 2.9 is about exactly that.

2.4 How the circuit becomes $M(f_n, Q)$

Every gain curve in every LLC note comes from three simplifications, applied in order. They are worth doing once, because each one throws something away, and it is useful to know what.

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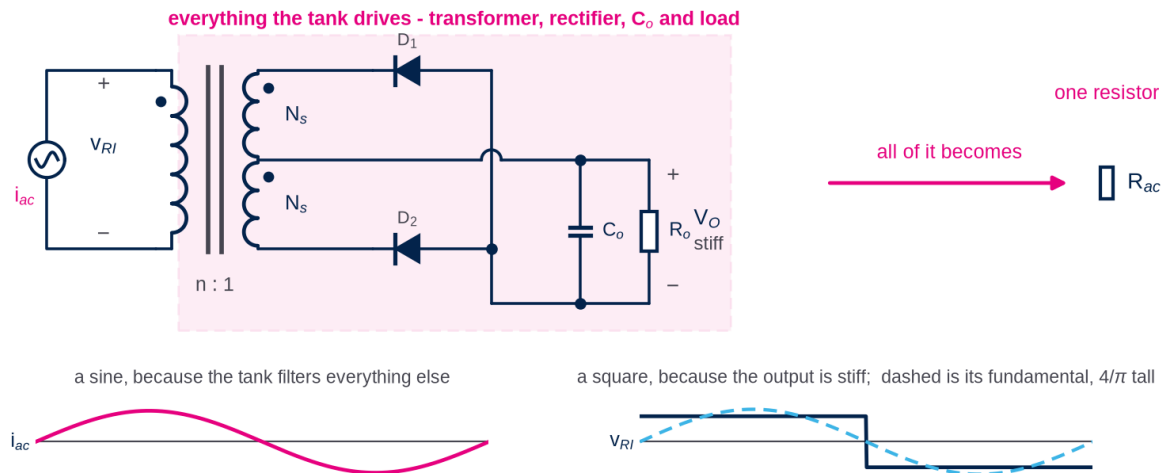


Figure 3: Why the rectifier and the load can be replaced by one resistance. The output is stiff, so the voltage the tank sees is a square wave; the tank filters everything except the fundamental, so the current is a sine wave. A sine of current in phase with a square of voltage delivers the same fundamental power as a resistor would take.

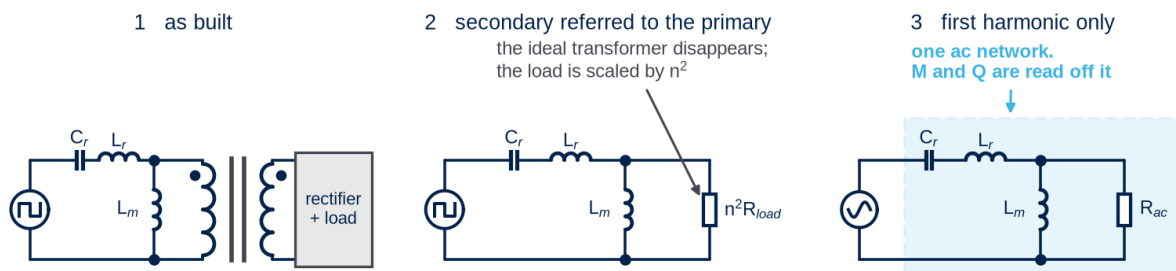


Figure 4: The reduction, in three steps: as built; the secondary referred to the primary; the fundamental only. What is left is one ac network whose voltage transfer is the gain M and whose damping is Q .

- **Refer the secondary to the primary.** The ideal transformer disappears and the load resistance is multiplied by n^2 .
- **Replace the rectifier and the output filter by one resistor.** The rectifier draws a square current wave and the output is stiff, so at the fundamental frequency the whole of it looks like a resistor.
- **Keep only the fundamental of the drive.** The tank is a filter centred near the switching frequency, so the harmonics of the square wave contribute little. This is the *first harmonic approximation*, and it is why every LLC design ends with a simulation or a measurement rather than with the equations.

What is left is one ac network: a voltage divider whose ratio changes with frequency. Two numbers describe it, and both are used on every page from here on:

- **The gain M :** the voltage across R_{ac} divided by the fundamental that drives it. Because C_r and L_r cancel at the series resonance f_r , **$M = 1$ there at any load**. Below f_r the network can step up ($M > 1$); above it, it steps down.
- **The quality factor $Q = Z_0/R_{ac}$,** where $Z_0 = \sqrt{L_r/C_r}$ is the characteristic impedance of the tank. Q says how heavily the tank is loaded: **Q rises with output power**. At $Q = 0$ (no load) the gain curve is tall and peaked; loading flattens the curve and pulls the peak down.

In symbols, with f_n the frequency normalised to the series resonance so that one family of curves serves every tank:

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Equation 1

$$f_n = \frac{f_{sw}}{f_r}, \quad Z_0 = \sqrt{\frac{L_r}{C_r}}, \quad Q = \frac{Z_0}{R_{ac}}, \quad \lambda = \frac{L_r}{L_m}$$

$M(f_n, Q)$ always means the gain of the right-hand circuit above.

2.5 The two resonances

Every LLC has two. While the secondary conducts, the reflected output voltage clamps L_m and it drops out of the circuit, leaving the series resonance

Equation 2

$$f_r = \frac{1}{2\pi\sqrt{L_r C_r}}$$

At no load nothing clamps L_m , and the tank resonates with both inductances:

Equation 3

$$f_o = \frac{1}{2\pi\sqrt{(L_r + L_m) C_r}}$$

At f_r the gain is exactly 1 for any load, because the load term disappears. At f_o the no-load gain is unlimited. Between the two the tank can boost; above f_r it cannot. An LLC fed from a fixed dc bus runs close to f_r nearly all the time, and drops into that band only when the bus sags.

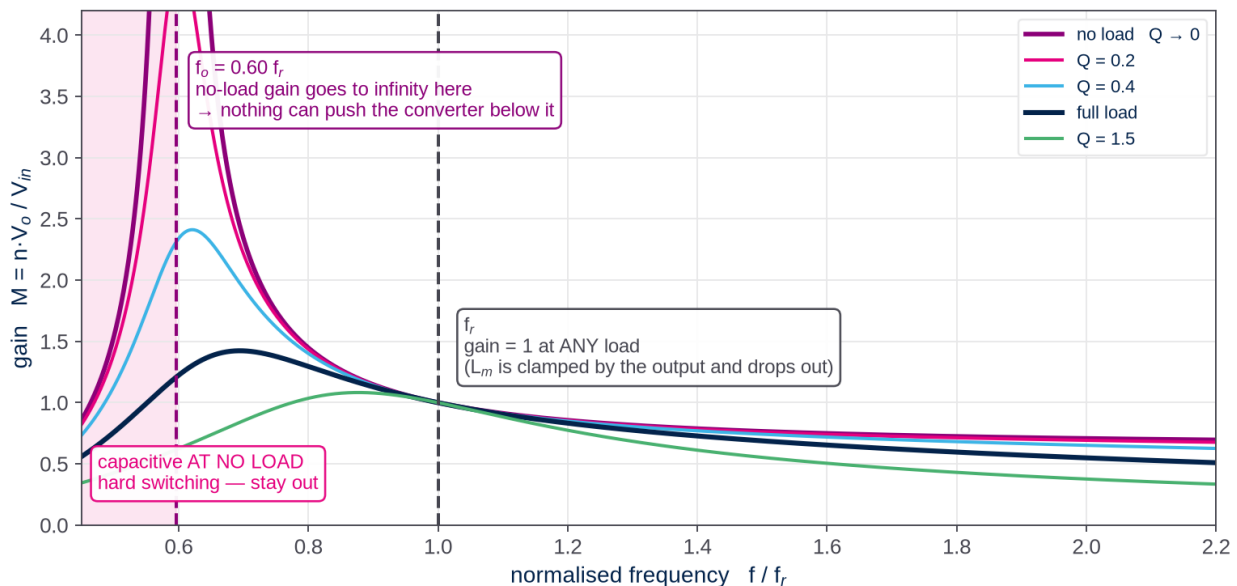


Figure 5: The two resonances, and the band between them in which the tank can boost. The shaded edge is the no-load position of the capacitive boundary; under load it moves up.

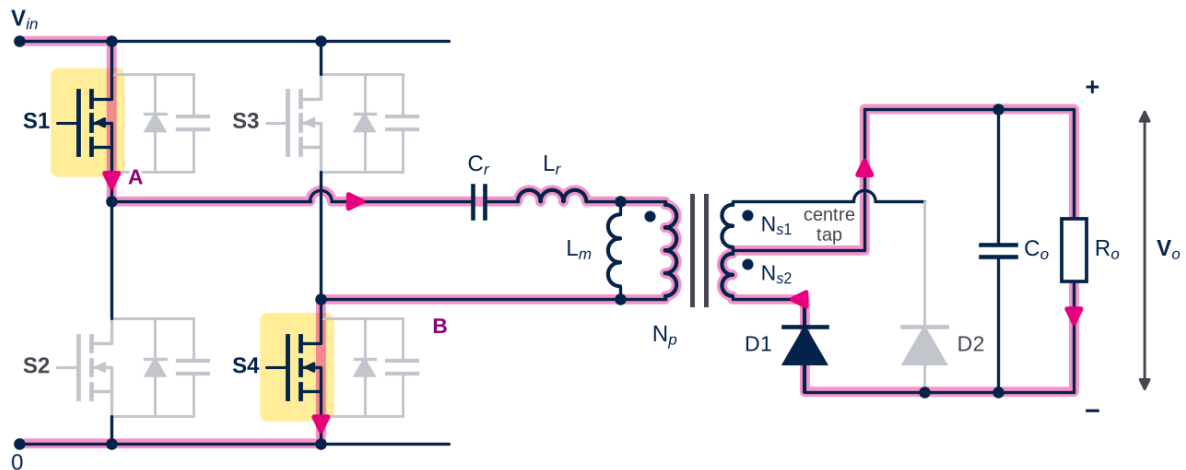
The two expressions are not much use without knowing what each one does to the current, because the current is where the difference shows.

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First, the two things the circuit can be doing. Below resonance a half cycle contains both, in this order: power delivery first, then freewheeling, **with the same pair of switches on throughout**. At resonance the half cycle is power delivery only; above resonance the power delivery is cut short.

1 POWER DELIVERY

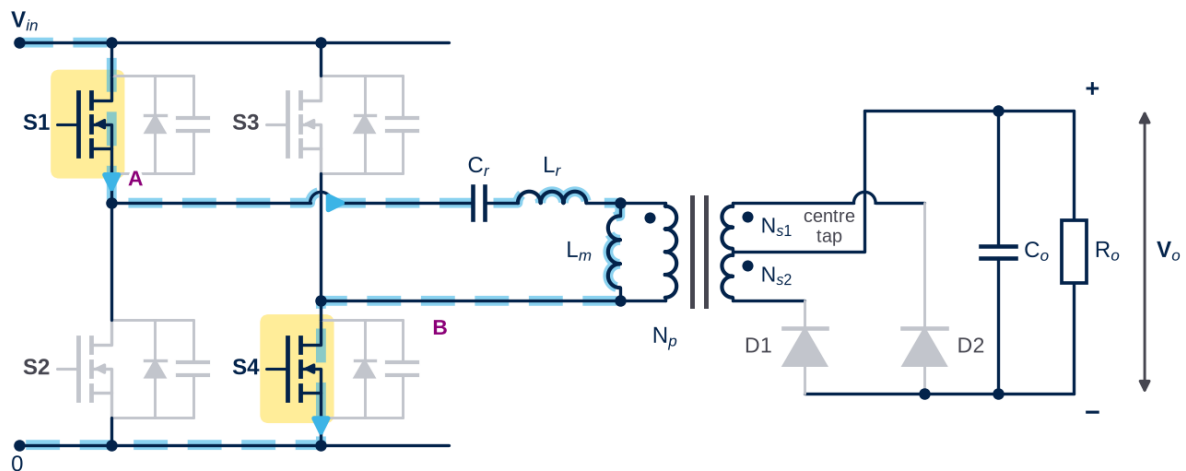
S1, S4 on · D1 conducts · lower half of N_s



L_m is clamped, so i_{Lm} is a straight ramp and $i_{Lr} - i_{Lm}$ is the half sine that crosses. What is traced is the load component; i_{Lm} flows in L_m too, but it reverses inside this interval.

2 FREEWHEELING

S1, S4 STILL on · secondary off



i_{Lr} has fallen to i_{Lm} . Every rectifier is off, L_m is free again, and the tank rings at f_o .

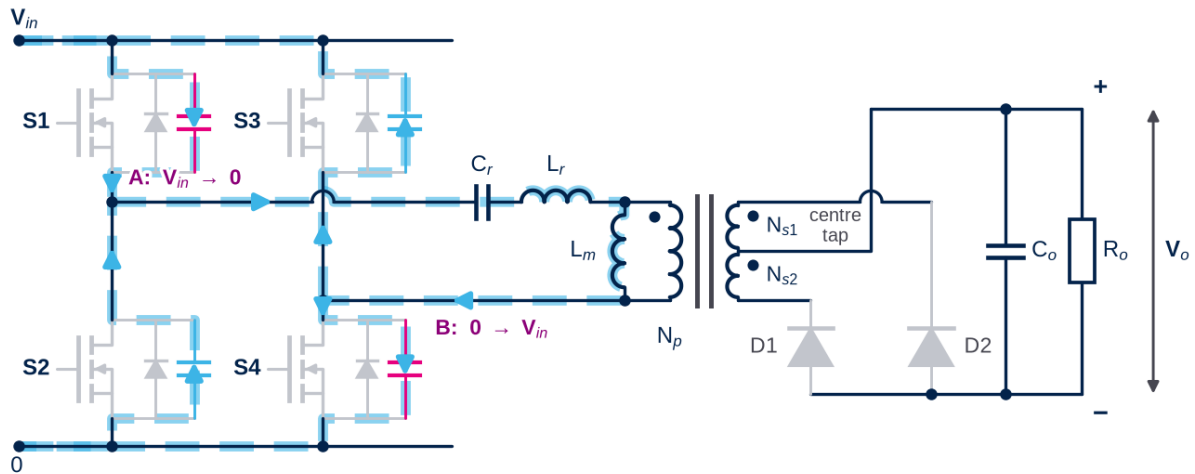
— load current — magnetising current only gated on
 body diode conducting = the ZVS window C_{oss} charging, V_{ds} rising C_{oss} discharging

Figure 6: One switching period, first half, steps 1 and 2. 1 Power delivery — S_1, S_4 are on, the resonant current exceeds the magnetising current and the difference passes through the transformer to D_1 and the load. L_m is clamped to the reflected output, so this interval resonates at f_r . 2 Freewheeling — the resonant current has fallen to the magnetising current, nothing is left for the secondary, the rectifiers are off and L_m joins the resonance: this interval rings at f_o . The same pair of switches is still on. Each switch is drawn with its body diode and its C_{oss} beside it, greyed while they do nothing.

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3 DEAD TIME (a)

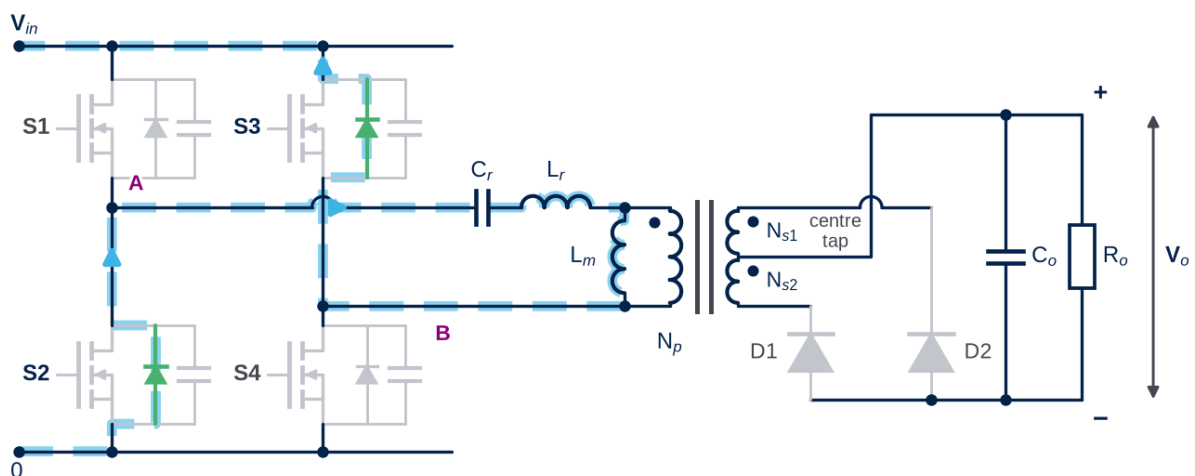
all four off · C_{oss} swaps the midpoints over



All four channels are off, so the ONE tank current can reach the rails only THROUGH the four C_{oss} . It arrives at A from both, so A is pulled down; it leaves B into both, so B is pushed up. That swap IS the transition.

4 DEAD TIME (b) — ZVS

body diodes of S2 and S3 clamp



$V_{ds} = 0$ on S2 and S3 now, because their own body diodes are clamping them. Gate them on HERE.

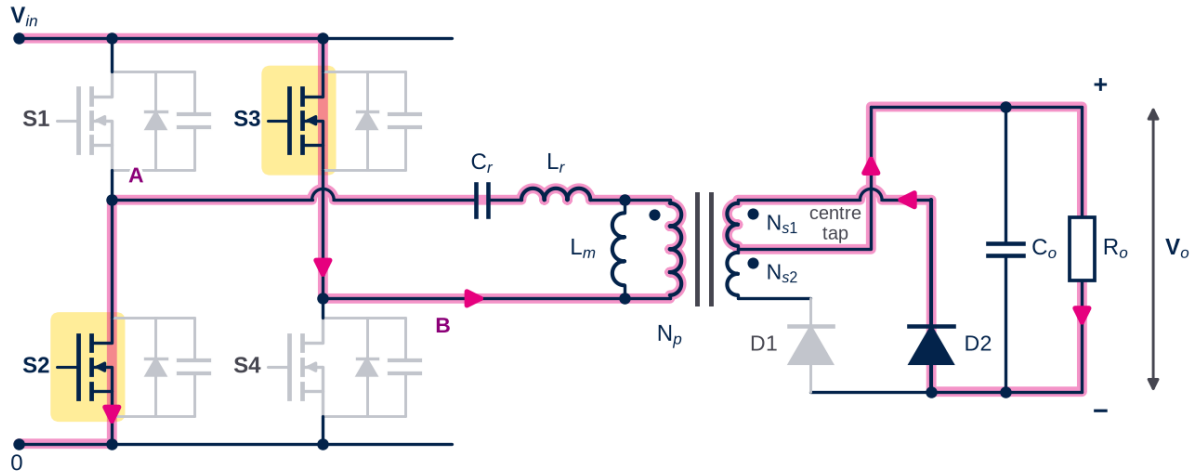
— load current — magnetising current only gated on
 body diode conducting = the ZVS window — C_{oss} charging, V_{ds} rising — C_{oss} discharging

Figure 7: Steps 3 and 4: the dead time. 3 All four switches are off, so the tank current — now the magnetising current alone — can only reach the rails through the four C_{oss} . It arrives at A from both, so A is pulled down, and leaves B into both, so B is pushed up: that swap is the transition. **4** The swing is complete and the body diodes of S_2, S_3 clamp their V_{ds} at zero. This is the ZVS window: the gate that follows has to arrive inside it.

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5 POWER DELIVERY

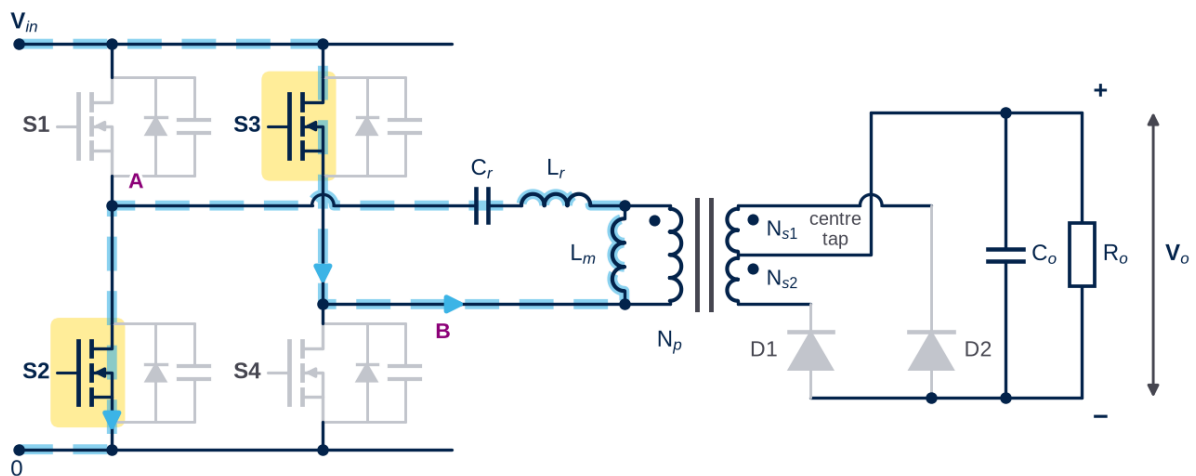
S2, S3 on · D2 conducts · upper half of N_s



The mirror of 1. Just after turn-on the old current still runs back through the two channels. Again only the load component is traced - i_{Lm} reverses inside this interval as well.

6 FREEWHEELING

S2, S3 STILL on · secondary off



The mirror of 2, and the easy one to miss: it ends at the right-hand edge of a waveform plot.

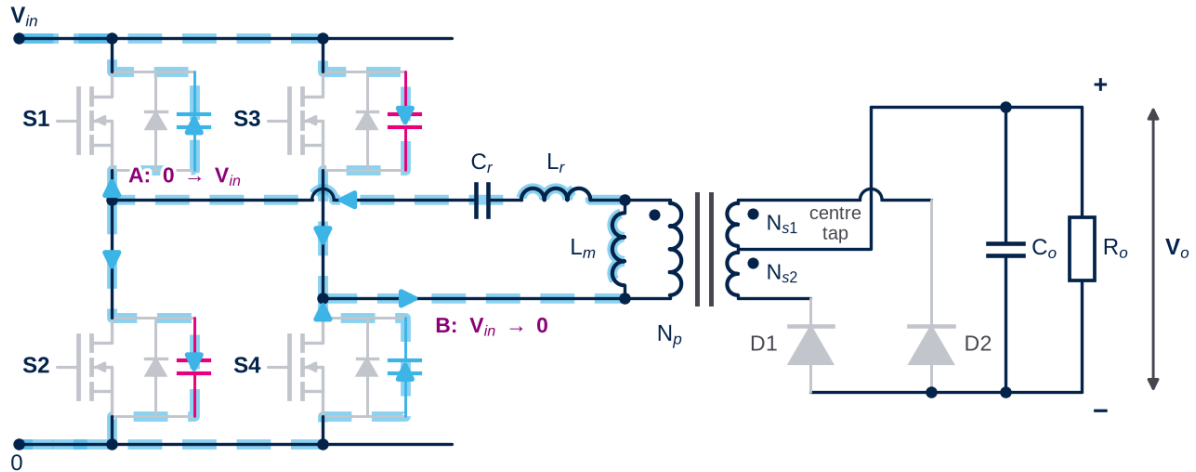


Figure 8: Second half, steps 5 and 6, the mirror image of 1 and 2. S_2, S_3 carry the power delivery (5, through D_2) and then the freewheeling (6), again with the same pair on throughout.

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7 DEAD TIME (a)

all four off · C_{oss} swaps the midpoints back

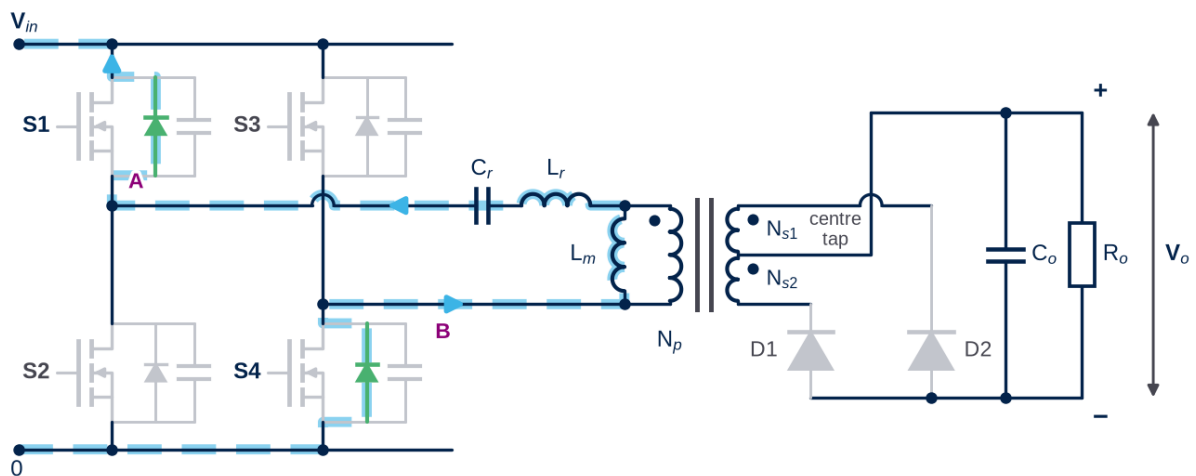


The mirror of 3: the current now arrives at B and leaves A, so B falls and A rises.

Through the capacitances again, never a channel. Magenta: C_{oss} charging, V_{ds} rising; cyan: discharging.

8 DEAD TIME (b) — ZVS

body diodes of S1 and S4 clamp



The same for S1 and S4, and the period closes. ZVS is in 4 and 8 - and in none of the other six.



Figure 9: Steps 7 and 8, the second dead time. The mid-points swing back (7) and the body diodes of S_1, S_4 clamp (8), which is where S_1, S_4 then close on zero volts and step 1 begins again. ZVS happens in steps 4 and 8, and in none of the other six.

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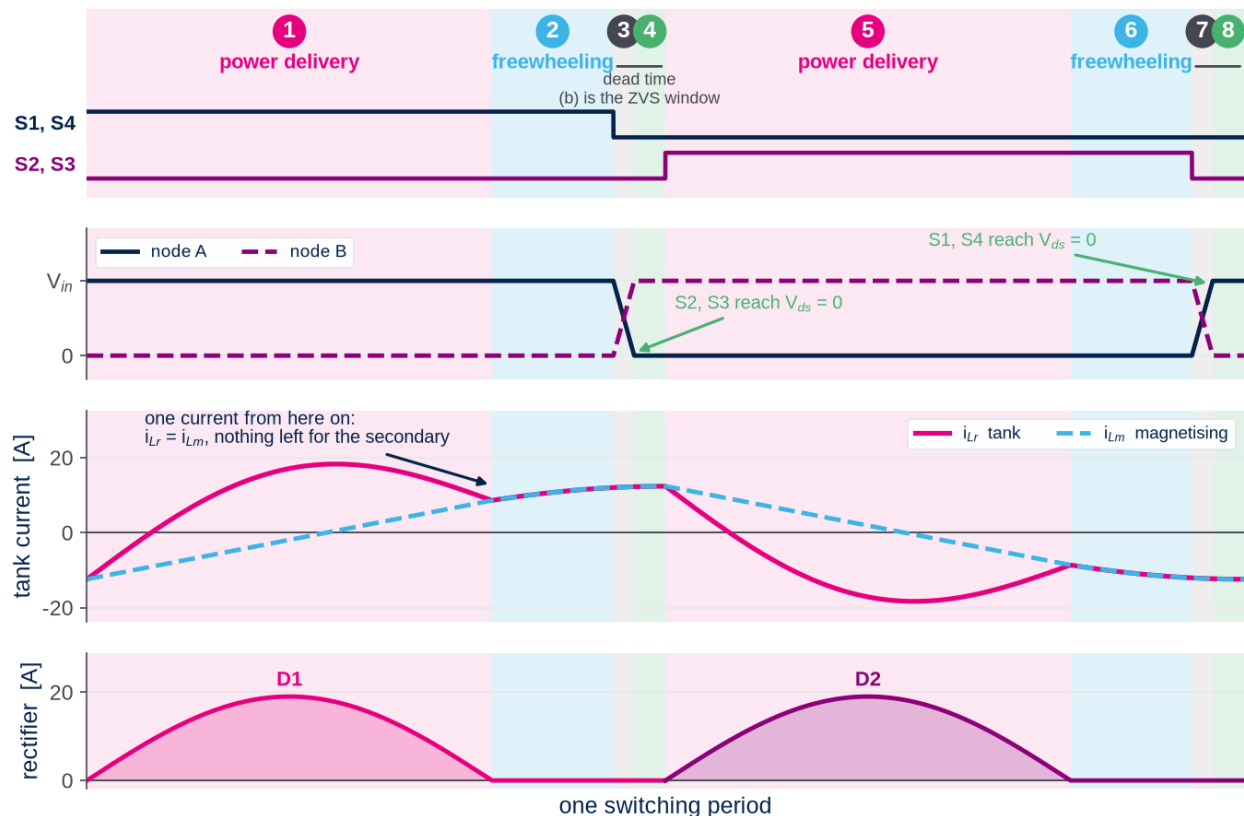


Figure 10: Where each step sits on the waveforms. The numbered bands are the eight panels above, in time order across one switching period: the gates, the two mid-point voltages, the tank and magnetising currents, and the two rectifier currents. Read a panel, then find its band here. Drawn below resonance at $f_{sw} / f_r = 0.70$, with the dead time wider than scale so that steps 3, 4, 7 and 8 have room to be seen; the current amplitudes are this design's own $I_{Lr,pk}$ and $I_{Lm,pk}$ at the low-line corner.

Steps 1 and 2 are the two resonances, and they are the two parts of *one* half cycle: the same pair of switches stays on through both. **ZVS happens in neither**. It happens in the dead time, steps 3–4 and 7–8, when both switches of a leg are off and the tank current carries the mid-point to the other rail — the step in v_d between the gate pulses in Figure 2.

The word “freewheeling” is used for two different intervals. Here it means steps 2 and 6: the secondary is off, L_m has rejoined the resonance, and the same pair of switches is still on. Other documents, among them Toshiba's *Resonant Circuits and Soft Switching* (2019), use the word for the **dead time**, when the current circulates in the body diodes. Both intervals carry current without delivering power, but the dead time exists at every frequency and steps 2 and 6 exist only below resonance. Check which one a source means before comparing numbers.

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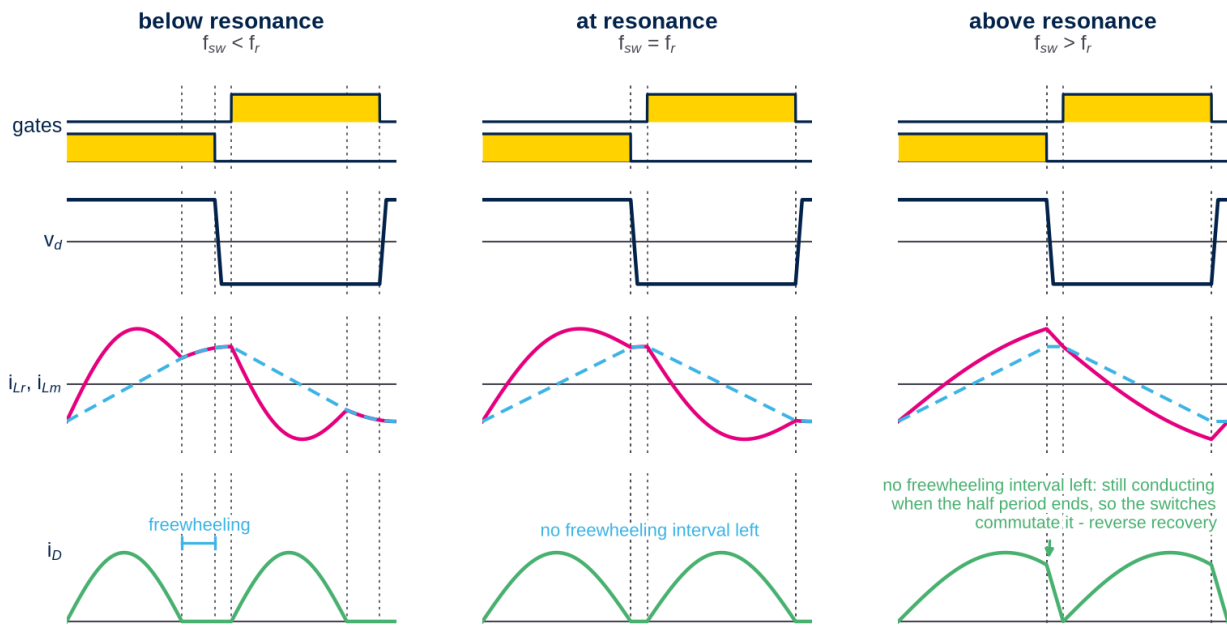


Figure 11: The three cases side by side. Gate signals, the bridge voltage, the resonant current i_{Lr} with the magnetising current i_{Lm} dashed over it, and the rectifier current. What changes across the three columns is how the resonant half period compares with the switching half period. Below resonance there is time left over, and it is spent freewheeling; at resonance the two coincide; above resonance the half period ends first and cuts off the rectifier current while it is still flowing.

- **Below f_r (left).** The resonant half period is shorter than the switching half period. i_{Lr} finishes its half sine early and **lands on i_{Lm}** . From that instant the two are one current and no power crosses to the secondary: that is the freewheeling interval.
- **At f_r (centre).** The two half periods are the same length. i_{Lr} meets i_{Lm} exactly when the bridge switches, and the rectifier current reaches zero at the same instant. This is the most efficient point an LLC has, and it is where a conventional design puts its nominal input.
- **Above f_r (right).** The switching half period ends first, so the resonant half sine is **cut off part way**: i_{Lr} is still well above i_{Lm} when the bridge switches. The primary switches turn off more current, and the rectifier is interrupted while it is still conducting: hard commutation on the secondary.

That freewheeling interval is what f_o is. While it lasts nothing clamps L_m , so the tank is $L_r + L_m$ with C_r , ringing at f_o . It looks flat only because f_o is so much lower than f_r . Lower the switching frequency towards f_o and the interval grows until it fills the whole half period, and the converter delivers no power. So the two frequencies are the two ends of one range: at f_r the flat interval has just disappeared, at f_o it has taken over, and everything useful happens between them.

Symbols are not standard across the literature. Before using a manufacturer's application note alongside this one, check which frequency that document calls f_o . At least one widely used guide gives that name to the series resonance, which is the opposite of the meaning here.

A conventional LLC picks one side and stays there, usually just above f_r at nominal input, where the gain is close to 1 and the circulating current is smallest. Which side the converter designed here runs on is a different question. It is answered in Section 4.7, once the rest of the method is in place.

2.6 The gain function and the three regions

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With the usual first-harmonic approximation the tank gain is

Equation 4

$$M(f_n, Q, \lambda) = \frac{1}{\sqrt{\left(1 + \lambda - \frac{\lambda}{f_n^2}\right)^2 + Q^2\left(f_n - \frac{1}{f_n}\right)^2}}$$

with f_n , Q and λ as defined in the conventions. Two checks can be made on it by eye before trusting any curve drawn from it. At $f_n = 1$ the second term under the root disappears and the first becomes 1, so **$M = 1$ at any Q** . That is the series resonance, and it is why every gain curve passes through the same point. **At no load** ($Q = 0$, so the second term is gone) raising f_n without limit only brings M down to $1/(1+\lambda)$ and no further; that floor is the subject of Section 4.10. Under load the Q term grows with f_n and the gain keeps falling towards zero.

The curve has three regions. Below the capacitive/inductive boundary the tank is capacitive and the bridge hard-switches, which must never be allowed. Between that boundary and f_r the tank is inductive and boosts; above f_r it is inductive and bucks. In the figure the lower boundary is drawn for the full-load curve. At no load it would sit at f_o ; it moves up as load is added, and the next sections are about that.

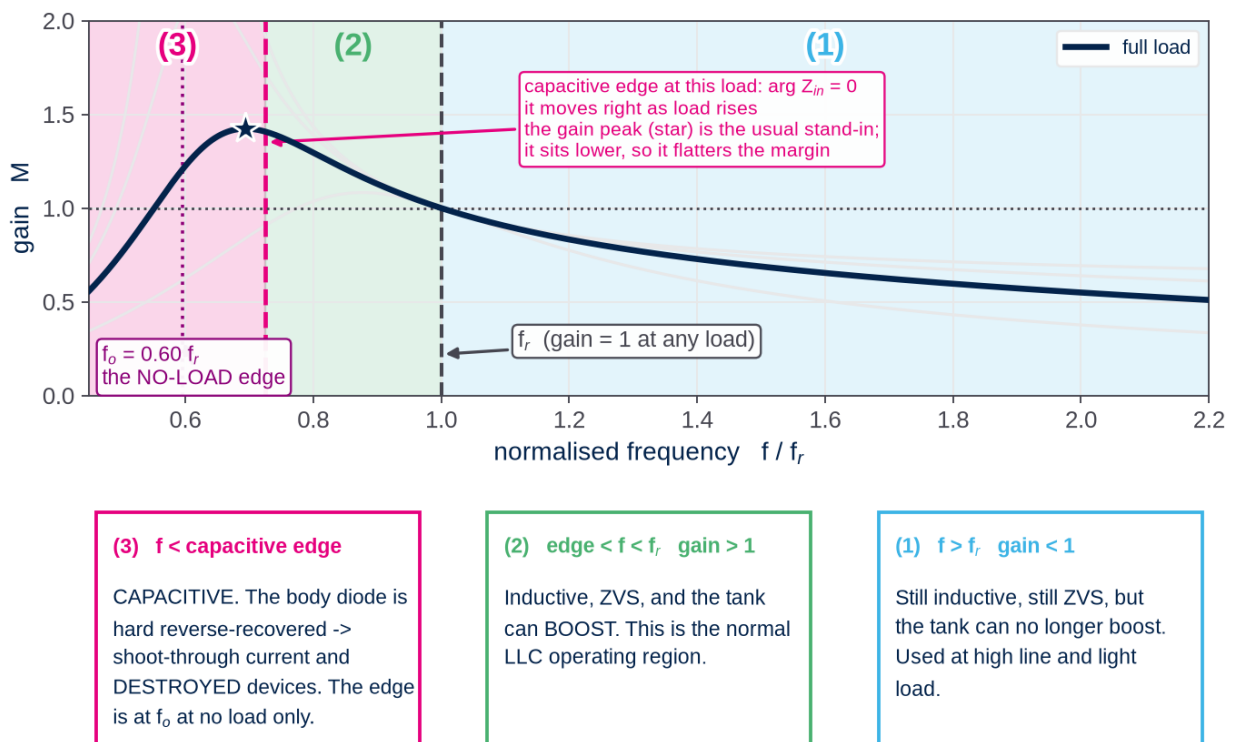


Figure 12: The three operating regions, shaded for the full-load curve drawn. Only the inductive ones are usable, and the boundary between them moves up as the converter is loaded. Region framing after ROHM TechWeb.

2.7 Capacitive and inductive, and why those words

The names have nothing to do with which component dominates. They describe **what the bridge sees**: the phase of the tank input current relative to the square wave that drives it. The tank is a series network. At low frequency C_r dominates and the current *leads*: the bridge is driving something that behaves like a capacitor. At high frequency the inductances dominate and the current *lags*: the bridge is driving something that behaves like an inductor. The

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word describes the load the bridge works into, not a part.

That phase decides whether the switching is soft or hard, which is why the distinction matters:

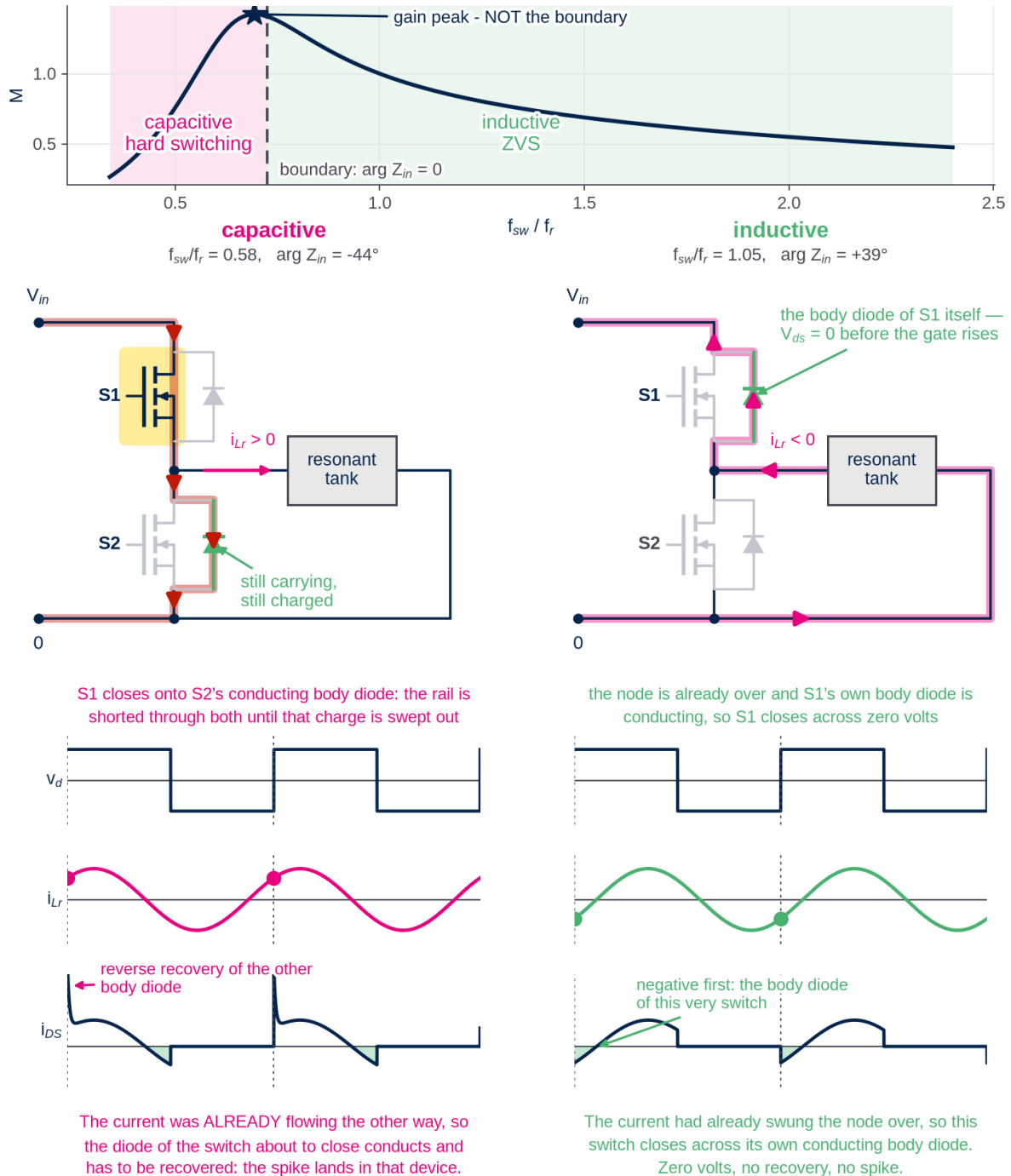


Figure 13: The same converter on either side of the boundary, in three rows. Top: where the boundary sits on the gain curve — note that it is not at the peak, but a little above it. Middle: the leg at the instant S_1 is gated on. This row is the answer to why one side recovers a diode and the other does not: on the left the tank current is already positive, so it is flowing in S_2 's body diode and S_1 closes onto a conducting diode; on the right it is negative, it has carried the mid-point up to the rail during the dead time, and S_1 finds its own body diode conducting at zero volts.

Bottom: what that costs, in the bridge voltage v_d , the tank current, and the drain current of the switch that is closing. The two drain currents are drawn to one scale.

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- **Inductive: the current lags.** When a switch turns off, the current is still flowing in the direction that pushes the bridge node towards the other rail. The dead time lets it do exactly that, and the next device turns on at zero volts. **ZVS.**
- **Capacitive: the current leads.** By the time the switch turns off, the current has already reversed, so it pushes the node *back* where it came from. The next device then turns on into the full rail voltage. Worse, the body diode of the device that was conducting is forced off with reverse recovery through a low impedance. **Hard switching, and the failure mode that destroys parts.**

This is why the capacitive region is not a performance trade-off to be balanced against something else. It is a boundary the design stays on one side of, and the controller has a dedicated protection for the case where it does not.

2.8 Why a diode has to recover on one side and not on the other

The drain-current row above is the difference that reaches the hardware, so it is worth following one switching instant through in both cases. Take the leg alone: a high-side switch, a low-side switch, and the tank hanging off the mid-point. Whichever device has just been turned off, **the tank current does not stop** — it is the current in an inductor. The only question is what carries it during the dead time, and the answer decides everything.

That instant is the middle row of Figure 13, and the rest of this section is what those two pictures mean. The framing follows ON Semiconductor AN-4151.

- **Inductive.** At turn-off the current is flowing in the direction that carries the mid-point towards the *other* rail. It charges one C_{oss} and discharges the other, the node arrives, and the body diode of the incoming device picks the current up and holds the node there. That device is then gated on with V_{ds} already at zero. When its channel takes over, the current in its own body diode falls away with no reverse voltage applied to it, so **there is nothing to recover.**
- **Capacitive.** At turn-off the current has already reversed. It pushes the mid-point back onto the rail it came from, so the body diode of the device that just turned off picks it up and clamps the node there. **The node never moves.** The incoming device is then gated on across the full rail, straight onto a conducting diode.

A diode carrying forward current holds stored charge in its junction, and it cannot block reverse voltage until that charge has been swept out. For as long as that takes, both devices in the leg are conducting and the input rail is short-circuited through them. Three things follow, and all three are bad:

- The current is limited only by the stray inductance of the loop, so it is a **spike, not a current** — many times the tank current, as the drain-current row shows.
- It flows in the incoming device while that device still stands at the full rail voltage, so it is dissipated there, in a few tens of nanoseconds, every cycle.
- When the diode finally does snap off, the di/dt in the loop inductance appears across it as overshoot, which is what actually breaks the part.

This is the reason this note asks for a fast-recovery body diode on the primary (Section 6.24). A superjunction device stores a great deal of charge in a body diode that was never designed to be commutated, and in a full bridge that diode conducts before every turn-on even when the converter is behaving. It is also the reason the oscillator floor exists: f_{Min} above f_o is what keeps the converter out of the left-hand picture, and the anti-capacitive protection is what catches it if that fails.

2.9 The two boundaries are not the same boundary

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This is the point that is most easily confused, because both pairs of words describe positions on the same frequency axis. They are different lines on it.

	capacitive / inductive	below / above resonance
the boundary	where $\arg Z_{in} = 0$	f_r , the series resonance
does it move?	yes — with load	no, it is fixed by L_r and C_r
what it decides	ZVS or hard switching, on the primary	boost or buck, and ZCS or not, on the secondary
which side is allowed	inductive only, always	both sides are legitimate

Table 1: Two classifications, two boundaries, two consequences.

The peak-gain frequency sits between f_o and f_r , and it **rises as the converter is loaded**. At no load it coincides with f_o ; at full load it has moved some way up towards f_r . So the capacitive region is not a fixed band below f_o . It grows as load is added, and a frequency that is safely inductive at light load can be capacitive at overload.

The edge is not quite the gain peak. The tank is capacitive while the phase of its input impedance is negative, so the edge is where $\arg Z_{in} = 0$, and that sits a little *above* the peak of the gain curve. The peak is the usual stand-in because it needs only the gain curve, but it is optimistic in the dangerous direction: it calls a band inductive while the bridge is still hard-switching. Section 7.5 gives both frequencies for the worked design; the verdict there comes from Section 6.5, which measures the condition directly as the time the tank current takes to reach zero after the bridge changes state.

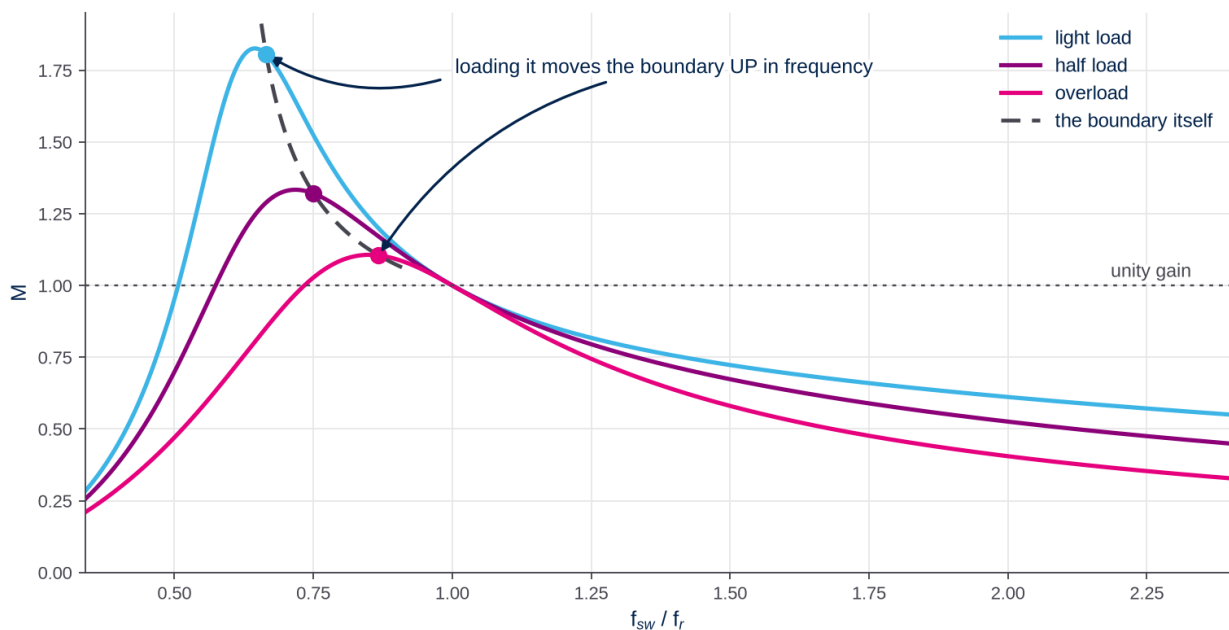


Figure 14: The boundary is not fixed. Loading the converter pushes it to a higher frequency, so a frequency that is inductive at light load can be capacitive at overload. The dashed curve is the boundary itself, traced out as Q varies.

- A converter can be **below f_r and still inductive**. That is the ordinary boosting region, and it is where an LLC normally runs.

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- It can be **below f_r and capacitive**, which is the fault. Going below f_r is not the danger; going below the boundary is.
- It cannot be **above f_r and capacitive**, because the boundary is always below f_r . Everything above f_r is inductive at every load.

What this means for the design rule. Keep the oscillator floor f_{Min} above f_o . That is the *no-load* position of the boundary, so it is the weakest form of the requirement: necessary, but not enough on its own. The real check is the ZVS sweep in Section 6.5, which is run under load and at every phase of the line cycle. Behind that, the controller has an anti-capacitive protection.

2.10 ZVS and ZCS are not the same thing

Both are ways of switching a device while one of its two quantities, voltage or current, is zero, so that their product, the loss, is zero. Here they apply to different devices, for different reasons, and only one of them is guaranteed.

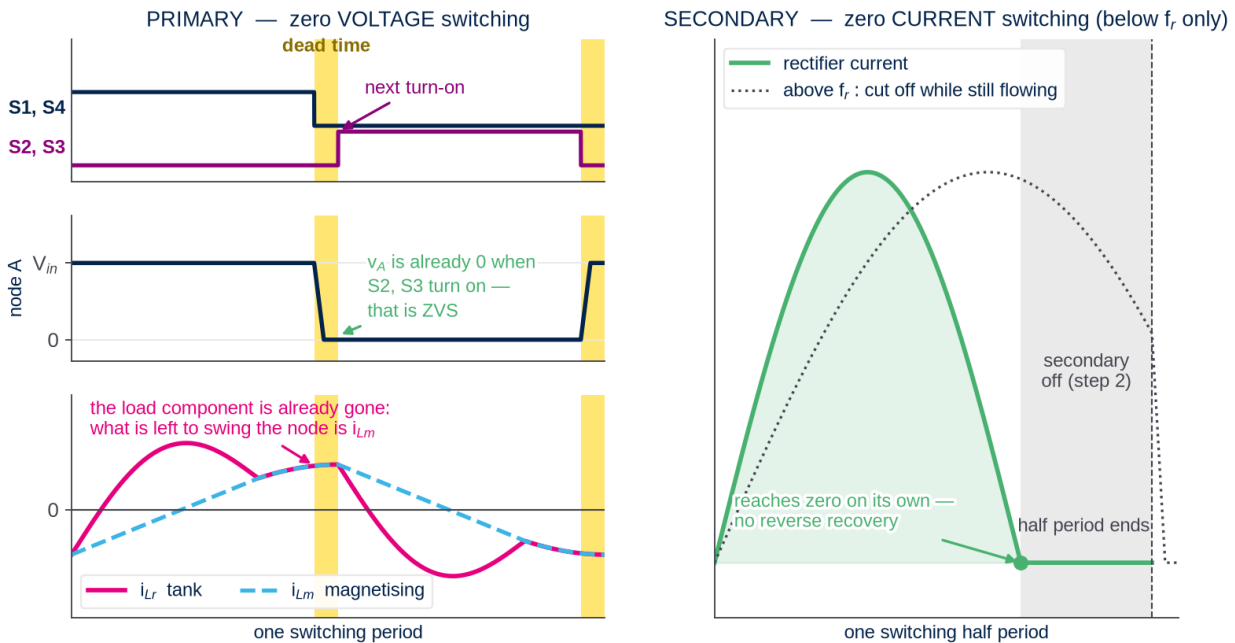


Figure 15: Zero-voltage switching on the primary and zero-current switching on the secondary. Different devices, different mechanisms, different conditions.

- **ZVS, zero-voltage switching, on the primary switches.** The device turns on with no voltage across it. It works because the tank current is still flowing when the previous device turns off, and that current discharges the bridge node during the dead time. It needs **inductive operation**, that is, anywhere above the capacitive/inductive boundary. So it is available on both sides of f_r , and f_o is that limit only at no load. It is required everywhere, and Section 6.5 says how much margin this design has.
- **ZCS, zero-current switching, on the secondary rectifiers.** The device turns off with no current through it, so a diode is not forced out of conduction and a synchronous rectifier has no reverse recovery. It happens because the resonant current reaches zero by itself, which only occurs **below f_r** . Above f_r the rectifier is cut off while it is still conducting, and reverse recovery returns.

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So ZCS is not a property of the circuit but of the operating point. Below f_r the secondary rectifiers turn off at zero current and reverse recovery never appears; above f_r they do not, and it does. A converter that stays on one side settles the question once. A converter whose input moves far enough to cross f_r has to be checked on both sides: its secondary body diode and synchronous-rectifier dead time matter whenever it runs above f_r .

2.11 Sizing the dead time so that ZVS actually happens

ZVS is what makes an LLC efficient, and it is a charge problem rather than a voltage problem. During the dead time the magnetising current must move enough charge to swing the bridge node across the rail before the opposite device turns on. The design quantity is T_{ZC} , the time the tank current takes to reach zero after the bridge transition, and the requirement is simply that it must be longer than the dead time:

Equation 5

$$T_{ZC} > t_D$$

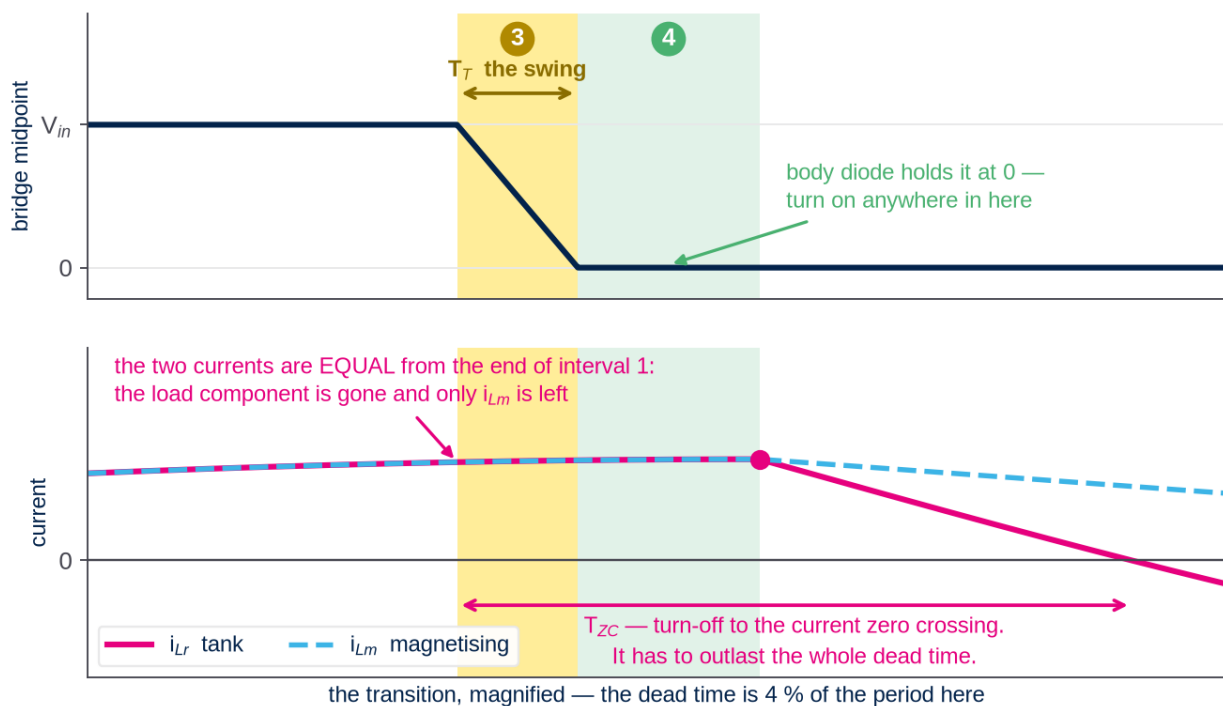


Figure 16: How ZVS happens: the magnetising current, not the load current, discharges the bridge node during the dead time. Two conditions are needed: the current must still be flowing when the gates drop ($T_{ZC} > t_D$), and the swing must finish inside the dead time.

Use the output charge, not the small-signal capacitance. A MOSFET datasheet quotes C_{oss} , $C_{o(er)}$ and $C_{o(tr)}$, and they differ by three to five times. ZVS is a charge question, so $C_{o(tr)}$ (equivalently Q_{oss}) is the one to use. Sizing the dead time from the headline C_{oss} is optimistic by a large factor.

2.12 Not a flyback, and why that changes the core

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Almost everybody meets a transformer first in a flyback, and almost every habit formed there is wrong here. Two of them cost money: **why a core area matters at all in a converter that stores none of the energy it delivers**, and **why the saturation test current is nothing like the peak winding current**. Both come from one difference, which is when the two windings conduct.

In a flyback they never conduct together. The secondary polarity dot is inverted, so the rectifier can only conduct while the switch is off. At every instant exactly one winding is carrying current, and therefore at every instant the whole winding current is magnetising current. Energy goes into the core during the on-time and comes out of it during the off-time, and every joule that reaches the output was first stored in the core. The flux follows the *current*, the core is gapped so that it can hold that energy without saturating, and A_e is sized from the peak current.

In an LLC they conduct together. While the secondary is rectifying, both windings are carrying current and their ampere-turns oppose. Only what is left after the subtraction magnetises the core:

Equation 6

$$N_p i_p - N_s i_s = N_p i_\mu$$

The load current passes *through*, by transformer action; it never enters the core as stored energy. What the core does hold is the magnetising energy $\frac{1}{2}L_m i_\mu^2$, and in this converter that is not a by-product but the mechanism: i_μ is the current that charges and discharges the bridge node during the dead time, which is to say it is what makes ZVS possible (Section 2.11). An LLC transformer is gapped too, but the gap is there to hold L_m at the value the tank asked for, not to store what goes to the load.

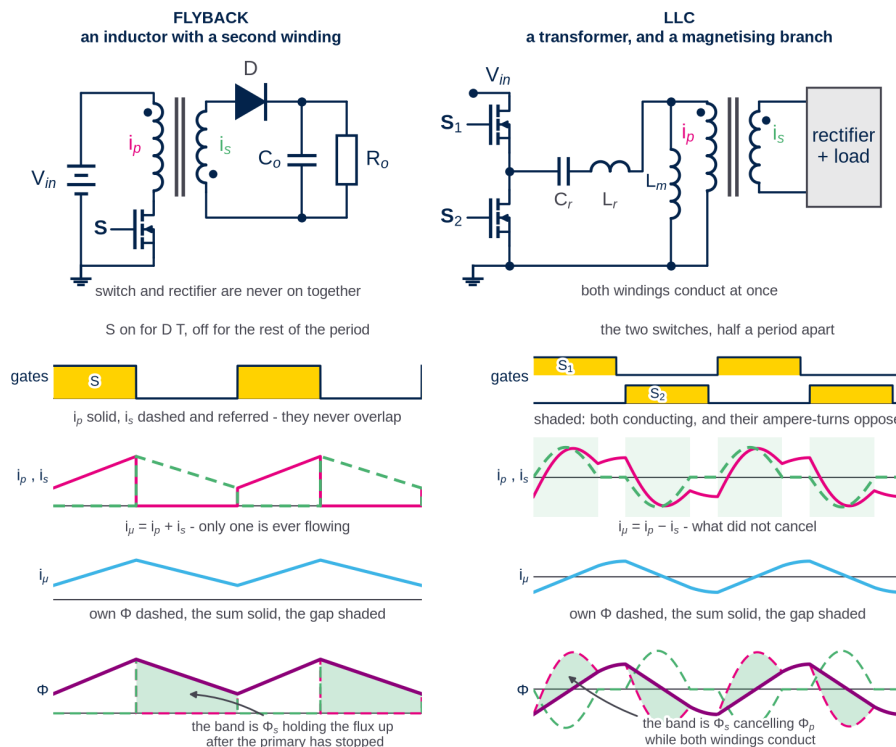


Figure 17: The two transformers from the same four rows. Left: the switch and the rectifier are never on together, so whichever winding conducts carries all of the magnetising current, and the flux is a one-sided ramp that follows it. Right: both windings conduct at once, their ampere-turns oppose, and only the difference $i_\mu = i_p - i_s$ magnetises the core. The i_s traces are referred to the primary, with the sign each polarity dot gives. In the bottom row the dashed curves are what each winding would drive on its own, the solid one is their sum, and the shaded band is what the secondary did to the core: taken away at the same instant on the right, handed over afterwards on the left.

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“An LLC transformer stores no energy” is a slogan, not a statement. It stores the magnetising energy, twice every switching period, and the design leans on it. What it does not store is the energy delivered to the load.

So why does A_e come into it at all? Because Faraday's law does not ask whether anything is being stored:

Equation 7

$$B(t) = \frac{1}{NA_e} \int v dt$$

with v the voltage across the winding, N its turns and A_e the effective area of the core it is wound on. A winding with a voltage across it makes flux; flux in a finite area is a flux density; and ferrite saturates at a flux density. That is equally true of a flyback, an LLC, a mains transformer and a plain choke. **What differs is only the input to the calculation.** In a flyback the volt-seconds are the input voltage times the on-time, which is another way of writing the peak current. In an LLC the conducting secondary clamps the winding to the output for a resonant half period, so the volt-seconds — and with them the flux — are set by the **output voltage alone**. Section 6.11 turns that into an equation with neither the input voltage nor the load in it.

Three consequences follow, and they are worth having in mind long before a core is chosen:

- **The flux does not rise with load.** Twice the output current is twice i_s , and the primary has to carry that extra current as well — so both grow together and their difference, i_μ , does not move at all. (i_p itself does not simply double: it is i_μ plus the reflected load current, and only the second term grows.) A core that is adequate at full load is adequate in overload; what overload threatens is the copper and the semiconductors, not the core.
- **The flux does rise with output voltage.** Anything that lets V_{out} climb raises the flux in proportion, so the ceiling to design against is the over-voltage threshold and not the nominal output.
- **The flux does not fall with switching frequency**, below resonance. The rectifier clamps the winding for a resonant half period $T_r/2$ whatever f_{sw} is doing, which is why f_r and not f_{sw} appears in the flux equation.

The same difference decides what the vendor is asked to test. A DC-overlap measurement is made with the secondary open, which removes the cancellation and puts the transformer back into the flyback condition — all of the test current is magnetising current. Section 6.19 takes that to a number.

2.13 How an LLC is normally designed

The sequence below is the standard one, and every step exists because of something already established above. It is worth having in mind before Section 6, which follows the same order and departs from it in exactly two places.

- **1 Turns ratio from the nominal point.** Choose n so that the converter runs at or near $M = 1$ at nominal input, that is, at f_r . That is where the circulating current is lowest and where the converter should normally run.
- **2 The gain range the tank must cover.** From the input range and the output tolerance, find M_{min} and M_{max} . M_{max} is needed at minimum input and full load, M_{min} at maximum input and minimum load.
- **3 Load resistance referred to the primary, R_{ac} ,** which gives Q its meaning at full load.
- **4 m (equivalently λ) and Q , together, from the peak-gain chart.** They are not independent: for each m there is a curve of attainable peak gain against Q , and the design has to sit under that curve with margin. A larger m gives less circulating current and less peak gain; a smaller m gives the opposite.
- **5 The component values.** Q and R_{ac} give Z_0 ; Z_0 with the chosen f_r gives C_r and L_r ; m then gives L_m .
- **6 Verify.** Check that the gain at the worst corner is still under the peak with margin, that ZVS holds at the worst point rather than at the nominal one, and that the currents and the flux are ones real parts can carry.

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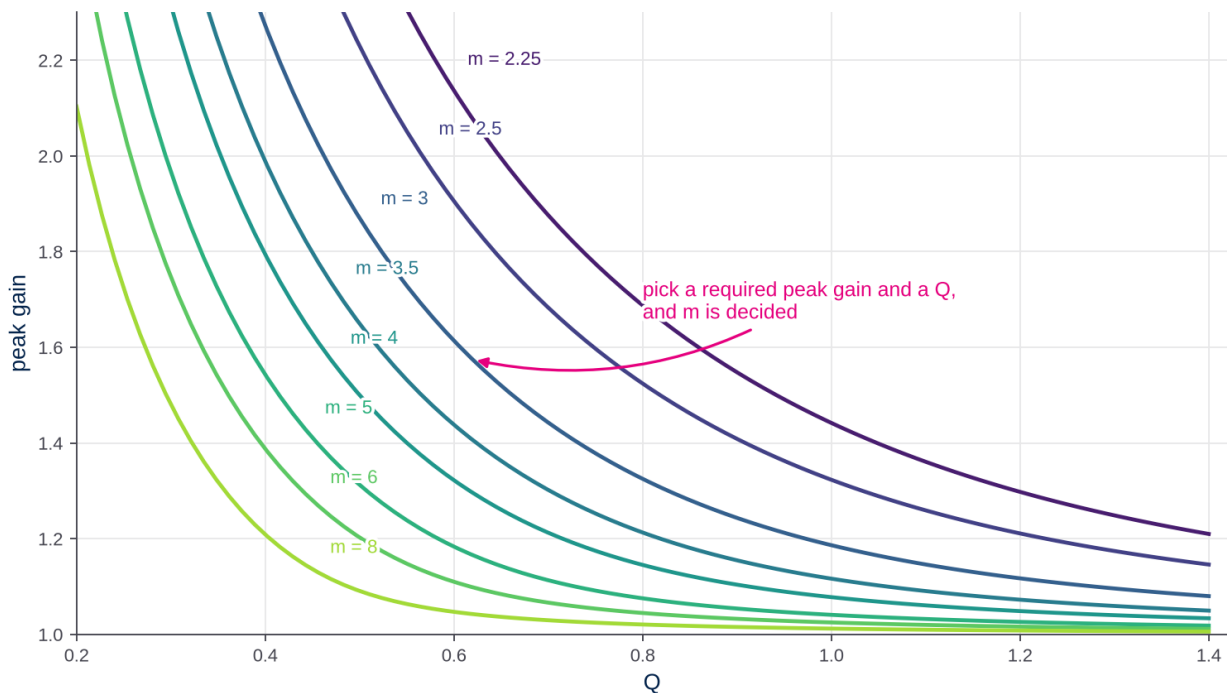


Figure 18: Step 4, and the reason m and Q cannot be chosen separately: the peak gain a tank can reach depends on both. Choosing a required gain and a Q leaves only a band of usable m.

That is the first half. Everything above is the LLC on its own: a tank, a gain curve, and a frequency that moves the operating point along it. None of it involves the mains. The rest of this section is the other half: what a power factor corrector is for, and how the ordinary one works. It can be read without reference to any of the above. Section 3 is where the two halves are put together.

2.14 What power factor correction is

Start with what happens if nothing is done. A bridge rectifier feeding a capacitor holds the bus near the peak of the mains, so the diodes can only conduct during the short window in which the mains voltage is above the capacitor voltage. All the charge the load will use over the whole cycle has to arrive inside that window.

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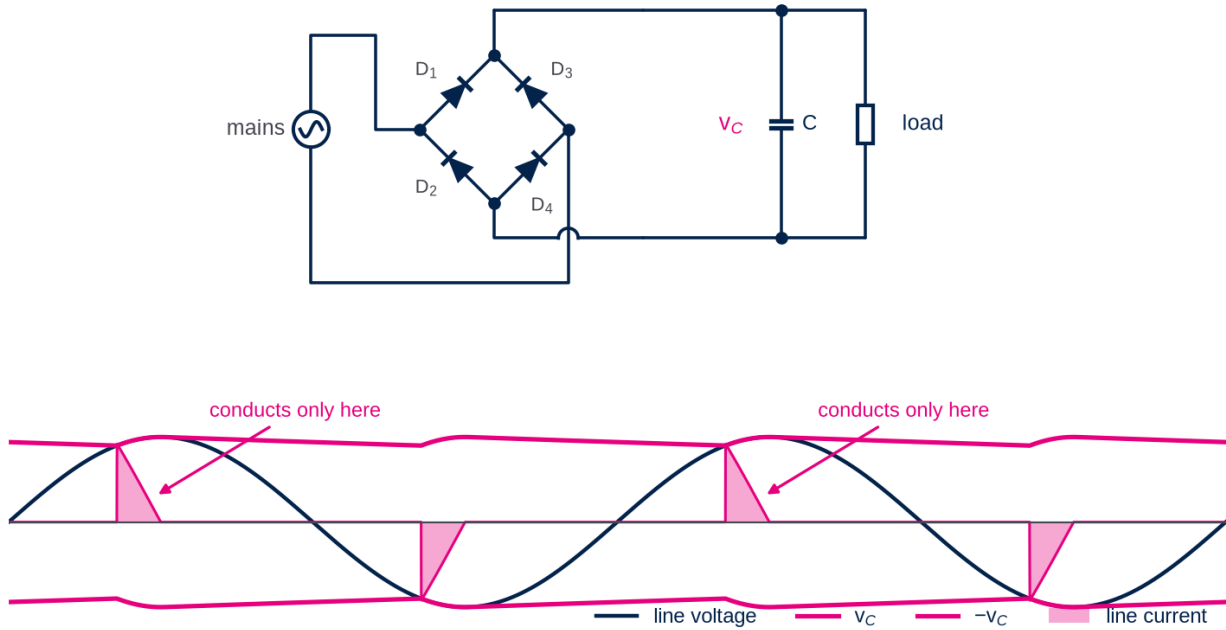


Figure 19: A capacitor-input rectifier and what it draws. v_C stays near the crest, so the diodes conduct only in the narrow windows where the mains is above it, and the current inside them is correspondingly tall.

All the consequences follow from that one fact. The peak current is several times what a resistor of the same average power would draw, so the wiring, the fuse, the bridge and the source all have to be sized for it. And a pulse train that narrow is mostly harmonics.

2.15 Power factor, and why it is not the same as distortion

Power factor is the ratio of real power to apparent power:

Equation 8

$$PF = \frac{P}{V_{rms} I_{rms}} = \cos \varphi_1 \times \frac{I_{1,rms}}{I_{rms}}$$

It has two independent factors, and they fail for different reasons. **Displacement** is the phase between the voltage and the fundamental of the current; it is what a motor gets wrong. **Distortion** is how much of the rms current is at the fundamental at all; it is what a rectifier gets wrong. The current in the figure above is roughly in phase, so its displacement factor is near 1, and it still scores only about 0.6, because the rest of the rms is harmonics.

The related figure of merit is **total harmonic distortion**, the harmonic content as a fraction of the fundamental:

Equation 9

$$THD = \frac{\sqrt{I_{rms}^2 - I_{1,rms}^2}}{I_{1,rms}}, \quad \frac{I_{1,rms}}{I_{rms}} = \frac{1}{\sqrt{1 + THD^2}}$$

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The two are not interchangeable. A converter can hold PF above 0.99 and still fail a harmonic limit, because the standard limits *individual* harmonics in amperes rather than one ratio. Harmonics do not deliver power; they heat the neutral, saturate distribution transformers and disturb other equipment on the same supply, which is why they are regulated. IEC 61000-3-2 sets the limits, and the limit depends on the class. A mains rectifier of this kind is Class D: exempt below 75 W, and limited in milliamps per watt above it. **But Class D only reaches 600 W**; a supply above that falls under the absolute Class A limits. Settle the class first.

2.16 How a corrector fixes it

Force the input current to follow the input voltage, and the converter looks like a resistor to the mains: both factors go to one at once. The standard implementation is a **boost converter placed directly after the bridge**, switching fast enough that the mains looks constant within one switching cycle.

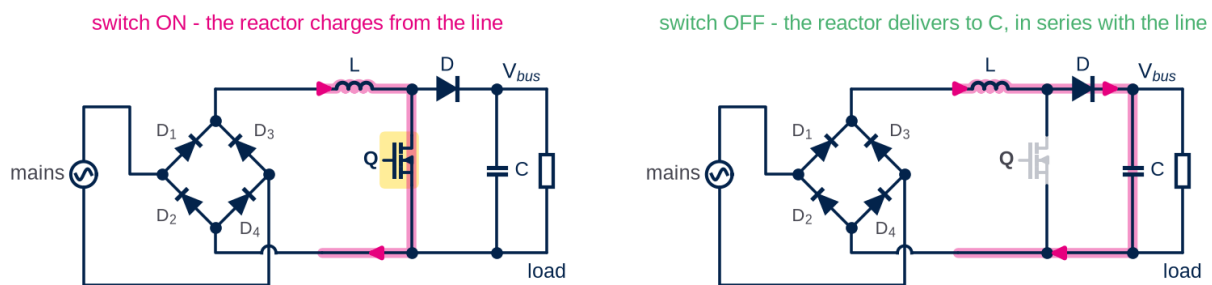


Figure 20: A boost corrector, with the current path traced for each half of the switching period: the reactor charges from the line while the switch is on, and delivers to the bus in series with the line while it is off.

- **Why a boost.** The input is a rectified sine that passes through zero, so the stage must be able to step up by an unlimited ratio near the zero crossing; only a boost can. Its input current is also continuous, which is what makes shaping it possible at all.
- **The inner loop** regulates the inductor current to a reference. It has to be fast compared with the switching frequency.
- **The outer loop** regulates the bus voltage, and it is deliberately made **slower than $2f_s$** . If it were fast, it would fight the ripple on the bus, modulate the current level within the line cycle, and flatten the very shape the corrector exists to make. A slow voltage loop is not a compromise here; it is a requirement.
- **The multiplier** joins them: the shape comes from the mains, the level from the output. That is the whole control law.

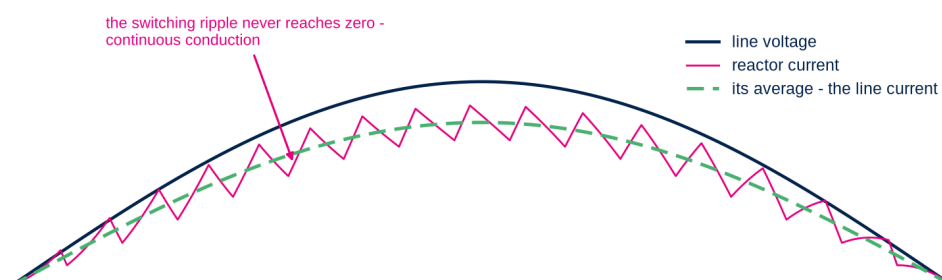


Figure 21: The result, in continuous conduction mode. Switching ripple rides on the reactor current and never brings it to zero; its average follows the input voltage, which is what the two loops were arranged to produce.

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This comes back later. Take the boost stage away and the same conflict returns in a sharper form: the outer loop is then the *only* loop, its output is a power command, and any $2f_l$ ripple that gets through it becomes input current distortion directly. That is Section 6.28.

The corrector delivers what was asked for, and creates a new problem in doing so. A current that follows the voltage means an input power that follows $\sin^2\theta$: zero twice per cycle, twice the average at the crests. The load still wants constant power. **Something has to store the difference**, and where that something sits is the subject of the next section.

2.17 Why they are normally two stages

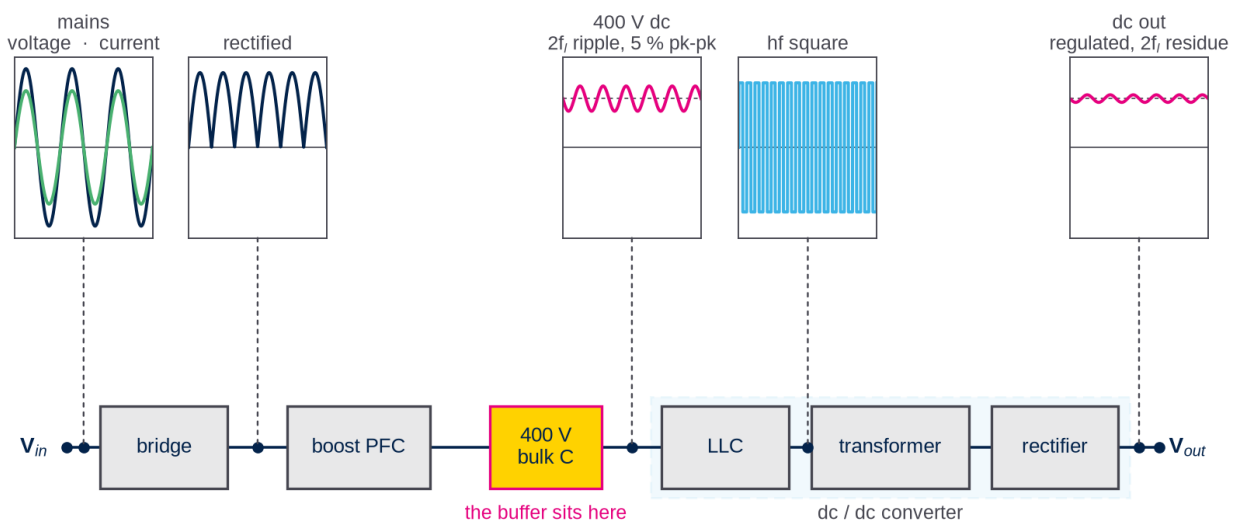


Figure 22: The usual two-stage arrangement, with the waveform at every node. The correction happens in the first block and the LLC works from the dc bus it produces — and the buffer that absorbs the twice-line-frequency pulsation sits between them, at 400 V. The ripple on the dc nodes is drawn larger than it is.

The bus capacitor between a boost PFC and an LLC does two jobs at once, and they are often confused with each other:

- it **buffers** the difference between the pulsating input power and the constant output power;
- it **isolates** the LLC from the mains waveform, so the tank sees a nearly constant input and has to cover only a narrow gain range.

Removing the boost stage removes both services at once. The buffering has to be done somewhere else, and the tank now has to work directly from a rectified sine that goes to zero a hundred times a second. What that costs, and why it works anyway, is the rest of this note.

3 Why single stage, and what it costs

Two steps of that standard sequence are the ones this converter breaks. Step 1 has no nominal point to settle at, because the input sweeps a half sine; and step 2 asks for a gain range that no single tank can cover, which is why the bridge itself has to change. Sections 4.1 and 5 are those two departures.

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3.1 The energy a unity power factor cannot deliver

Drawing a sinusoidal current in phase with a sinusoidal voltage makes the instantaneous input power a raised sine squared. It swings between zero and twice the average, at twice the line frequency:

Equation 10

$$p_{in}(t) = V_{ac}I_{ac} [1 - \cos(2\omega_l t)]$$

The load wants constant power. The difference has to be stored and returned a few milliseconds later. This requirement does not depend on topology; it follows from asking for unity power factor at all. Figure 23 shows the imbalance that must be buffered.

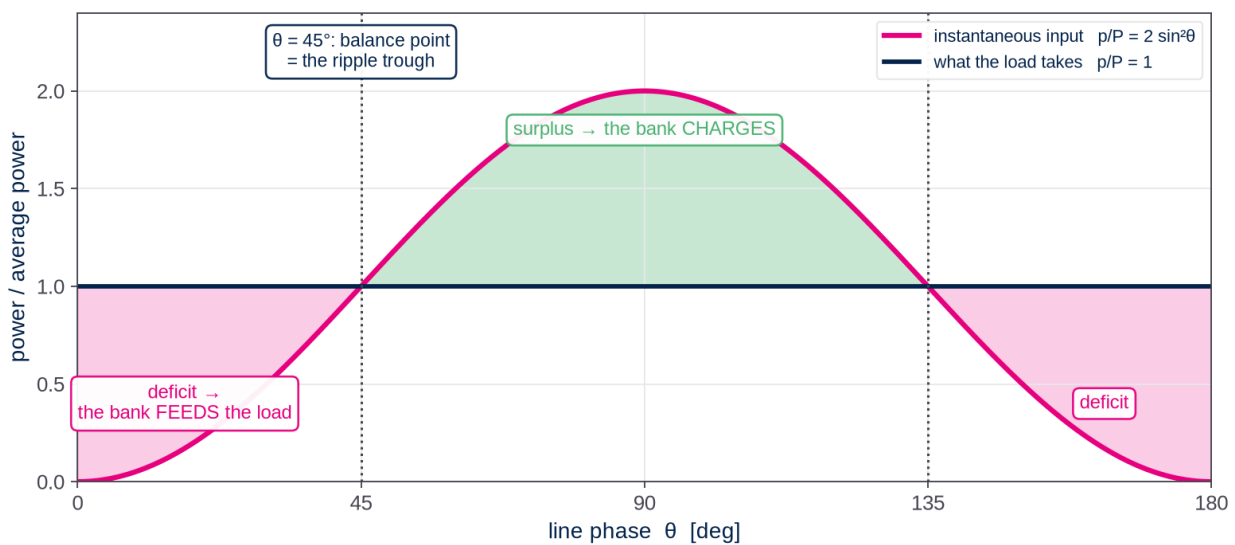


Figure 23: Unity power factor forces a $2f_l$ energy imbalance. The two shaded areas are equal, and they are what a capacitor somewhere must absorb and give back.

3.2 Moving the buffer from 400 V to the output

In the two-stage converter that buffer is the bus capacitor, at 400 V. In the single-stage converter there is no bus, so the buffer moves to the output. **The joules do not change — the voltage they sit at does**, and a capacitor only ever returns the energy between its starting voltage and the lowest the load will accept:

Equation 11

$$E = \frac{1}{2}C(V^2 - V_{min}^2)$$

Hold-up is the clearest way to see the price, because both architectures must ride out the same T_{hold} at full power, so they must store the same energy. On a 400 V bus that may fall to about 320 V the usable window is $400^2 - 320^2$; on an output of a few tens of volts that may fall to $V_{o,min}$ it is $V_{out}^2 - V_{o,min}^2$, smaller by a factor in the hundreds. The capacitance grows by the same factor. That single ratio is the price of the architecture, and it should be settled before anything else is designed: if the enclosure cannot house the bank, the output voltage is what has to change. Section 7.11 works it out for the worked design, where ripple, not hold-up, finally sets the bank.

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The trade in one line. Removing the boost stage removes a switch, an inductor, a diode and a 400 V electrolytic, and their losses with them. It adds a very large low-voltage capacitor bank, a $2f_l$ ripple on the output that the load must tolerate, and a resonant tank that has to work over a much wider range. Whether that is a good trade depends almost entirely on how much output ripple the load will accept.

3.3 What the load must tolerate

Because the buffer is on the output, the output carries the $2f_l$ ripple directly; there is no second stage to reject it. The ripple is set by the bank and the load, not by the control loop, and it cannot be regulated away — attempting that puts distortion on the input current instead (Section 6.28). **How tight the figure has to be is a property of the load, not of the converter**, and it has to come from the specification rather than from a design rule; the worked design allows a few per cent peak-to-peak.

4 Operating principle

4.1 Gain is a boundary condition, not a control variable

In a two-stage LLC the controller commands gain: it moves the switching frequency until the output is right. In a single-stage PFC LLC it cannot, because **both ports are voltage sources**. The output is a millifarad-class capacitor whose voltage cannot move within a switching cycle, and the input is the rectified mains, which the converter does not control either. The instantaneous gain is therefore forced:

Equation 12

$$M(\theta) = \frac{n V_{o, eff}}{V_{drive}(\theta)} = \frac{n V_{o, eff}}{\sqrt{2} V_{ac, eq} \sin \theta}$$

with θ the line phase angle. In a two-stage converter the output voltage is free to move — the loop moves it — so commanding a gain is the same as commanding an output. Here both voltages are pinned from outside, so their ratio is fixed moment by moment and the controller cannot change it.

So what does moving the switching frequency do? It moves **Q** — the loading — and Q is power. The tank is given the gain it must produce, and the frequency decides how much current it draws while doing so. **In this converter frequency is a power command, not a voltage command.**

4.2 Two divergences that cancel

Near the mains zero crossing that equation looks impossible to satisfy: as $\theta \rightarrow 0$ the required gain rises as $1/\sin\theta$ without limit. The converter survives because the load disappears at the same time. A unity-power-factor input draws power proportional to $\sin^2\theta$, so

Equation 13

$$Q(\theta) = Q_{pk} \sin^2 \theta$$

and an unloaded (lossless) LLC has unlimited gain at its lower resonance f_o . The two infinities cancel, and the operating point moves down towards f_o as the mains voltage approaches zero (Section 7.7 draws it for the worked design). **f_o is therefore the frequency floor of the entire design**, and putting the oscillator clamp below it is one of the few mistakes that destroys hardware.

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4.3 Frequency modulation is the power factor correction

Putting these together gives the control law, and Figure 24 follows it round one line half cycle in four steps. Nothing in the first three steps is a choice: they are forced by the two voltage sources. Only the fourth is something the controller does.

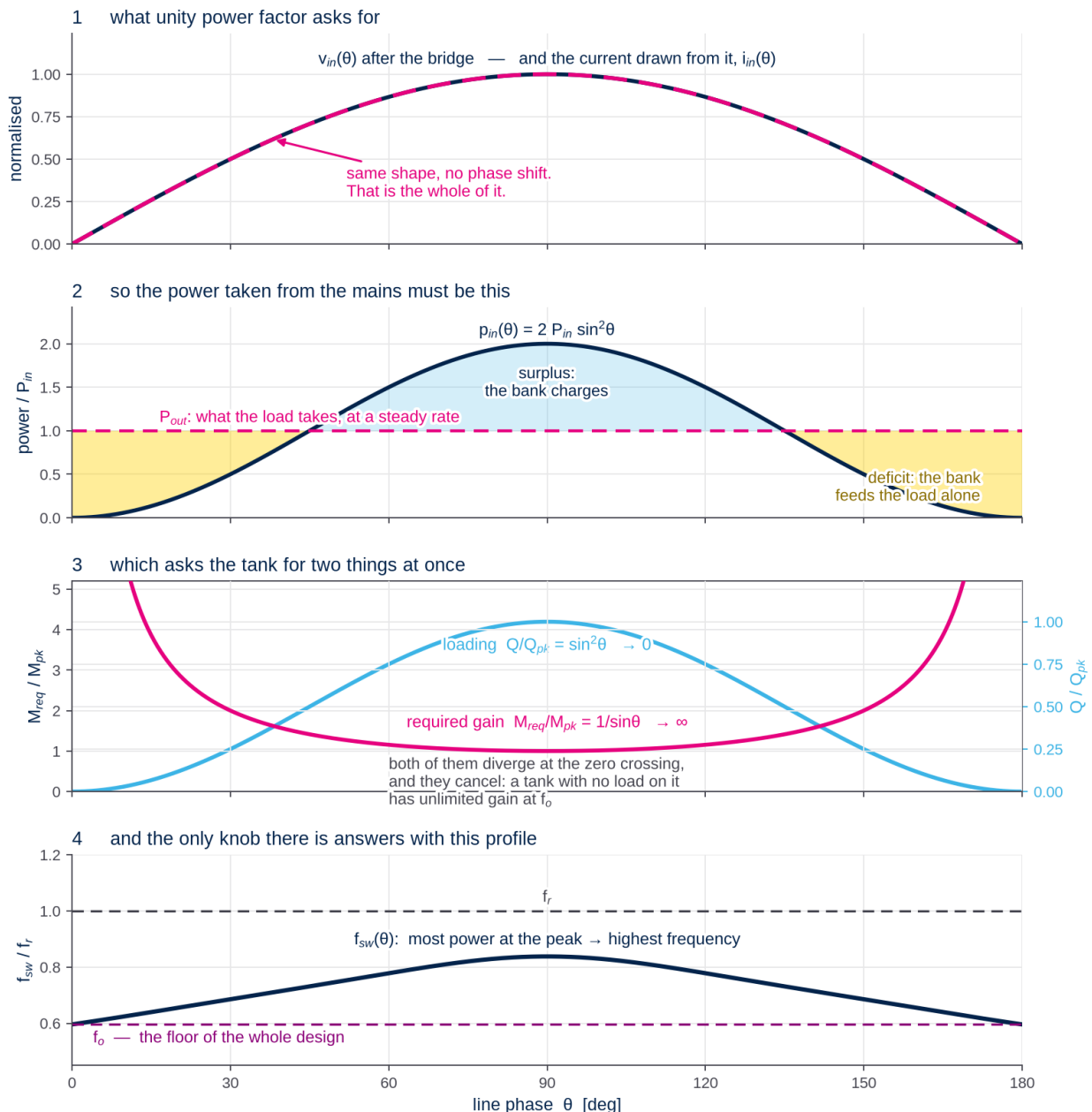


Figure 24: How the power factor gets corrected, over one line half cycle. 1 Unity power factor means only that the current drawn has the same shape as the rectified mains voltage and no phase shift between them. **2** A sine voltage times a sine current is a $\sin^2$ power, so the power taken from the mains swings between zero and twice the average twice per line cycle, while the load takes a steady P_{out} : the shaded areas are what the output bank has to buffer. **3** Because both ports are voltage sources the tank is handed a gain it must produce and a load it must carry, and both of them run away at the zero crossing. **4** The switching frequency is the only free variable, and this is the profile that satisfies 1 to 3. Drawn normalised; the same profile with numbers on it is in Section 7.7.

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Read the four panels as one sentence. **The mains offers** a voltage that goes to zero twice per cycle, and there is no bulk capacitor to hide that, so the tank sees it raw. **Unity power factor demands** that the current drawn follow that voltage exactly, which fixes the instantaneous power at $2 P_{in} \sin^2\theta$. **The tank is therefore told**, at every instant, both what gain to produce (Equation 12) and how hard it is being loaded (Equation 13). **And the controller answers** with the one variable it has: the switching frequency. There is no current loop, no multiplier against a rectified reference and no separate corrector stage. **The frequency profile $f_{sw}(\theta)$ is the power factor correction.**

At the line phase	$\sin\theta$	instantaneous power p/P_{in}	required gain M_{req}/M_{pk}	loading Q/Q_{pk}
$\theta = 90^\circ$, the peak	1.000	2.000	1.000	1.000
$\theta = 60^\circ$	0.866	1.500	1.155	0.750
$\theta = 45^\circ$	0.707	1.000	1.414	0.500
$\theta = 30^\circ$	0.500	0.500	2.000	0.250
$\theta = 10^\circ$	0.174	0.060	5.759	0.030
$\theta \rightarrow 0$	0	0	∞	0

Table 2: The same half cycle at three instants, in the normalised form of Figure 24. Only trigonometry: no part value enters until a tank has been chosen.

The last two rows are where the topology earns its reputation. The gain demand runs away, but so does the ability to meet it, because the load is vanishing at the same time and an unloaded tank has unlimited gain at f_o . The frequency simply walks down towards f_o and the current drawn goes to zero with the voltage, which is exactly what unity power factor asked for.

What the controller actually has to work with. Two signals, on two different time scales:

- **The FB pin says how much.** It carries the output of the error amplifier through the optocoupler, and it commands an input power (Section 4.6). The voltage loop that drives it crosses well below the line frequency, so over any one line half cycle this number is effectively a constant: it sets the *height* of the profile in panel 4, not its shape.
- **The HVSU pin says where in the cycle.** The same pin that starts the IC is also its input-voltage sense, so the controller knows the instantaneous rectified mains without a separate divider. That is the fast signal, and it is what the *shape* of the profile is built from.
- **The internal PFC and THD block combines them** and drives I_{EA} , the one current that moves the oscillator (Equation 59). Everything in panel 4 reaches the power stage through that single term.

The draft datasheet does not publish the law inside that block, only that it exists and what it is for. So this note does not state it. What it does instead is compute, from the tank, the profile that *has* to come out if the current is to follow the voltage — which is what Figure 24 and every $f_{sw}(\theta)$ curve here are. Measuring $f_{sw}(\theta)$ on the first board is therefore not a nice-to-have: it is how the block gets characterised.

The zero-crossing dead zone. Very near $\theta = 0$ the frequency the profile asks for falls below the oscillator floor f_{Min} , and the converter stops drawing current until the mains comes back up. This is inherent to the topology and shows up as third-harmonic distortion on the input current. It is one of two mechanisms that set the achievable THD; the other is the voltage loop (Section 6.26).

4.4 Solving for the profile without a root search

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Computing that profile seems to need a numerical root search: at each θ the gain equation has to be solved for f_n , and it has two roots: a capacitive one and an inductive one. **It does not.** Substituting $M_{\text{req}} = M_{\text{pk}}/\sin\theta$ and $Q = Q_{\text{pk}}\sin^2\theta$ and writing $x = 1/f_n^2$ turns the gain equation into a cubic in x :

Equation 14

$$\lambda^2 x^3 + (q - 2\lambda(1 + \lambda))x^2 + ((1 + \lambda)^2 - 2q - \frac{u}{M_{\text{pk}}^2})x + q = 0, \quad u = \sin^2 \theta, \quad q = Q_{\text{pk}}^2 u^2$$

One root is always negative: the cubic equals $+q > 0$ at $x = 0$ and goes to $-\infty$ as x goes to $-\infty$, so it crosses zero somewhere below $x = 0$. The other two roots are where one curve is crossed twice. For $x > 0$ the inverse gain $1/M^2$ of the tank, as a function of x , starts at infinity, falls to a single minimum and rises again. So a required gain that the tank can produce is met at **exactly two** positive values of x : the capacitive crossing and the inductive one. All three roots are then real. The physical, inductive root is the **middle** one, because $x = 1/f_n^2$ decreases as frequency increases. There is no branch to choose: in the trigonometric form of Cardano it is always the $k = 1$ branch. If the required gain is below that minimum there is no positive root at all, which is the no-solution case of Section 4.10 rather than a numerical failure.

Why this matters beyond elegance. A grid search needs an upper bound on f_n , and a bound taken from the ac maximum instead of the morphing corner quietly leaves out the real operating point. The closed form has no grid, no bracket and no starting guess, so that failure cannot happen. The inverse problem — finding the $\sin^2\theta$ that gives a required f_n — is a quadratic and is also exact.

4.5 Reading the gain chart of a single-stage converter

Every LLC note opens with a gain chart, and the one in Section 2.5 (Figure 5) is the standard picture: a family of curves on an f_{sw}/f_r axis, one horizontal line for the gain the converter must produce, and a crossing where the two meet. A single-stage converter is drawn the same way and has to be **read differently**, and that is the commonest source of confusion about this topology. Figure 25 puts the two side by side.

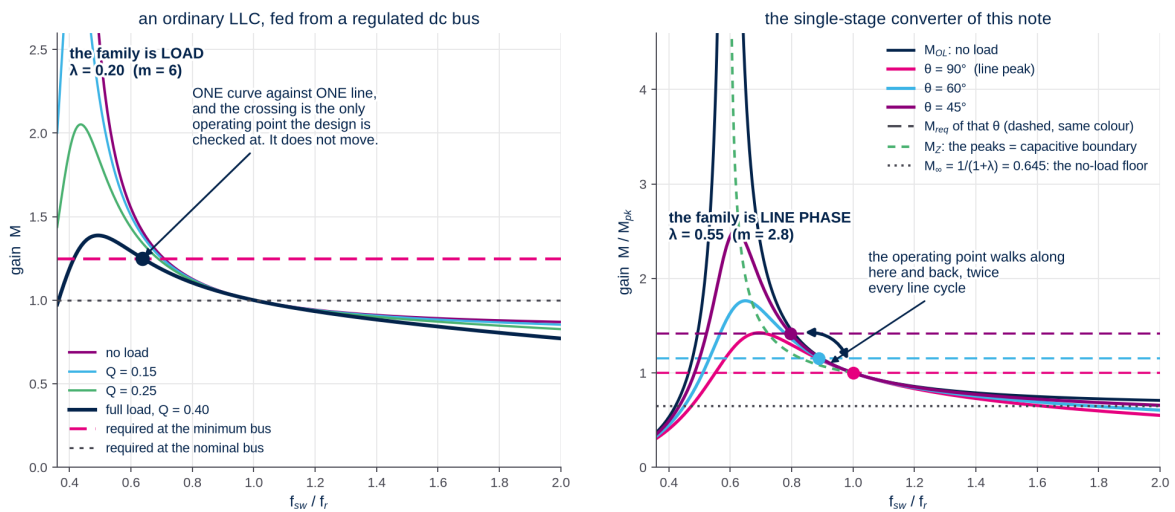


Figure 25: The ordinary gain chart and the single-stage one. Same axes, same gain equation, read differently. On the left the family of curves is load and the converter sits still at one crossing. On the right the family is line phase, every curve has its own required-gain line in its own colour, and the converter travels the whole family and back twice in every line cycle. The right-hand panel is normalised to M_{pk} , the gain required at the line peak, so that it holds for any input voltage.

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Six things are different, and each one changes what the chart is being used for.

- **The family parameter is not load, it is line phase.** On an ordinary chart the curves are $Q = 0$ (no load) through Q at full load, and a design sits on *one* of them. Here the load is fixed at full and the curves are $\theta = 90^\circ, 60^\circ, 45^\circ$... because $Q = Q_{pk} \sin^2\theta$ falls as the mains voltage falls. The same equation is being used to plot a different sweep.
- **The operating point moves, and fast.** On the left it is a point: it stirs only when the load steps or the bus sags, and slowly. On the right it is a **trajectory** that is traversed 100 or 120 times a second. There is no single operating point to check, so a design is checked along a path.
- **There is not one required gain, there is a ladder of them.** $M_{req}(\theta) = M_{pk}/\sin\theta$ gives a horizontal line per phase, and they climb without limit towards the zero crossing. They are drawn in the colour of the curve they belong to for a reason: **each curve must be compared with its own line and with no other.**
- **The curves get taller as the lines get higher.** That is the only reason the topology works. Halve $\sin\theta$ and the requirement doubles, but Q falls by four, and a gain curve with a quarter of the damping is far more than twice as tall. The margin grows towards the zero crossing rather than shrinking.
- **M_z , the locus of the peaks, is drawn.** It is the capacitive boundary of Section 2.9, and every crossing has to lie to the **right** of it. On an ordinary chart one boundary for one curve is enough; here the boundary has to be cleared at every phase, so it is plotted as a curve of its own.
- **$M_{\infty} = 1/(1+\lambda)$ is on the chart at all.** It is the floor the no-load curve flattens onto at high frequency, and an ordinary LLC never goes near it because it never needs a gain that small. A single-stage converter at high line does: if the required gain falls below this line there is **no solution at any frequency**, and burst mode has to take over. Section 4.10 is about that bound.

The mistake this chart invites. Look at the right-hand panel and it is tempting to check the tallest curve against the lowest line, or the full-load curve against the 45° line, and conclude there is plenty of margin. There is no such operating point. θ sets the curve *and* the line together, so the only meaningful comparisons are the three same-coloured pairs. This is also why this note does not draw an overview frame with several line voltages on one set of curves: at the line peak every condition looks comfortable, and the phases where it is not comfortable are not on the picture.

What the chart is **for** follows from that. Read at the lowest equivalent input it answers “can the tank make the gain, and does it stay inductive doing it?” Read at the highest equivalent input it answers “how high does the frequency have to go, and is there still a solution at light load?” Those are two different questions, and they are why Section 7.6 draws this chart twice.

4.6 The feedback pin is a power command

Combining the burst-mode expression in the datasheet with the multiplier in the block diagram gives the physical meaning of the feedback voltage. It is not an error signal in the usual sense:

Equation 15

$$V_{FB} = V_{os} + \frac{2K_{HV}}{K_M K_{FF}} R_{CS} P_{in} = 0.5 \text{ V} + 0.167 \frac{\text{V}}{\Omega \cdot \text{W}} R_{CS} P_{in}$$

Substituting the 2.8 V feedback span gives $2.8/0.167 = 16.8 \Omega \cdot \text{W}$, which is exactly the maximum-power rule in the datasheet. **Maximum power limiting, burst entry and overload detection therefore all sit on one scale**, and that scale is R_{CS} . The outer optocoupler loop commands an *input power*; the internal loop distributes that power over the line cycle as $\sin^2\theta$.

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This is why the current-sense pin tolerates no filter and no series resistor. The maximum-power law and the over-current threshold read the same pin, so anything that shifts it shifts both.

4.7 Which side of resonance this converter runs on

Section 2.5 left this question open. Now it can be answered: a single-stage converter does not pick one side. The equivalent input follows a half sine every 10 ms, the required gain follows it, and the operating point crosses f_r and comes back within one line cycle. How much of the cycle it spends on each side depends on the mains voltage.

Section 7.7 shows the profile of the worked design at four inputs.

Both sides have to be designed for. The boosting side sets the gain requirement and the ZVS margin. The bucking side sets the top switching frequency, and it takes the secondary out of zero-current switching — so the rectifier body diode and the SR dead time matter here in a way they do not in a conventional LLC. A design checked only at the line peak, or only at low line, has checked only one of the two.

4.8 Why λ must be about 0.5

A classic LLC uses $m = L_p/L_r$ between 5 and 10, that is λ between 0.11 and 0.25. A single-stage converter cannot: it has to produce very high gain near the zero crossing, and peak gain falls as λ falls. The evaluation board ST publishes for this part runs $\lambda = 0.500$, and the worked design comes out close to it (Section 7.4): the value comes from the topology, not from one particular design.

The price is circulating current. A small L_m means a large magnetising current that carries no power but does carry conduction loss, at every load. **A single-stage PFC LLC is less efficient than a two-stage LLC, and that is inherent rather than a design defect.**

4.9 $f_{sw,max}$ does not check the tank — it computes it

Two of the four candidates for λ carry the specified maximum switching frequency in a denominator:

Equation 16

$$\lambda_2 = \frac{\lambda_1}{1 - \left(\frac{f_r}{f_{sw,max}}\right)^2} \quad \lambda_{TD} = \frac{\lambda_1}{1 - \frac{\pi^2}{8} \left(\frac{f_r}{f_{sw,max}}\right)^2}$$

λ_1 is the plain minimum-gain condition. The other two are the same condition with the frequency ceiling folded in: the tank has to reach the lowest required gain *before* it runs out of frequency. With the usual $f_{sw,max} = 1.5 f_r$ the squared ratio is 0.44, both denominators fall well below one, and **the number typed as $f_{sw,max}$ roughly doubles the λ the tank is asked for**. Set it close to f_r and the denominators go to zero and the required λ to infinity.

So where does $f_{sw,max}$ come from? The common rule is $1.5 \times f_r$. It is a **choice**, not a limit — and it sits upstream of the tank, so changing it changes L_m . A maximum in a design sheet usually only checks a result; this one is an input, so a round number typed into it changes the tank without anyone noticing.

It should not be confused with f_{Max} , the oscillator ceiling. That one *is* a limit: R_T , C_T and the idle time fix it in silicon, the control loop cannot push past it however much gain it wants, and it is checked after the fact by k_{ceil}

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(Section 7.3). The two sit at opposite ends of the design: one decides the tank, the other reports whether the oscillator can serve it.

4.10 The other bound on λ , and where it has no solution

The condition above sets a floor under λ . The ceiling comes from the opposite end: the tank must also be able to reach the *lowest* required gain, which happens at the highest equivalent input and no load. That is where the usual design rule quietly breaks down, because the no-load gain of an LLC does not fall without limit. Raising the frequency only brings it down towards a limit (an asymptote):

Equation 17

$$M_{\infty} = \lim_{f \rightarrow \infty} M_{no\ load} = \frac{1}{1 + \lambda}$$

If the required minimum gain lies below M_{∞} , no frequency satisfies the condition at all. A spreadsheet solving for that frequency returns an error rather than a number, and the error is easy to mistake for a broken formula. It is not: the question simply has no answer.

Whether a given tank sits above or below the asymptote is a matter of a per cent or so, and no full-load check shows it; Section 7.4 reports it for the worked design.

Losing the solution is not a failure. No load at high line is the region **burst mode** owns, not frequency control. Under load the gain curve falls further and the corner arrives sooner, so every full-load check is unaffected — which is exactly why this can go unnoticed. What matters is to know which of the candidate turns ratios can still regulate by frequency alone at no load, because no full-load number shows it.

5 Topology morphing

A resonant tank has a usable gain range of roughly 1.5:1 at full load. A universal mains asks for 2.93:1. No single tank covers that and keeps ZVS, so the L6790A changes the bridge instead.

Below a mains peak of 235 V the part runs a **full bridge**, which drives the tank with twice the rail. Above 245 V_{pk} it runs a **half bridge**. The tank therefore sees 173.2 to 332.3 Vac equivalent, a range of 1.92:1, and one tank can cover it.

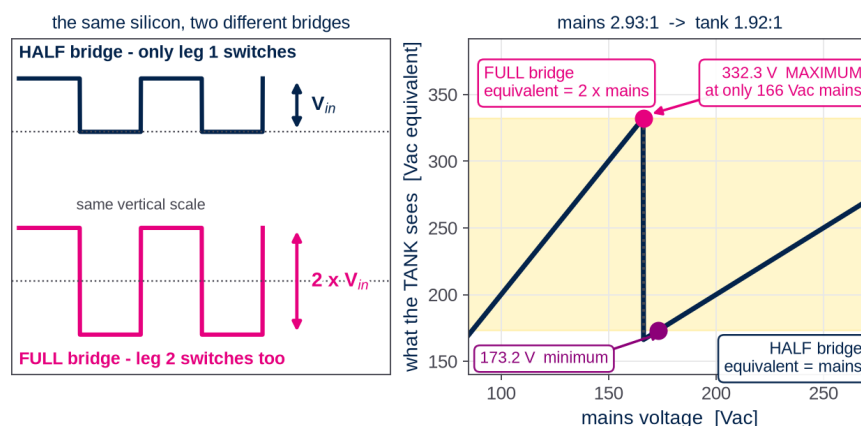


Figure 26: Morphing collapses a 2.93:1 mains range into a 1.92:1 range at the tank. The worst corners are the morphing edges, not the ends of the mains range.

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5.1 Full bridge

Diagonal pairs conduct together. S1 and S4 put $+V_{in}$ across the tank, S2 and S3 put $-V_{in}$ across it, so the drive is a square wave of amplitude V_{in} and its fundamental is $(4/\pi)V_{in}$. All four devices switch, and each carries the tank current for half of every period.

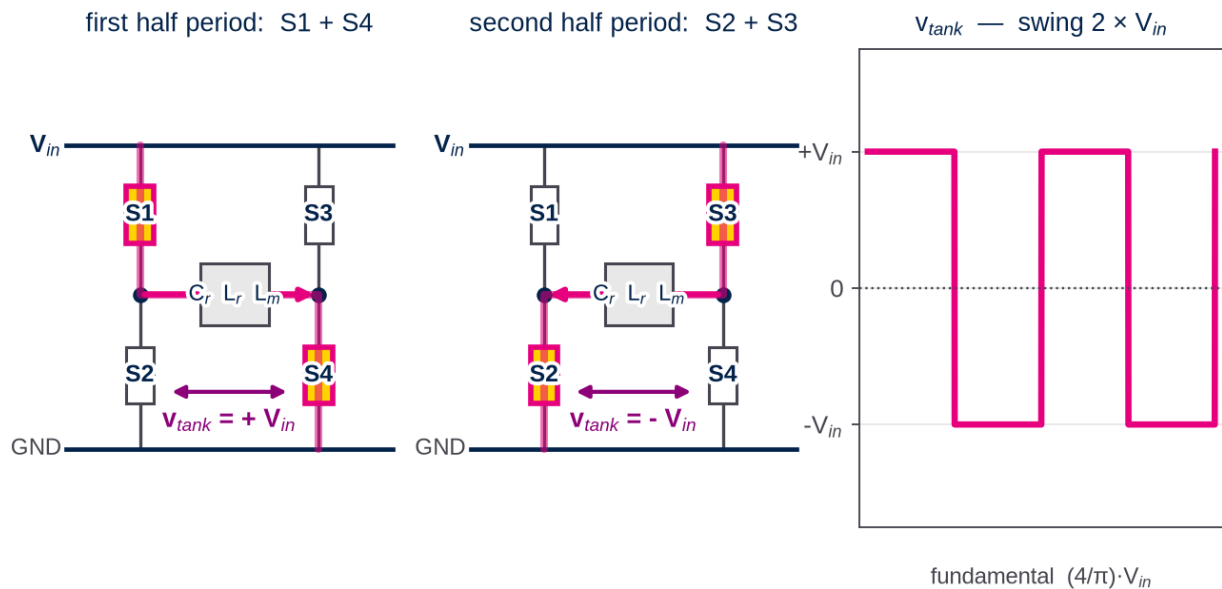


Figure 27: Full bridge: conduction path in each half period, and the resulting tank drive.

5.2 Half bridge

LOUT2 is held statically high. S3 never turns on and S4 never turns off, so the leg-2 midpoint becomes ground and the bridge output swings between 0 and V_{in} . C_r is already in series with the tank, so it blocks the $V_{in}/2$ of dc for free and the tank sees $\pm V_{in}/2$ — a fundamental of $(2/\pi)V_{in}$, exactly half. No extra hardware is involved.

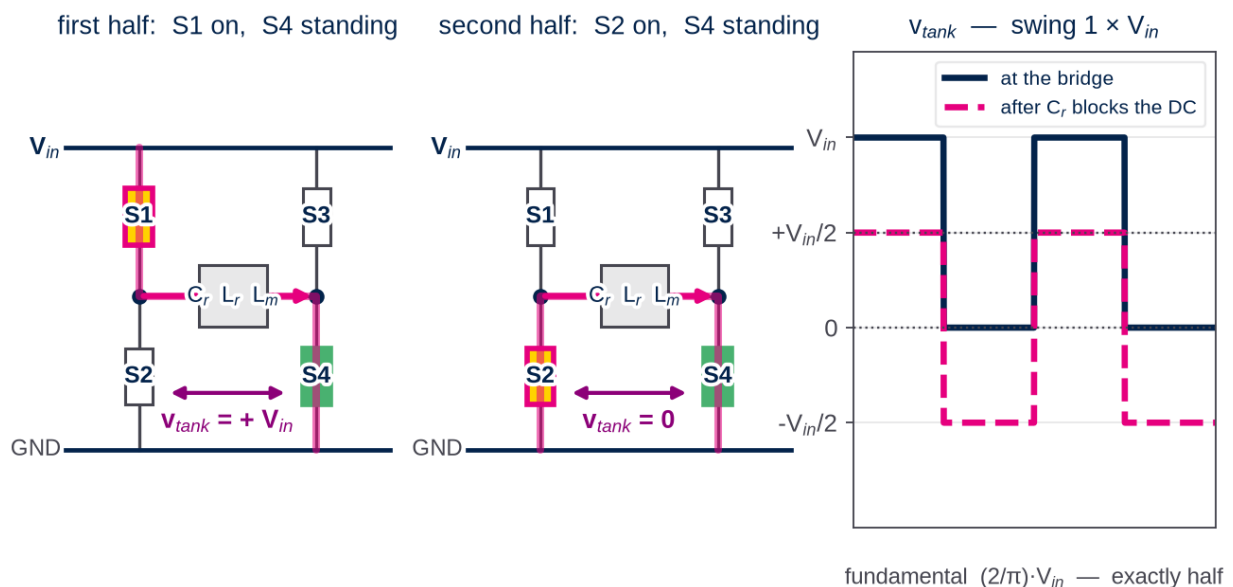


Figure 28: Half bridge: leg 2 stops switching, and C_r removes the dc that the asymmetric drive creates.

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The standing device is the hottest one. S4 never switches, but it conducts the full tank current continuously. It is easy to overlook exactly because it is not switching, and in the worked design it is the single largest loss item (Section 7.12).

5.3 How the controller does it

Only one gate signal changes between the two modes, which is why morphing costs nothing in hardware. The thresholds themselves are **fixed constants inside the IC** and cannot be moved by the designer; R_{CFG} decides only whether morphing is enabled at all and where the brown-out threshold sits.

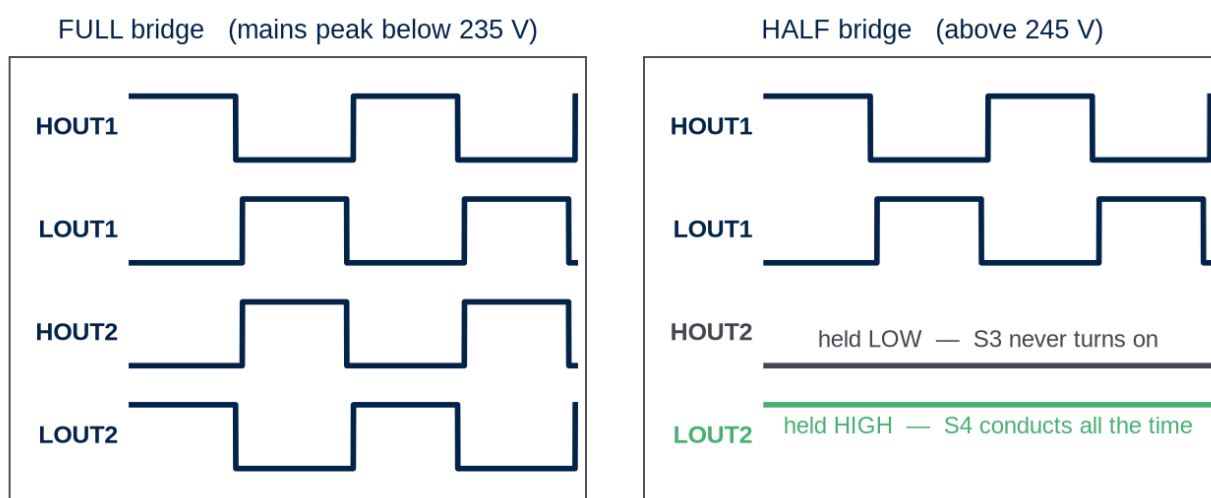


Figure 29: The four bridge drive signals in each mode. LOUT2 held high is the entire mechanism.

5.4 The hysteresis band, and the range it really implies

The two thresholds are not the same number: the part drops to half bridge at $245 V_{pk}$ going up and returns to full bridge at $235 V_{pk}$ coming down. The 10 V of hysteresis stops the mode from toggling back and forth at the threshold, but it also means that **between 235 and 245 V_{pk} the mains voltage alone does not determine the mode** — the direction of travel does.

Taking each threshold on the side where its mode is guaranteed gives the design range 173.2 to 332.3 Vac (1.92:1). Including the band gives **166.2 to 346.5 Vac (2.08:1)**. This design passes every check at the wider range as well, but a design with less margin would not, and the narrower figure must not be used when tightening a specification.

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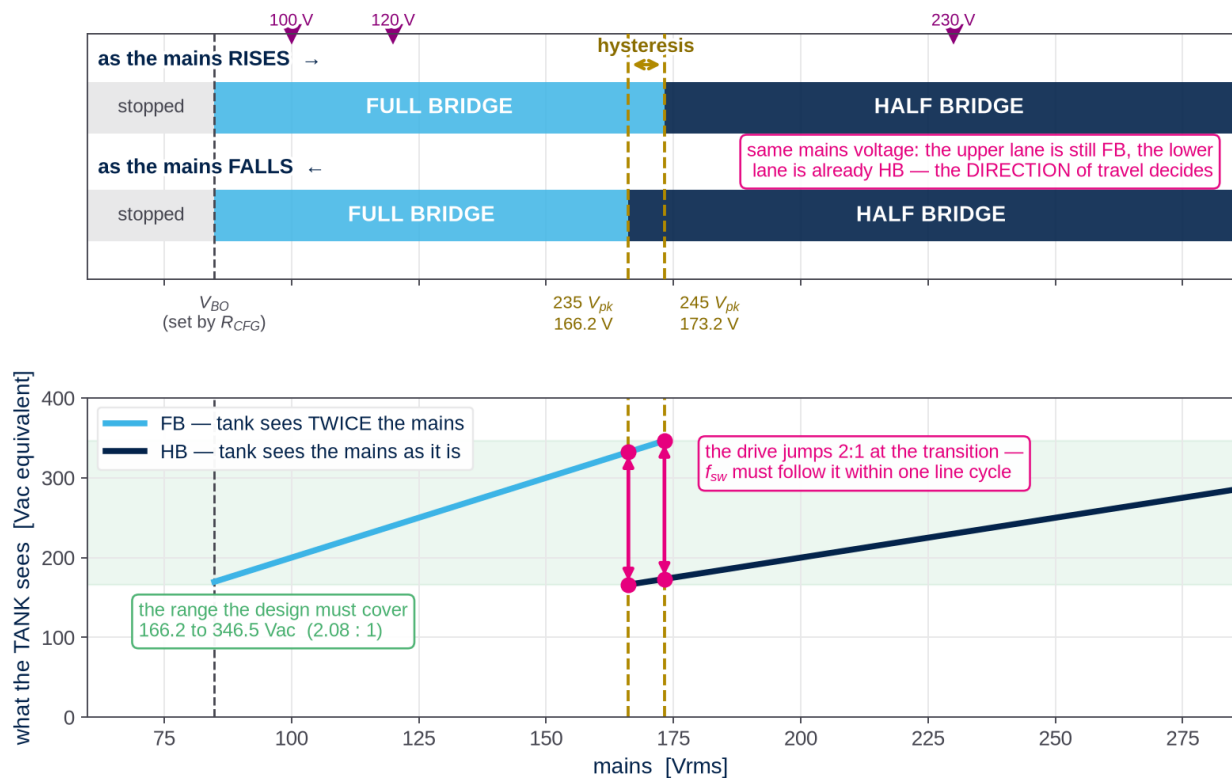


Figure 30: Where each mode applies. Inside the band the mode depends on which direction the mains arrived from.

Where you will actually meet the band. Real mains never sits at 166 to 173 Vrms, so in service the mode is settled at start-up and stays. The band is met on a programmable ac source and in line dip and surge testing. Test it with a **step** across the threshold, not a ramp — a ramp gives the voltage loop time to follow, and a step does not.

Transition transient. Crossing a threshold changes the tank drive 2:1 within one line cycle, and f_{sw} must follow it against a voltage loop that crosses in the tens of hertz. The output droops until the loop recovers. This event is **less severe than the hold-up case**, which removes power entirely for T_{hold} and is already met, so it is not a reason to enlarge C_{out} or speed up the loop. That comparison has not yet been checked in the time domain, by simulation or by measurement.

6 Design procedure

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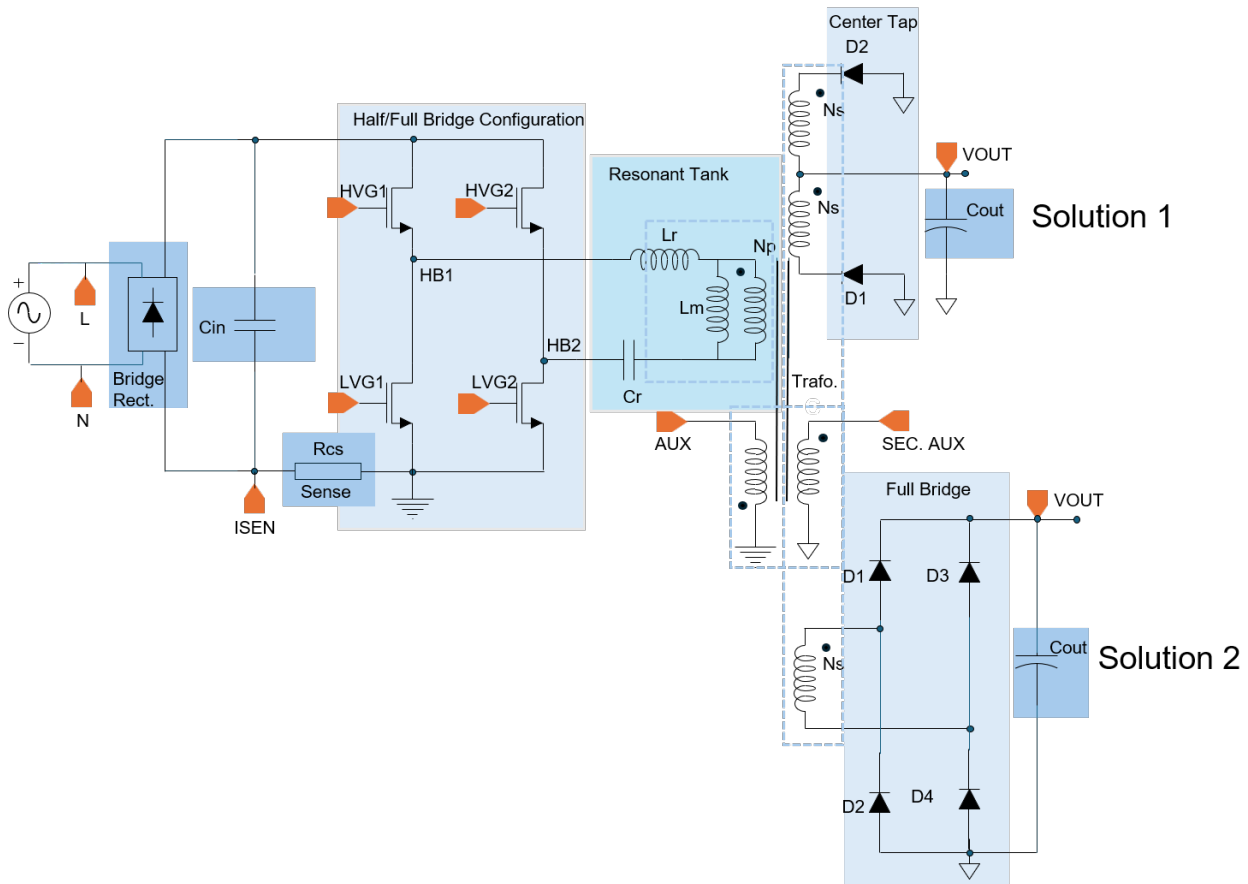


Figure 31: The complete power stage. The bridge rectifier feeds the half/full-bridge leg pair directly — there is no bulk capacitor and no boost stage. The secondary is drawn both ways: centre tap (Solution 1) and full bridge (Solution 2).

Pick the secondary before the tank: it sets N_{rect} , and N_{rect} sets the reflected voltage. A centre tap puts one device in the conduction path, a full bridge puts two, so at a low output voltage the centre tap halves the secondary conduction loss. The price is twice the reverse voltage on each device, and a secondary winding that works only half of the time. This design uses the centre tap.

The sequence below is the order in which the quantities actually depend on one another. Working out of order will produce a tank that has to be redone.

6.1 The specification

The worked example in Section 7.1 states the specification in full, sorted by what kind of number each one is; the procedure below refers to it but does not repeat it. Two qualifiers matter to every step here: **the worst line frequency is the lowest**, so everything that depends on f_l uses $f_{l,min}$, and **hold-up is specified at the worst line phase**, the ripple trough.

6.2 The equivalent input range

This is the step that is most often done wrong. With morphing the tank does not see 90 Vac at the bottom and 264 Vac at the top. It sees an **equivalent** input given by the bridge mode:

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Equation 18

$$V_{ac,eq} = 2 V_{ac} \text{ (full bridge)} \quad V_{ac,eq} = V_{ac} \text{ (half bridge)}$$

At the half-bridge edge the mains is 245 V_{pk} and the tank sees 173.2 Vac equivalent; at the full-bridge edge the mains is 235 V_{pk} in full bridge and the tank sees 332.3 Vac equivalent. **These two edges are the design corners, not the ends of the mains range:** the low edge is the worst case for gain and ZVS, the high edge for switching frequency.

6.3 Turns ratio and reflected voltage

The turns ratio only ever appears multiplied by the effective output voltage, so what matters is the **reflected voltage**:

Equation 19

$$V_{refl} = n V_{o,eff} = n(V_{out} + N_{rect} V_f)$$

Two ratios have to be kept apart: **n** is the equivalent-model ratio in the gain equation, **n_T** the physical turns ratio with the leakage folded into L_r. At λ ≈ 0.5 they differ by more than 20 %; Section 6.9 has the relation, and the transformer drawing must carry open- and short-circuit inductance as well as turns.

6.4 The resonant tank

The equivalent ac load resistance for a centre-tapped secondary is

Equation 20

$$R_{ac} = \frac{4}{\pi^2} \frac{n^2 V_{o,eff}^2}{P_{in,LLC}}$$

with P_{in,LLC} the power into the resonant stage. The 4 is the usual 8 with the *line-peak* power 2P in the denominator rather than the average — see the note below.

The gain requirement at the low equivalent corner and the ZVS requirement together cap the quality factor at Q_{ZVS}, and R_{ac} Q_{ZVS} is then the impedance the tank is *sized* from: it fixes C_r through equation 2, and C_r with the target f_r fixes L_r. Three rules govern what happens next, and they matter more than the arithmetic:

- The three *calculated* figures are not one tank. C_r comes from the design impedance; L_r is then whatever pairs with the **selected** C_r, which is why it is not R_{ac} Q_{ZVS} / 2πf_r; and the L_m figure is L_r divided by the λ that *no load at the high corner* asks for — a condition a design may knowingly not meet (Section 4.10), so it is not a value to round towards.
- **Never round L_m up.** That lowers λ and takes the ZVS margin with it. C_r above its calculated value and L_m below have the opposite effect: Q_{pk} falls and λ is held.
- **Stay below the Q_{ZVS} cap, not on it.** That gap is where most of a design's ZVS margin comes from, and it is a deliberate choice, not an accident of rounding. L_m is chosen together with n so that n_T = n√(1+λ_{act}) lands on a ratio that can actually be wound.

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The same expression, evaluated at the line peak. Every LLC text writes $R_{ac} = (8/\pi^2)n^2V_o^2/P$ with the one output power a two-stage converter has. Here the drawn power is $p(\theta) = 2P\sin^2\theta$, so R_{ac} has to be pinned to one instant. Pinned to the **line peak**, where $p = 2P$ and the tank is most heavily loaded, the 8 becomes a 4. It is not a different formula and it has nothing to do with half against full bridge — the bridge factor belongs to the equivalent input voltage. Q then means the quality factor at the line peak, with $Q(\theta) = Q_{pk}\sin^2\theta$ elsewhere. Using the two-stage value understates the loading by two, and every tank value that follows is wrong.

6.5 ZVS verification

The closed-form ZVS estimate in most design guides is a fitted approximation evaluated at the design Q_{ZVS} , not at the Q the converter runs at. Against a full sweep it errs in **either** direction by tens of per cent, and the sign depends on the tank; optimistic is the dangerous direction, because it reports ZVS margin that is not there.

And it has to be told which λ to use. The shortcut is written with a bare λ , and it means the *design* λ — the largest of the four candidates — together with the design Q_{ZVS} . Read the same expression with λ_{act} , which is what the symbol means everywhere after the tank is chosen, and the phase can come out **negative** — a capacitive answer for a tank that is well inside the inductive region. The sweep carries no such ambiguity: it uses λ_{act} and the Q of the moment, which is what the hardware has.

The reliable procedure is to sweep θ , input voltage and load with the **selected** tank and take the minimum. Section 7.5 runs it for this design, with numbers on both directions of the closed-form error, and the worst point is not where one would expect.

6.6 Which side of resonance the converter runs on

Below f_r the secondary current is a truncated sine with a dead interval and the conduction ratio $d = f_{sw}/f_r$ is less than one; above f_r the current is continuous and d clamps at one. The rms figures that ratings and losses are built on contain d , so the difference is real, not a matter of words.

Which side of resonance the converter runs on is not a separate choice. The turns ratio decides it. The demand is $M_{req} = 2nV_{o,eff}/(\sqrt{2}V_{eq})$, so a larger n asks for more gain, and more gain means further below resonance. Two candidate transformers on the same tank can end up on opposite sides of f_r at the same mains voltage.

Worth settling before the transformer is ordered. Comparing candidate turns ratios on gain margin alone hides it, and the two sides do not stress the secondary the same way: below resonance the rectifiers turn off at zero current, above it they do not.

6.7 Currents over the line cycle

Ratings come from the worst switching cycle; losses come from the line-cycle rms. They are different numbers and must not be confused. Two further rules apply to a single-stage converter in particular.

- The **composite** tank current is what the primary devices and the sense resistor see. It is the sum of the reflected load current and the magnetising current, and its peak is neither the sum nor the larger of the two peaks, because they occur at different instants.
- Below resonance the secondary conducts for only $d = f_{sw}/f_r$ of each half period. Omitting that conduction ratio over-estimates the rms, and the loss by the square of that.

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Section 7.8 draws the composite current of the worked design.

6.8 The transformer

The tank has fixed L_r , L_m and a ratio; the transformer has turns, an open-circuit inductance and a leakage. Getting from one to the other is where most of the errors in a resonant design are made, because **two different turns ratios and two different inductances carry almost the same names**. Section 2.12 has already set out why this transformer is sized the way it is; this section turns that into the numbers a supplier can measure.

Every number on the drawing is read off one of three waveforms — the primary winding current, the current in each secondary winding, and the secondary winding voltage — or off one bench test. Figure 42, in the design example, marks where on the worked design's waveforms, and Table 3 says what each mark sets and over which interval it is taken. The sections that follow derive the entries.

Mark	Item	Read from — and over what	What it sets
1	Primary peak current	Top of i_p : the half sine of load current riding on the magnetising ramp. Worst switching cycle: line peak, minimum equivalent input, full load.	Primary switch and OCP1 headroom. Not the saturation test.
2	Primary rms current	i_p^2 averaged over the whole switching period, then over the line half cycle. Every unit carries the same current: the primaries are in series.	Primary copper: $A_{cu} = I_{rms}/J$ per unit
3	Magnetising peak	The dashed i_{Lm} at the switching instant. Below resonance it depends on neither f_{sw} nor load, only on V_{out} , so it is the same on every cycle.	The peak flux; and, scaled to V_{OVP2} , the DC-overlap test current (mark 7)
4	Secondary peak current	Top of the pulse in either secondary winding, same worst cycle. Per unit it is $1/N_x$ of the rectifier-leg value: the secondaries are in parallel.	Rectifier peak rating; the foil termination
5	Secondary rms current, each winding	One pulse per period in each winding, so the rms is taken over the whole period with the idle half counted, then over the line half cycle.	Secondary copper: $A_{cu} = I_{rms}/J$, per winding and per unit
6	Winding volt-seconds	The secondary winding voltage is V_{out} for the resonant half period $T_r/2$ while the rectifier conducts. That area is the flux swing $2N_s A_e B_{pk}$; f_r is the worst case because the interval never gets longer than $T_r/2$.	Core area and N_s : $A_e \geq V_{out}/(4 f_r N_s B_{max})$
7	DC-overlap (saturation) test	Not a waveform. Secondaries open, dc current in the primary, inductance at the primary read against current. The test current is mark 3 scaled to the highest output the controller allows, V_{OVP2}/V_{out} .	Core saturation margin; the number the supplier tests to
—	L_{open} , L_{short}	LCR meter at the primary with the secondaries open, then shorted. Terminal quantities, so a supplier can test them.	$L_m + L_r$ and L_r : the tank itself

Table 3: What each mark in the figure sets. Currents are per unit of the assembly: the primaries are in series and the secondaries in parallel.

6.9 Two ratios, two inductances

When all of the leakage is referred to the primary, the tank sees an ideal transformer of ratio n , a series L_r and a shunt L_m . That n is **not** the wound ratio. With k the coupling coefficient,

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Equation 21

$$n = k n_T, \quad k = \sqrt{\frac{L_m}{L_m + L_r}} = \frac{1}{\sqrt{1 + \lambda}}$$

so $n_T = n \sqrt{1 + \lambda}$. At $\lambda \approx 0.5$ the two differ by more than 20 %, which is far more than any gain margin.

The same split appears in the inductances. The tank model uses L_r and L_m ; the physical transformer is described by a magnetising inductance L_μ with leakage on both sides:

Equation 22

$$L_\mu = \sqrt{L_m (L_m + L_r)}, \quad L_{L1} = L_m + L_r - L_\mu, \quad L_{L2} = \frac{L_{L1}}{n_T^2}$$

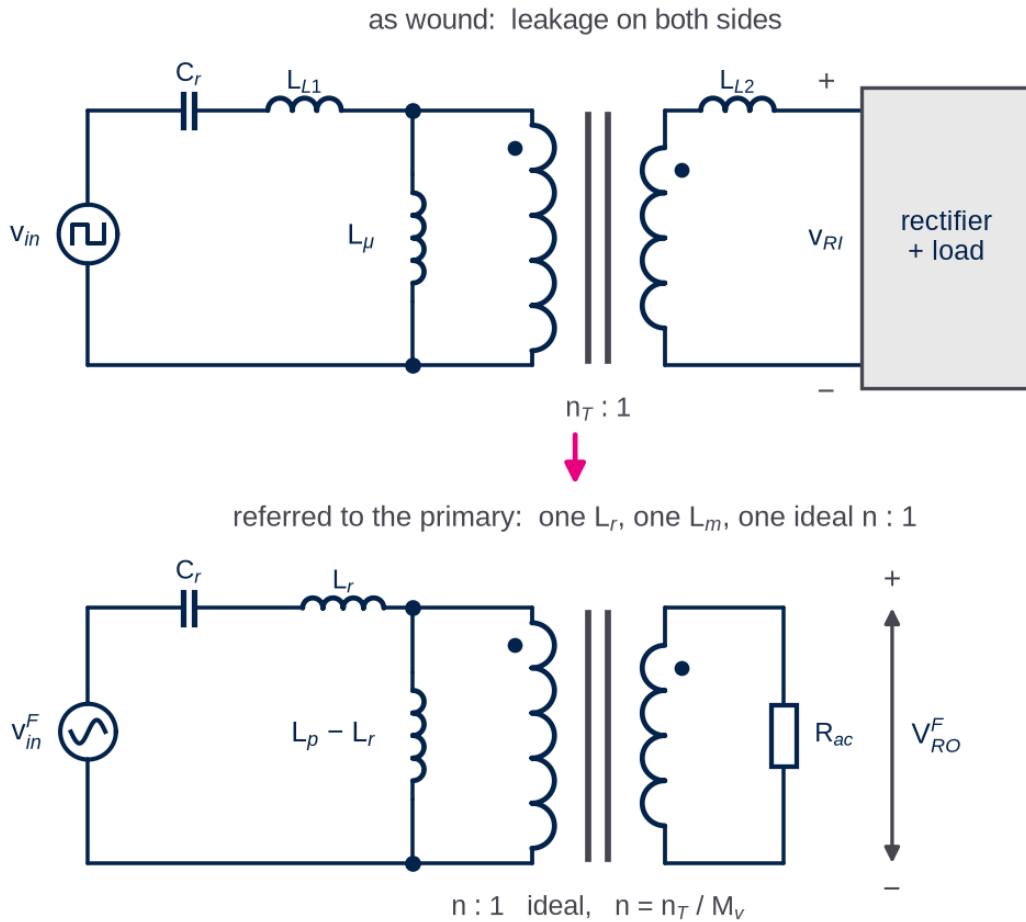


Figure 32: The same transformer drawn both ways. Above, as wound: the leakage split between the two sides, the physical magnetising inductance L_μ in shunt, and the wound ratio $n_T : 1$. Below, referred to the primary: one L_r , one L_m , and an ideal ratio $n : 1$ that is no longer the wound one. The two shunt elements are not the same number — $L_\mu = \sqrt{L_m (L_m + L_r)}$ — and neither are the two ratios.

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That three-element form assumes the leakage splits evenly between the two sides. The real split is set by the winding arrangement rather than by the design, and it does not matter: the tank sees only the two quantities a meter can read.

6.10 What the transformer specification must say

Only two inductances are measurable at terminals, and they are exactly the two the tank needs:

Equation 23

$$L_{open} = L_{\mu} + L_{L1} = L_m + L_r, \quad L_{short} = L_{L1} + \frac{L_{\mu} n_T^2 L_{L2}}{L_{\mu} + n_T^2 L_{L2}} = L_r$$

L_{open} is read at the primary with every other winding open, L_{short} with the secondary shorted. **Specify those two and the turns, never L_{μ}** — no terminal pair exposes L_{μ} , so a supplier cannot test it, and a part that meets it can still miss the tank. Also state the turns as numbers: with one or two secondary turns, the inductance of that winding is dominated by the lead-out loop, so $\sqrt{(L_1/L_2)}$ — the open-circuit inductance measured at the primary and at one secondary — does not give the turns ratio.

6.11 Peak flux is set by the secondary, not the primary

Mark 6 of Figure 42: the secondary winding voltage is a rectangle of height V_{out} and width $T_r/2$, and the flux swings from $-B_{pk}$ to $+B_{pk}$ across it. The equation is that area divided by $2N_s A_e$, and nothing else enters.

Below resonance the secondary conducts for a resonant half period and the output voltage stands across the secondary winding for all of it. Above resonance it conducts for a shorter switching half period, so f_r is the worst case everywhere on the line cycle:

Equation 24

$$B_{pk} = \frac{V_{out, eff}}{4f_r N_s A_e} \implies A_e \geq \frac{V_{out, eff}}{4f_r N_s B_{max}}$$

N_p does not appear. Adding primary turns does not reduce the flux; only adding secondary turns does. That is an uncomfortable result for a low-voltage, high-current output, where the secondary wants to be a single turn: halving N_s doubles the core area needed. That is why a single-turn secondary is expensive in core area even though it is the obvious choice for the current.

Computing the same flux from an inductance is correct only with the **physical** L_{μ} : $B_{pk} = L_{\mu} i_{\mu, pk} / (N_p A_e)$ reduces exactly to the expression above. Using the tank model L_m with the real N_p understates the flux by $\sqrt{(1+\lambda)}$ — about a quarter at $\lambda \approx 0.5$ — because the L_m branch carries $n V_{out}$ while the winding reflects $n_T V_{out}$.

6.12 Choosing the core: two areas, and the one that usually decides

A core has to satisfy two independent conditions, and they have nothing to do with each other. The magnetic one is the equation above: enough cross-section A_e to carry the flux at the turns chosen. The electrical one is the winding window A_N : enough room for the copper the currents need. Catalogues quote both, and their product is what handbooks call the **area product**. Either can be the binding one.

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Read the flux check against $A_{\min}$, not A_e . A_e is the effective area for inductance; $A_{\min}$ is the narrowest section the flux actually has to pass. On a PQ core the two differ by ten to twenty per cent, and it is $A_{\min}$ that saturates first: the real peak is $B_{pk} A_e / A_{\min}$. Both numbers are in the datasheet, and only one of them is usually used.

In a **single-stage PFC LLC** there is a **third condition**, and it is normally the one that decides: the leakage inductance has to come out at L_r . Written as a fraction of what a meter reads at the primary,

Equation 25

$$\frac{L_{short}}{L_{open}} = \frac{L_r}{L_m + L_r} = \frac{\lambda}{1 + \lambda}$$

At $\lambda \approx 0.5$ that is **about a third**. A conventional transformer, wound to couple as well as it can, leaks a few per cent; this one has to leak an order of magnitude more, *on purpose*. Leakage that large is not a winding tolerance, it is a winding **geometry** — and geometry costs window area. A core that passes on A_e and on copper can still fail here.

6.13 The winding arrangement is the leakage

Leakage is the flux that links one winding and not the other, so it is set by how far apart the two windings are and by how much of the window separates them. That gives the designer one knob and a well-known ordering:

- **Interleaved** — primary, secondary, primary, all concentric. Lowest leakage, typically one or two per cent. This is what a conventional forward or flyback transformer wants, and it is the opposite of what is needed here.
- **Concentric, not interleaved** — the whole primary, then the whole secondary, over it. A few per cent. Still far short.
- **Side by side** — the primary on one half of the winding width, the secondary on the other. Tens of per cent, and adjustable by the gap left between them. **This is the arrangement a single-stage tank asks for**, and it is why the window has to be generous: half of it goes to each winding, and part of what is left is deliberately empty.
- **A magnetic shunt** — a ferrite bar in the window between the two windings, which carries leakage flux on purpose. It buys leakage without spending winding width, at the cost of a part and of a tolerance that is harder to hold.

This inverts the usual advice. Every transformer text says to interleave, because in every other topology leakage is loss and overshoot. Here L_r is a design value that the tank needs, and a supplier who "improves" the coupling has broken the converter — f_r moves, λ collapses and the gain curve goes with it. Say so on the drawing, in words, next to L_{short} .

6.14 The gap, and why the drawing must not name it

L_m is two orders below what an ungapped core of this size gives, so the core is gapped. The inductance factor that follows from the turns is

Equation 26

$$A_L = \frac{L_{open} / N_x}{N_p^2} \quad g \approx \frac{\mu_0 A_e}{A_L}$$

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with N_x the number of units the assembly is built from, μ_0 the permeability of free space and g the total gap in the centre leg. The second expression is only an estimate: it assumes the gap carries the whole reluctance and it ignores fringing, which always makes the real gap larger than it predicts. **Specify A_L , or better L_{open} itself, and leave the gap to the supplier** — that is the number they grind to, and it is the number a meter can check.

6.15 The window: wire, current density and what actually fits

Marks **2** and **5** of Figure 42. The copper is sized on the **line-cycle** rms: the rms of each switching period (the bracket under the trace), averaged in I^2 over the line half cycle (the lower-left panel). The secondary value is per winding and per unit — each winding carries one pulse per period, and its idle half is inside the average.

Each winding needs a conductor cross-section set by its own rms current and the current density the design can cool:

Equation 27

$$A_{cu} = \frac{I_{rms}}{J} \quad \sum_w N_w A_{cu, w} \leq k_u A_N$$

J is an assumption, not a constant: 4 to 5 A/mm² is usual for a transformer of this size in free air, less if it is enclosed. k_u is the window utilisation, and it is small — round wire in a round bundle, insulation, the bobbin wall, the margins and the layer-to-layer tape leave **0.3 or less** of the window as copper when the wire is Litz. Sizing on bare copper alone overstates what fits by three times.

The primary current does not divide between units. With the primaries in series every unit carries the whole primary current, and only the secondaries share. So the primary copper is the same in each unit as it would be in one big transformer, while the secondary copper is divided — which is exactly why splitting helps a low-voltage, high-current output.

6.16 Skin depth, and why the wire is not a wire

At the switching frequency current does not fill a conductor. It crowds into a surface layer of depth

Equation 28

$$\delta = \sqrt{\frac{\rho}{\pi f \mu_0}}$$

with ρ the resistivity of copper at the temperature the winding runs at and f the frequency it carries, which for a winding in this converter is f_r . That depth, for a series resonance in the 100 to 200 kHz range is about **0.2 mm** at 100 °C. A conductor thicker than about 2δ carries no more current than one of 2δ — it only adds weight. Worse, the field from the *other* turns drives circulating current in each conductor (the proximity effect), and in a side-by-side winding, where the two windings sit in each other's leakage field, that term can dominate the skin term.

So a winding is not made of one wire. There are two ways out of it, and this note uses both.

- **Litz wire** on the primary. A Litz conductor is a bundle of many fine wires, twisted and transposed so that each one takes a turn at every position in the bundle, and each one insulated from its neighbours by its own enamel. One of those fine wires is a **strand**: d_s is its diameter and a_s is its copper cross-section, the area of that one little circle. Because the strands are separately insulated they cannot share a skin, the rule to satisfy is $d_s \leq 2\delta$ **per strand**, not for the bundle as a whole.

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- **Copper foil** on a low-voltage, high-current secondary: a thin strip whose thickness t_f is held near δ while its width does the carrying. It also terminates well into a centre tap.
- Whatever is chosen, the ac resistance is what heats the winding. The dc resistance $\rho l_N N/A_{cu}$ is a floor, not an answer.

Sizing the Litz. Pick a standard strand diameter at or below 2δ ; its copper area is then a circle, and the number of strands is simply the copper the winding needs divided by what one strand provides:

Equation 29

$$a_s = \frac{\pi d_s^2}{4}, \quad n_s = \left\lceil \frac{A_{cu}}{a_s} \right\rceil$$

with A_{cu} from Equation 27. The finished bundle is wider than that copper, because round strands in a round bundle leave gaps between them and every strand carries enamel and the bundle a serving. That is collected into a fill factor k_{litz} , the fraction of the bundle's area that is copper — about 0.5 to 0.6 for a served bundle — and the outside diameter the bobbin has to take is

Equation 30

$$d_{litz} = \sqrt{\frac{4A_{cu}}{\pi k_{litz}}}$$

Sizing the foil. The thickness is chosen first, from the skin depth and from what the supplier stocks, and the width follows. If that width is more than the bobbin can take, the turn is made of several narrower strips connected in parallel and stacked radially rather than one wide strip laid axially:

Equation 31

$$w_f = \frac{A_{cu}}{t_f}, \quad n_f = \left\lceil \frac{w_f}{w_{f,max}} \right\rceil$$

Neither equation is a loss calculation. They size the copper so that the current density and the skin depth are both respected, and they say what will fit. The ac resistance that follows — skin plus proximity — is a separate question, and on a side-by-side winding the proximity term is usually the larger of the two.

6.17 Isolation, margins and what they cost in window

The transformer is the isolation barrier of the whole supply, so the winding that looks like a copper problem is also a safety one. Two arrangements are common, and they spend the window differently. **Margin tape** leaves a creepage margin at each end of the bobbin — typically 3 mm each side for reinforced isolation from a universal mains — so it takes 6 mm off the usable winding width and costs nothing in material. **Triple-insulated wire** needs no margin, so the winding uses the full width, but the wire is thicker for the same copper and it costs more.

On a side-by-side winding the margin is taken twice over: once for the isolation, and once again for the gap that sets the leakage. Work the window budget with both in it, or the part that is ordered will not be the part that was calculated.

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6.18 Loss, and the temperature the drawing has to survive

- **Core loss** comes off the material curve at the operating flux and frequency, not off the headline figure. A datasheet quotes P_v at one point — 100 or 200 mT at 100 kHz — and a design at a different flux and a different frequency has to be read from the curve.
- **Copper loss** is $I^2 R_{ac}$, and R_{ac} is the number the previous section is about. Both windings count, and in a side-by-side arrangement the proximity term is not small.
- **The temperature rise** is what actually limits the design, and it depends on the surface area and the airflow, neither of which is in any of the equations above. Treat the computed loss as an input to a thermal measurement, not as an answer.

Flux and loss do not move together here. B_{pk} is set at f_r by the secondary volt-seconds, so it does not change when the converter runs faster — but core loss does, and in a single-stage converter the frequency sweeps over the line cycle. The core sees its worst flux at the bottom of the sweep and its worst loss per cycle count at the top.

6.19 The saturation test is not the peak winding current

Mark **3** and the bench panel of Figure 42. The current to ask for is the peak of the dashed magnetising trace at the switching instant, scaled to the over-voltage ceiling — not the peak of the winding current at mark 1.

A DC-overlap test leaves every other winding open, so none of the primary ampere-turns is cancelled by the secondary and the same current makes far more flux than it does in operation. It is the flyback condition of Section 2.12, reached deliberately. Ask for the current that reproduces the operating flux in that test:

Equation 32

$$I_{eq} = \frac{B_{pk} N_p A_e}{L_\mu}$$

That denominator is L_μ , not L_{open} . In the open-circuit test the primary current is entirely magnetising current, so what links the core is the L_μ part of the flux linkage and the rest is primary leakage, which goes round no core at all. Dividing by L_{open} understates the current by $1/\sqrt{1+\lambda}$ — about a fifth at $\lambda \approx 0.5$. There is a check that costs nothing: the answer **must** come out equal to $i_{\mu,pk}$, because in that test they are the same current.

Giving the supplier the peak tank current instead asks for a flux density no ferrite reaches, and the supplier will either oversize the core or ask why. Nor is an arbitrary margin wanted on top of I_{eq} : the flux ceiling the core has to survive is already defined, by the highest output voltage the controller allows before it shuts down. Scale I_{eq} by that ratio and there is no number left to invent.

Equation 33

$$I_{sat} = I_{eq} \frac{V_{OVP2}}{V_{out, eff}}$$

6.20 The specification, and what it is not allowed to leave out

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The open-circuit inductance tolerance cannot simply be the usual round number. f_o goes as $1/\sqrt{L_{open}}$, so a low part raises the frequency floor towards the oscillator clamp. The fall a tank can tolerate follows from k_{floor} and is typically a few per cent, so the usual $\pm 10\%$ can allow a part that meets its specification to hard switch. Section 6.25 derives the figure and gives the two ways out. Ask the supplier for a tighter low-side limit, or first make room in the oscillator setting and then ask for $\pm 10\%$.

Core, bobbin, wire and winding order are the supplier choice, and insulation and creepage follow the applicable safety standard. Nothing above constrains them; everything above is measurable at the terminals.

6.21 If one transformer is not practical

At a low output voltage and high current the secondary is a single heavy turn and the core comes out large, which may not fit a height or footprint limit. The transformer can then be built as N_x identical units with **primaries in series and secondaries in parallel**. Series primaries carry the same current, so the paralleled secondaries share automatically without balancing resistors.

- Open-circuit and leakage inductance divide by N_x , and so do the secondary currents. **The primary current does not divide.**
- The DC-overlap test current does not change: flux and primary ampere-turns scale together.
- The wound ratio of the assembly becomes $N_x N_p / N_s$, so **the achievable ratios are quantised in steps of N_x / N_s** . That has to be respected when the tank is chosen, not discovered afterwards.

6.22 The input capacitor

With no bulk capacitor, C_{in} is a film capacitor that absorbs switching ripple and nothing else. A large value is not the safe choice: it holds charge across the zero crossing and distorts the line current, so it is a THD term, not a filter term.

Equation 34

$$C_{in} = 3 \frac{\text{nF}}{\text{W}} \times P_{in}$$

That is a **lower** bound, so round up to the next standard value and rate it for the full line voltage.

6.23 The output capacitor bank

Two conditions apply, and the larger wins. The ripple condition is

Equation 35

$$C_{out} \geq \frac{P_{out}}{2\pi f_{l,min} \Delta v V_{out}^2}$$

and the hold-up condition is

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Equation 36

$$C_{out} \geq \frac{2P_{out}T_{hold}}{\left(V_{out} - \frac{1}{2}\Delta v_{pp}\right)^2 - V_{o,min}^2}$$

Hold-up starts at the worst line phase, and that belongs in the sizing. If the mains disappears at the trough of the $2f_l$ ripple the bank is already half a ripple below nominal when it starts, which is why V_{out} above carries the $-\frac{1}{2}\Delta v$ term. The difference is not small: it costs about a tenth of the ride-out time. Note also that Δv_{pp} there is the ripple the bank actually produces, not the allowance. The condition therefore depends on its own answer, which is why it is checked once the bank is chosen.

Which one wins is a question about the specification and not about the output voltage: both conditions scale as $1/V_{out}^2$, so V_{out} cancels when they are divided. Writing $k = V_{o,min}/V_{out}$, ripple dominates when

Equation 37

$$\left(1 - \frac{\Delta v}{2}\right)^2 - k^2 > 4\pi f_l \Delta v T_{hold}$$

This screening form uses the *allowed* Δv rather than the achieved ripple, because it is asked before the bank exists. Section 7.11 evaluates it for the worked design.

- Include the **$2f_l$ component** in the ripple current. The switching-frequency part alone is far too small: the $2f_l$ envelope carries a comparable rms of its own, and the two are orthogonal so they add in quadrature.
- With a centre-tapped secondary, judge the capacitor ripple current at the **output node**, not per winding.
- Use the ESR at the **switching frequency**. An electrolytic $\tan\delta$ is quoted at 120 Hz and is five to ten times larger there.

A risk this architecture creates. The controller start-up window is finite, and a bank of this size is a far heavier start-up load than the few millifarads a conventional design presents. Cold start into the full bank should be one of the first things measured on hardware.

6.24 Semiconductor requirements

State requirements rather than pick parts: what to pick depends on package, thermal design and cost. Four rules decide whether the requirement is even stated correctly.

- Rate the primary switches on the **composite tank peak**, not on the reflected load current. The load component alone under-rates the device by more than a tenth here, because the magnetising current is missing from it.
- Divide currents by the number of devices actually in parallel before judging stress. Tabulated currents are **per switch position** on the primary and **per leg** on the secondary; a centre-tapped leg carries the whole secondary current, not half of it.
- Compute loss from $R_{DS(on)}$ at $T_{j,max}$, not at 25 °C. The datasheet maximum is process spread at 25 °C; temperature is a separate multiplier, close to two for a superjunction device, read from its normalised $R_{DS(on)}$ curve at $T_{j,max}$. Using the 25 °C value to size a heatsink is optimistic by about a factor of two.
- A curve plotted against T_a may still be read as T_j **if it is a pulse test** — the die does not self-heat during the pulse. For any non-pulsed curve this is not valid.

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The standing device in half-bridge morphing is the one to watch (Section 5.2). It never switches and carries the whole tank current, and it is where the conduction-loss budget is met or knowingly missed — Section 7.12 shows which happened here.

6.25 Controller network

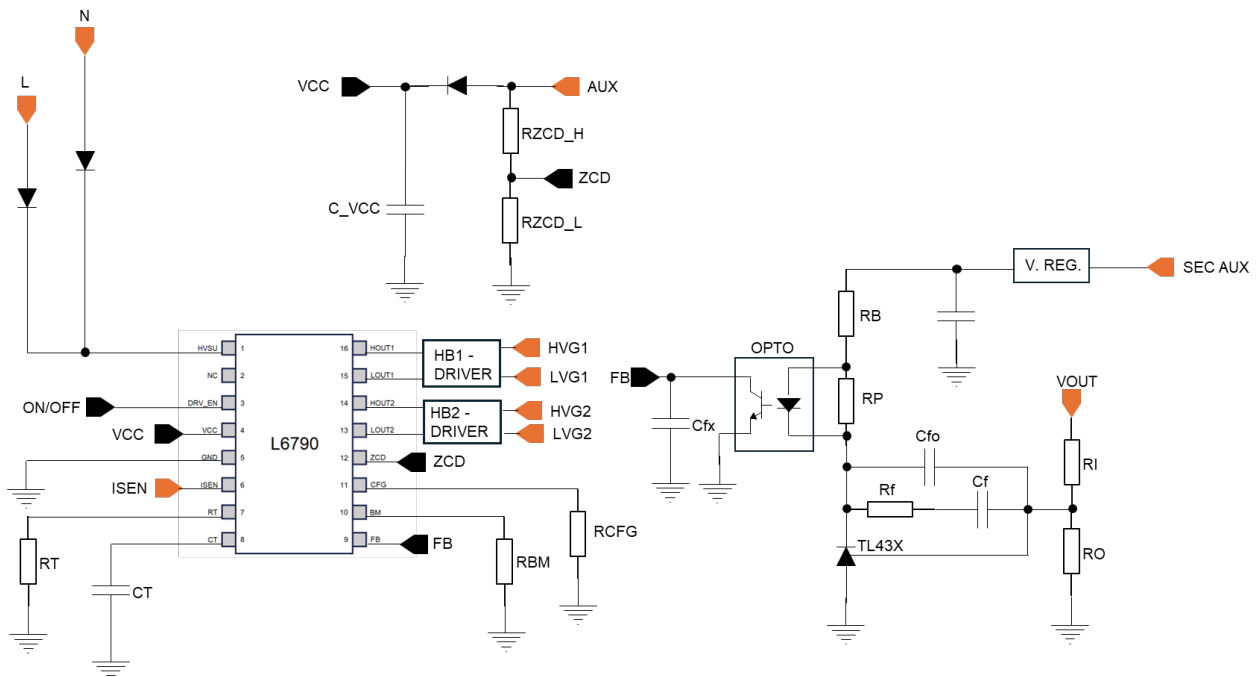


Figure 33: The controller and its passive network. HVSU is fed from the AC side of the bridge; the auxiliary winding drives ZCD through a divider; R_T and C_T set the oscillator limits; R_{CFG} and R_{BM} are read at power-up.

Decide first what the auxiliary winding is for. When it also supplies V_{CC} , the rectified auxiliary voltage at OVP2 must stay under the V_{CC} rating, and that caps its turns ratio. Here V_{CC} comes from elsewhere, so the winding only senses, the only constraint is that the turns come out whole, and the datasheet ratio check is not applicable (Section 7.20).

Each of these parts follows from something already fixed, and Section 7.15 sizes them one at a time in the order they have to be done. The one that deserves the most attention throughout is R_T , because it sets the oscillator floor and that floor must stay above f_o .

That floor is usually the thinnest margin in the design. f_{Min} must sit above f_o , or nothing but the anti-capacitive protection stands between the converter and the capacitive region. It also depends on the oscillator idle time, which the draft datasheet states inconsistently, so whatever it computes to has to be confirmed by measuring $f_{sw}(\theta)$ on hardware.

That ratio is also what the transformer tolerance has to fit inside, and this is easy to miss because the two numbers are in different documents. Since $f_o = 1/(2\pi\sqrt{L_{open}C_r})$, a **low** open-circuit inductance raises f_o . The fall that can be tolerated is

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Equation 38

$$\frac{\Delta L}{L} = 1 - \frac{1}{k_{floor}^2}$$

A floor margin of a few per cent allows a fall of a few per cent in inductance. A transformer drawing asking for the usual $\pm 10\%$ does **not** fit inside that: at the bottom of the tolerance the converter hard switches. There are two ways out — tighten the tolerance on the low side, or lower R_T to lift f_{Min} and make the room. **Check this before the transformer is ordered**, not after.

6.26 Voltage loop and compensation

The voltage loop holds the output at its set point. It is the chain of Figure 34. A divider measures the output. A TL431 compares the divided voltage with its reference and drives the LED of an optocoupler. The optocoupler transistor pulls the FB pin of the controller. The FB voltage is the power command of Section 4.6, the converter delivers that power into the output capacitor, and the output voltage closes the loop.

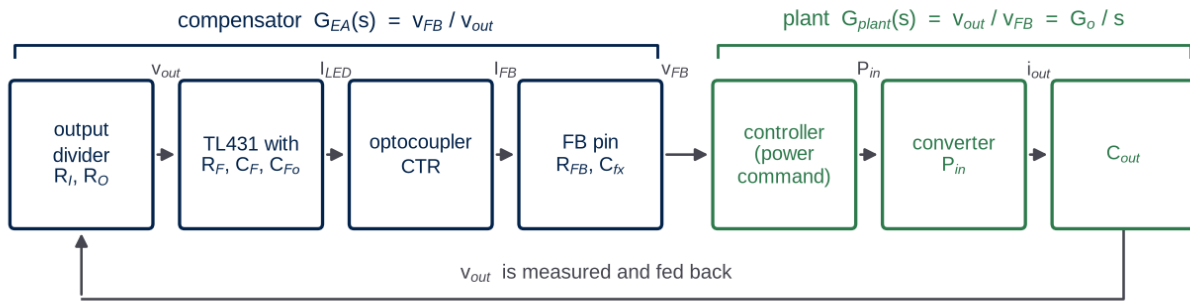


Figure 34: The voltage loop as blocks. Everything left of the FB pin is the compensator $G_{EA}(s)$; everything right of it is the plant $G_{plant}(s)$. The loop gain is $T(s) = G_{plant}(s) G_{EA}(s)$.

The plant. The converter delivers the power the FB pin commands, and that power goes into a capacitor. A change of the FB voltage changes the power in proportion. The power divided by the output voltage is the current into C_{out} . A current into a capacitor gives a voltage that keeps rising for as long as the current flows. So the plant is an **integrator**:

Equation 39

$$G_{plant}(s) = \frac{v_{out}(s)}{v_{FB}(s)} = \frac{G_o}{s}, \quad G_o = \frac{P_{out}}{V_{out} V_{FB} C_{out}} \quad [\text{rad/s}]$$

V_{FB} is the feedback voltage above its 0.5 V offset at rated power (Equation 15), so P_{out}/V_{FB} is the slope of the power command. An integrator has a gain that falls by 20 dB per decade and a phase of -90° at every frequency. On its own it would cross 0 dB at $f_{cto} = G_o/2\pi$. There is no second pole and no right-half-plane zero. That makes this loop easy to stabilise. What makes it hard to place is the output ripple, in Section 6.28.

6.27 What the loop must achieve, and what goes wrong when it does not

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Three numbers are read from the Bode plot of the loop gain $T(s)$: the **crossover frequency** f_c , where $|T| = 1$ (0 dB); the **phase margin**, which is $180^\circ + \arg T$ at f_c ; and the **gain margin**, which is how far $|T|$ is below 0 dB at the frequency f_{180} where $\arg T$ reaches -180° . Table 4 says what each one should be and what happens when it is not.

Quantity	Aim	If it is too low	If it is too high
Crossover f_c	15 to 20 Hz in this converter (Section 6.28). The general rules are below $f_{sw}/5$ and below the ripple frequency the loop must not follow.	The loop is slow. The output dips further and recovers later after a load step or a morphing transition.	The loop follows the $2f_l$ output ripple. The power command, and so the input current, is modulated at $2f_l$; third harmonic on the input current, and chatter across the burst threshold.
Phase margin Φ_M	45° is the floor; 50 to 60° is the target. At 76° the step response has no overshoot; at 45° it rings ($Q \approx 1.2$) [onsemi TND381].	Ringing after every disturbance. At 0° the loop oscillates at f_c .	The response is over-damped and slow. Not a fault, but no longer free: the crossover has to drop to get it.
Gain margin GM	6 dB is the floor; 10 dB is comfortable.	Part spread moves $ T $ up. The optocoupler CTR alone spreads by 2:1 over bins, current and temperature; a loop with a small gain margin oscillates at f_{180} on a warm board with a high-CTR part.	—
G_{EA} at $2f_l$	At or below the value that keeps the third harmonic inside its budget (Equation 42).	Nothing wrong with the loop; the crossover is lower than it needs to be.	Third harmonic above budget, burst chatter.

Table 4: What the loop must achieve.

6.28 Why the crossover must be low: the $2f_l$ ripple

The output carries a ripple at twice the line frequency, because the input power pulses at $2f_l$ and the capacitor buffers the difference (Section 3.1). Its peak-to-peak value is

Equation 40

$$\Delta V_{loop} = \frac{P_{out}}{V_{out}} \frac{1}{2\pi f_l C_{out}}$$

The compensator passes a fraction $G_{EA}(2f_l)$ of that ripple to the FB pin. The FB pin is a power command, so a ripple on it makes the input power, and the input current, swing at $2f_l$ over the line cycle. A sine wave whose amplitude is modulated at twice its own frequency gains a third harmonic. The ripple amplitude is half the peak-to-peak value, and a modulation of the command by a fraction m puts $m/2$ into the third harmonic, so

Equation 41

$$D_3 = \frac{G_{EA}(2f_l) \Delta V_{loop}}{4 V_{FB}}$$

Turned around, the third-harmonic budget D_3 fixes the largest compensator gain the loop may have at $2f_l$:

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Equation 42

$$G_{EA}(2f_i) \leq \frac{4 V_{FB} D_3}{\Delta V_{loop}}$$

The compensator gain falls with frequency, so a small gain at $2f_i$ means a crossover well below $2f_i$. With 5 % of third harmonic allowed this lands at 15 to 20 Hz. A faster loop would give less output ripple and a quicker load-step response, and it would put more distortion on the input current. The two pull in opposite directions, and the distortion limit wins. This is the ripple rejection every power-factor corrector needs; onsemi TND381 calls the failure “tail chasing”.

6.29 The compensator: TL431, optocoupler and the FB pin

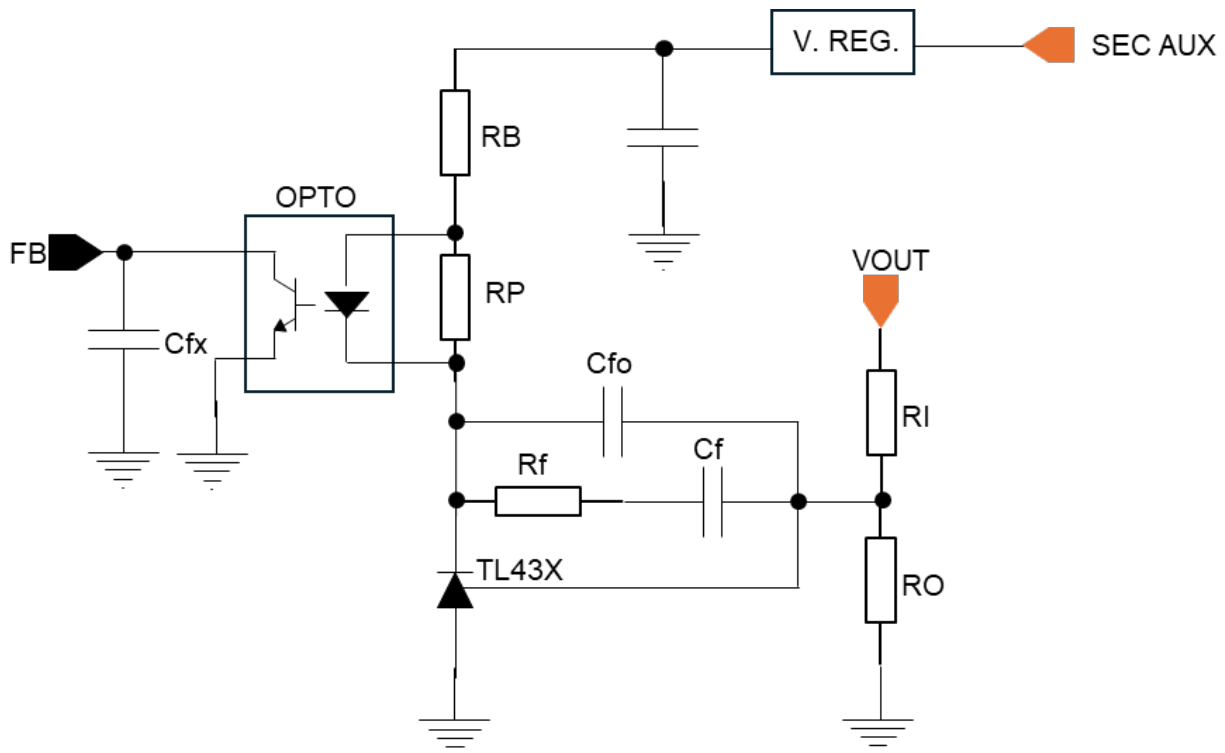


Figure 35: The compensator on the secondary side. R_I and R_O set the regulated output; the TL431 is the error amplifier; R_f , C_f and C_{fo} shape its gain; R_B feeds the optocoupler LED from the regulated V_Z rail; R_P keeps the TL431 above its minimum current; C_{fx} sits on the FB pin.

How it works. The divider R_I , R_O brings the output down to the TL431 reference $V_R = 2.495$ V:

Equation 43

$$V_{out} = V_R \left(1 + \frac{R_I}{R_O} \right)$$

When the output rises above the set point the TL431 sinks more current from its cathode. That current flows through the optocoupler LED, which is fed by R_B from a regulated rail V_Z . The optocoupler transistor then pulls the FB pin down against the pull-up R_{FB} inside the controller, the power command falls, and the output comes back. R_P in parallel with the LED carries the minimum cathode current the TL431 needs to work as an amplifier, so it has an upper limit:

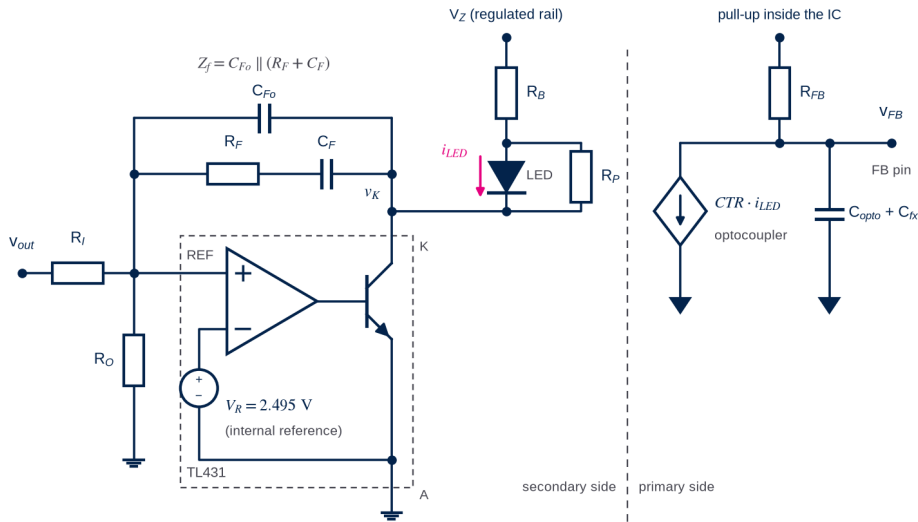
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Equation 44

$$R_P \leq \frac{V_{Fo}}{I_{min}}$$

Because the LED is fed from V_Z and not from the output, the output ripple reaches the LED only through the TL431. There is no second path (what TND381 calls the “fast lane”), and the compensator is the TL431 network alone.

The same network as an op-amp. Figure 36 redraws it with every part as a circuit symbol. Inside the TL431, as its datasheet draws it, the REF pin goes to the non-inverting input of an amplifier, an internal 2.495 V reference V_R to the inverting input, and the amplifier drives an NPN transistor whose collector is the cathode and whose emitter is the anode, here ground. The transistor inverts, so raising REF lowers the cathode: from REF to cathode the part is an inverting amplifier of high gain, and with feedback from the cathode back to REF it holds REF at V_R . R_I is the input resistor of an inverting amplifier, and C_{Fo} in parallel with $R_F + C_F$ is its feedback impedance Z_f , so the cathode moves by $-Z_f/R_I$ times the output change. R_O sets only the dc point: REF is held at V_R , so no signal current flows in it. The cathode current flows through the LED, which R_B feeds from the V_Z rail with R_P across it, so it is the LED current. The transistor side of the optocoupler is drawn as its small-signal model, a current-controlled current source: it sinks CTR times i_{LED} from the FB node, with the optocoupler capacitance across it. Everything left of the barrier returns to the secondary ground and everything right of it to the controller ground; they are different nodes. That current makes the FB voltage across the pull-up R_{FB} , with the pole of $C_{opto} + C_{fx}$. Multiplying the three blocks gives the transfer function below; the sign is negative, which is the negative feedback the loop needs.



$$\frac{v_K}{v_{out}} = -\frac{Z_f}{R_I}, \quad i_{LED} = -\frac{v_K}{R_B}, \quad v_{FB} = -CTR i_{LED} \left(R_{FB} \parallel \frac{1}{s(C_{opto} + C_{fx})} \right)$$

R_O carries no signal: the REF input is held at V_R

$$G_{EA}(s) = \frac{v_{FB}}{v_{out}} = \frac{Z_f}{R_I} \cdot \frac{CTR R_{FB}}{R_B} \cdot \frac{1}{1 + s R_{FB}(C_{opto} + C_{fx})}$$

Figure 36: The TL431 compensator drawn as an op-amp circuit. Inside the TL431, as in its datasheet: an amplifier with REF on its non-inverting input and the internal 2.495 V reference V_R on its inverting input, driving an NPN whose collector is the cathode K and whose emitter is the anode A, grounded. Around it: R_I and Z_f make an inverting amplifier; R_B , R_P and the LED turn the cathode voltage into the current i_{LED} ; the optocoupler is a current-controlled current source, $CTR i_{LED}$ sinking from the FB node; R_{FB} with $C_{opto} + C_{fx}$ turns it into v_{FB} . The three factors multiply to $G_{EA}(s)$. The optocoupler is also the isolation barrier, so the two sides return to different grounds and the symbols differ.

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Its transfer function from the output to the FB pin is a Type II network with one more pole:

Equation 45

$$G_{EA}(s) = \frac{EA_o}{s} \cdot \frac{1 + s/\omega_z}{(1 + s/\omega_p)(1 + s/\omega_{px})}$$

Equation 46

$$EA_o = \frac{CTR R_{FB}}{(C_F + C_{Fo}) R_I R_B} \quad [\text{rad/s}], \quad f_z = \frac{1}{2\pi R_F C_F}$$

$$f_p = \frac{1}{2\pi R_F C_{ser}}, \quad C_{ser} = \frac{C_F C_{Fo}}{C_F + C_{Fo}}, \quad f_{px} = \frac{1}{2\pi R_{FB} (C_{opto} + C_{fx})}$$

Each part has one job. C_F and C_{Fo} together make the pole at the origin: the gain keeps rising toward dc, so the static error is zero. R_F with C_F makes the zero f_z : above it the phase comes back up, and that is where the phase margin comes from. C_{Fo} with R_F makes the pole f_p : above it the gain falls at 40 dB per decade again, so the gain at $2f_l$ is small. The optocoupler capacitance C_{opto} with R_{FB} makes a third pole, and C_{fx} is added to put it at about a kilohertz where it removes switching noise from the FB pin. The optocoupler CTR multiplies the whole gain, which is why its spread matters.

The bias window. The datasheet gives a window for R_B . The LED must be able to drive the FB pin at its steady-state current with the lowest CTR (upper bound), and must not push the pin past its maximum current with the highest CTR when the TL431 is fully on (lower bound):

Equation 47

$$R_{B,max} = \frac{V_Z - (V_R + V_{Fo})}{V_{Fo}/R_P + I_{FB,steady}/CTR_s}$$

$$R_{B,min} = \frac{V_Z - (V_R + V_{Fo})}{V_{Fo}/R_P + I_{FB,max}/CTR_m}$$

Two currents, two CTR values. The gain EA_o is computed with the CTR at the steady-state LED current, CTR_s . The lower bound of R_B uses the CTR at the maximum LED current, CTR_m , which is higher because CTR rises with current. Take both from the optocoupler datasheet for the bin that will be fitted, and check the loop again with the CTR at the top of the bin: the crossover moves up with the square root of CTR and the phase margin comes down.

6.30 Placing the zero and the pole: the K-factor method

The compensator has three things to set: its gain EA_o , its zero and its pole. The Venable K-factor method sets them from two targets, the phase margin Φ_M and the gain the loop is allowed at $2f_l$. First a spread factor K_v is computed from the phase margin, with α_v a weighting constant of the method (1.2 in the ST tool):

Equation 48

$$K_v = \frac{1}{2\alpha_v} \left[(1 + \alpha_v^2) \tan \Phi_M + \sqrt{(1 + \alpha_v^2 \tan \Phi_M)^2 + 4\alpha_v^2} \right]$$

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The zero goes a factor K_v below the crossover and the pole the same factor above it. Above the pole the compensator gain is $EA_o K_v^2/\omega$, so the gain limit at $2f_l$ fixes EA_o :

Equation 49

$$EA_o = \frac{2\pi(2f_l) G_{EA}(2f_l)}{K_v^2}$$

The point the zero and pole are placed about follows from the plant gain, with $\Gamma_v = 0.744 V_{eq,max}/V_{eq,min}$ an input-voltage margin factor of the ST tool (its constant is listed in Appendix 10.3 as not derived here):

Equation 50

$$f_{MB} = \frac{1}{2\pi} \sqrt{\frac{G_o K_v EA_o}{\Gamma_v}}, \quad f_p = K_v f_{MB}, \quad f_z = \frac{f_{MB}}{K_v}$$

The parts then follow from Equation 46, solved for each one in turn. The high-frequency pole is placed at a chosen f_{pHF} , and C_{fx} is what is left after the optocoupler capacitance:

Equation 51

$$C_{Fo} = \frac{f_z}{f_p} \cdot \frac{CTR_s R_{FB}}{R_I R_B EA_o}, \quad C_F = C_{Fo} \left(\frac{f_p}{f_z} - 1 \right)$$

$$R_F = \frac{1}{2\pi f_z C_F}, \quad C_{fx} = \frac{1}{2\pi f_{pHF} R_{FB}} - C_{opto}$$

Each value is then rounded to a standard part, and the zero, pole and gain the standard parts actually give are computed back with Equation 46. The loop is checked on those, not on the targets.

6.31 Checking the loop: crossover, phase margin and gain margin

With the plant of Equation 39 and the compensator of Equation 45 the loop gain is two integrators, one zero and two poles. Its magnitude and phase at any frequency are

Equation 52

$$|T(\omega)| = \frac{G_o EA_o}{\omega^2} \cdot \frac{\sqrt{1 + (\omega/\omega_z)^2}}{\sqrt{1 + (\omega/\omega_p)^2} \sqrt{1 + (\omega/\omega_{px})^2}}$$

$$\arg T(\omega) = -180^\circ + \arctan \frac{\omega}{\omega_z} - \arctan \frac{\omega}{\omega_p} - \arctan \frac{\omega}{\omega_{px}}$$

The -180° is the two integrators. Everything after it is the compensator's own phase, so the **phase margin is simply the compensator phase at the crossover**. The crossover is where $|T| = 1$. Writing $A(\omega)$ for the fraction in Equation 52, the condition is $\omega_c^2 = G_o EA_o A(\omega_c)$. $A(\omega)$ changes slowly, so it is solved by repeating

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Equation 53

$$\omega_c \leftarrow \sqrt{G_o EA_o A(\omega_c)}, \quad \text{starting from } \omega_c = \sqrt{G_o EA_o}$$

a few times until it stops moving; no solver is needed. The phase margin is then

Equation 54

$$\Phi_M = \arctan \frac{\omega_c}{\omega_z} - \arctan \frac{\omega_c}{\omega_p} - \arctan \frac{\omega_c}{\omega_{px}}$$

Figure 37 shows where the three numbers are read on the Bode plot of such a loop. It is drawn in units of its own crossover, with the zero a factor K below f_c and the pole K above, so it is the shape of every loop of this kind and not of one design. The gain falls at 40 dB per decade where both integrators act, at 20 dB per decade between the zero and the pole, and steeper again above the pole and above f_{px} . The phase starts at -180° , is lifted by the zero, peaks near the crossover, and is pulled down again by the two poles until it crosses -180° at f_{180} . Section 7.14 puts this design's numbers on the same plot.

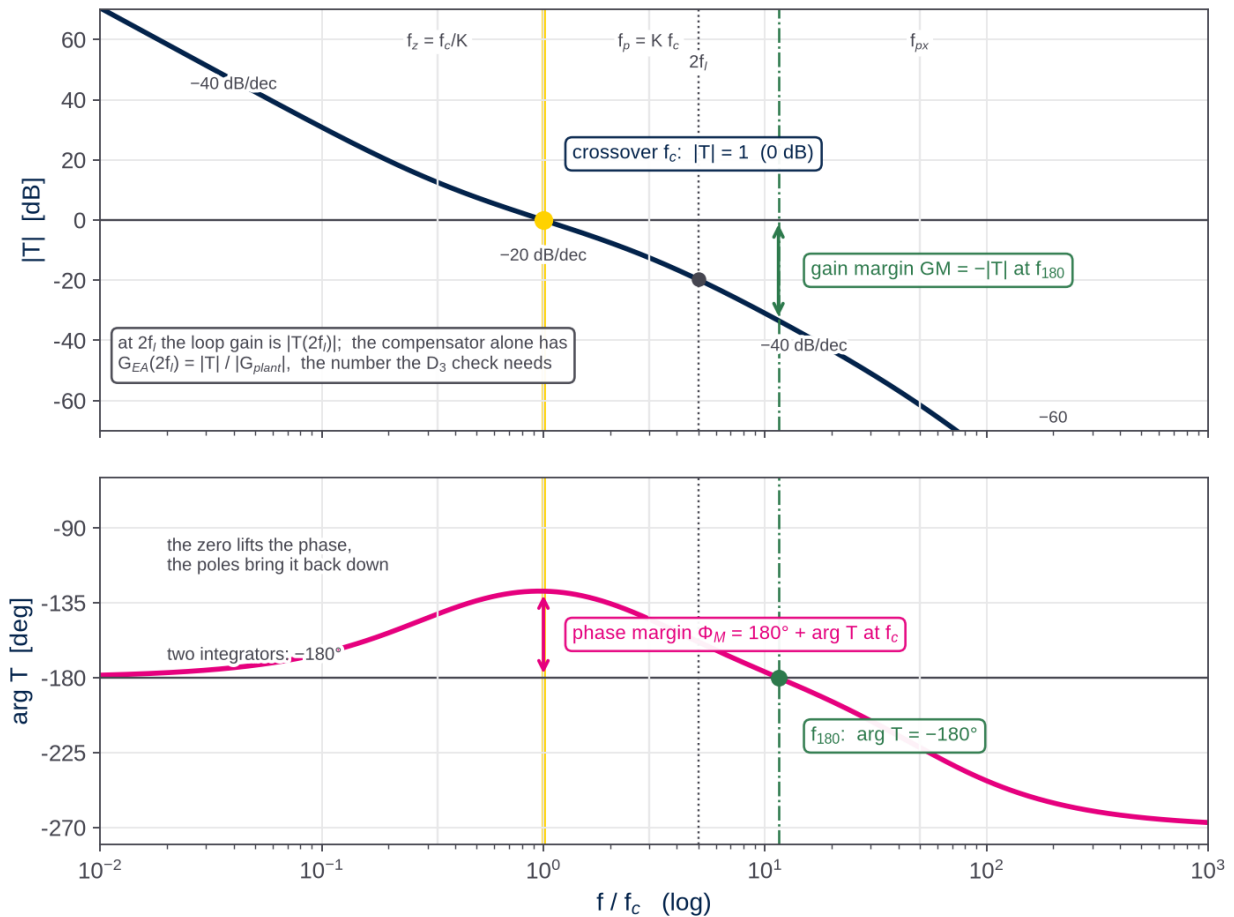


Figure 37: A two-integrator loop with a Type II compensator, in units of its own crossover f_c . The crossover is where $|T|$ crosses 0 dB; the phase margin is the distance from -180° there; the gain margin is how far $|T|$ is below 0 dB at f_{180} , where the phase reaches -180° . The dot at $2f_i$ is the loop gain the third-harmonic check reads.

6.32 Gain margin, and why it needs checking

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Phase margin alone does not prove the loop is safe. Above f_{px} this network has one zero against two poles, so the phase keeps falling toward -270° and **-180° is crossed at a finite frequency**. That frequency needs no search either. Setting the phase contributions of the zero and the two poles equal and taking the tangent of both sides leaves one square root:

Equation 55

$$f_{180} = \sqrt{f_p f_{px} - f_z(f_p + f_{px})}, \quad GM = -20 \log_{10} |T(f_{180})|$$

The value under the root is positive whenever the zero sits well below both poles, which is what the K-factor placement gives. If it came out negative the loop would not be extra safe: the two poles would pull the phase below -180° before the zero could lift it, at every frequency, so the phase margin at any crossover would be negative. Judge the gain margin against 6 dB as a floor and 10 dB as a comfortable target. With a crossover in the tens of hertz the margin is usually large in this topology. That is a reason to compute it, not a reason to assume it.

The gain the standard parts give at $2f_l$ is read from the same expression, and the third harmonic it causes follows from Equation 41. That closes the check: the loop is stable, and it also meets the distortion budget it was slowed down for.

6.33 Feedback ripple against the burst threshold

A crossover this low leaves $2f_l$ ripple on the feedback pin. Because that pin is a power command, the ripple moves the commanded power up and down every half line cycle. If it crosses the burst-entry threshold the converter chatters in and out of burst. The ripple amplitude at the burst point is

Equation 56

$$\Delta V_{FB} = \frac{P_{in, BM}}{V_{out}} \frac{1}{2\pi f_l C_{out}} G_{EA}(2f_l)$$

Lowering R_{BM} moves the burst entry point below that ripple. The burst threshold moves by 0.01 V per k Ω , and the ripple has to clear it on both sides:

Equation 57

$$\Delta R_{BM} = \frac{\Delta V_{FB}}{2 \times 0.01 \text{ V/k}\Omega}$$

Raising the crossover would shrink the ripple but worsen the distortion, so trading it against R_{BM} is the practical answer rather than speeding up the loop.

A loop this slow cannot recover from a large load step by itself. The controller's anti-saturation circuit does that, and it needs the FB pin to reach its full current range. That is why R_B must sit inside its window (Equation 47). Checking the window is not optional.

7 Design example

One design, from the specification to the component values. Every number below is either given, chosen, assumed, or computed from those, and each computed value shows the arithmetic it came from. The specification is **90 to 264 Vac in, 25 V / 26.3 A = 657.5 W out**, centre-tapped synchronous rectification, no bulk capacitor and

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no boost stage.

7.1 The specification, sorted by what kind of number it is

The only useful question about a number is whether it can be moved, so the kind of each one is stated: **given** by the load and the mains, **chosen** by the designer, or **assumed** in place of a measurement that has not been made yet.

Item	Symbol	Value	Kind	Note
Mains input	V_{ac}, f_l	90 to 264 Vac, 47 to 63 Hz	given	the worst line frequency is the lowest, so every f_l term uses 47 Hz
Output	V_{out}, I_{out}	25 V, 26.3 A (657.5 W)	given	$V_{o,eff} = V_{out} + N_{rect} V_f = 25.0 \text{ V}$ with $V_f = 0$
Output ripple, $2f_l$ pk-pk	Δv	$\leq 5 \%$ of V_{out} (1.25 V)	given	a property of the load, not of the converter
Hold-up	$T_{hold}, V_{o,min}$	12 ms down to 19 V, at 100 Vac	given	starts at the ripple trough, the worst line phase
Secondary rectifier	N_{rect}	1 (centre tap, SR)	chosen	halves the secondary conduction loss; doubles the device reverse voltage
Target series resonance	f_r	150 kHz	chosen	sets the magnetics size
Specified maximum f_{sw}	$f_{sw,max}$	225 kHz ($1.5 f_r$)	chosen	an input to λ , Section 4.9
Bridge dead time	t_D	220 ns	chosen	what the ZVS sweep has to beat
V_{CC} source	—	external 12 V rail	chosen	no turns-ratio ceiling on the auxiliary winding
LLC stage efficiency	η_{HB}	98 %	assumed	a 95 % outcome moves R_{ac} and R_{CS} by about 3 %; nothing downstream is sensitive
Oscillator idle time	T_{idle}	250 ns	assumed	the draft datasheet also says 700 ns, which would close the floor margin — the first thing to measure
$R_{DS(on)}$ temperature factor	k_T	1.8 / 1.9	assumed	primary / secondary, read off the datasheet curves at $T_{j,max}$
Burst entry point	r_{BM}	75 W (11 % of rated)	assumed	sets R_{BM} , then checked against the feedback ripple

Table 5: The specification. Everything else in this chapter is computed from these.

Harmonic class. IEC 61000-3-2 Class D reaches only 600 W. This design draws 685 W at the input, so the absolute Class A limits apply.

7.2 What the design came out as

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Block	Values
Tank	C_r 100 nF, L_r 11 μ H, L_m 20 μ H, λ 0.55, f_r 151.7 kHz, f_o 90.4 kHz
Transformer	$N_p:N_s$ 15:2, n 6.02, n_T 7.50, L_{open} 31.0 μ H, L_{short} 11.0 μ H, $A_e \geq 103$ mm ² at B_{pk} 0.20 T
Frequency	f_{sw} 127.1 kHz at the low corner, 249.0 kHz at the high corner
Output	470 μ F $\times$ 160 = 75.2 mF, ripple 1.18 V (4.74 %), hold-up 13.42 ms
Controller	R_T 11 k Ω , C_T 470 pF, R_{CS} 24.0 m Ω , R_{CFG} 30 k Ω , R_{BM} 30 k Ω
Loop	f_{cross} 17.57 Hz, Φ_M 49.64°, third harmonic 4.23 %

Table 6: Principal values. The rest of the chapter shows where each one comes from.

7.3 What a verification margin is

Every check has the same shape: what the design has, divided by what it needs.

Equation 58

$$k = \frac{X_{act}}{X_{req}} \implies \text{pass when } k > 1$$

$k = 1.05$ is five per cent of room, $k = 0.9$ is ten per cent short. The ratios have no units and all point the same way, so a frequency check and a power check can sit in one column, and the smallest k is the thinnest part of the design.

Check	$k = \text{has} / \text{needs}$	Substituted	k
ZVS at the worst point of the sweep	$T_{ZC,min} / t_D$	401 ns / 220 ns	1.824
Oscillator floor clears the lower resonance	f_{Min} / f_o	92.25 kHz / 90.39 kHz	1.021
Oscillator ceiling clears the operating maximum	$f_{Max} / f_{sw,max}$	319.61 kHz / 249.03 kHz	1.283
Over-current threshold clears the tank peak	$I_{OCP1} / I_{Lr,pk}$	22.92 A / 18.32 A	1.251
Hold-up achieved against hold-up required	t_{hold} / T_{hold}	13.42 ms / 12 ms	1.119
Secondary loss against its budget	$P_{budget} / P_{SR,leg}$	3.00 W / 2.83 W	1.063
Primary loss against its budget	$P_{budget} / P_{mos,dc}$	3.00 W / 9.65 W	0.311

Table 7: The verification margins.

Two rows need reading with care. $k_{Ploss} = 0.311$ is a result, not a failed calculation: the standing primary device spends 9.65 W against a 3 W budget, and Section 6.24 says why the design goes ahead anyway. $k_{floor} = 1.021$ looks like a safe 2 %, but the transformer tolerance and the oscillator idle time both move it, and neither is known to 2 % yet.

7.4 Following the numbers through

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The design in the order it was computed. Each step names the equation it uses, shows it again, and puts this design's numbers into it, so that nothing in the tables above appears without its arithmetic. Table 8 collects the results.

Step 1 — the power the tank has to pass. The output is P_{out} . The input bridge, the EMI filter and the LLC stage itself are each given a loss budget, the last one as η_{HB} of the power into the tank:

$$P_{in} = P_{out} + P_{LLC} + P_{EMI} + P_{BR} = 657.5 + 13.42 + 8.34 + 5.52 = \mathbf{684.8 \text{ W}}$$

$$P_{in, LLC} = P_{in} - P_{BR} - P_{EMI} = 684.8 - 5.52 - 8.34 = \mathbf{670.9 \text{ W}}$$

Step 2 — the two corners of the equivalent input. They are the morphing thresholds, not the mains limits:

Equation 18, repeated

$$V_{ac, eq} = 2 V_{ac} \text{ (full bridge)} \quad V_{ac, eq} = V_{ac} \text{ (half bridge)}$$

$$V_{ac, eq, low} = \frac{245 V_{pk}}{\sqrt{2}} = \mathbf{173.2 V_{ac}} \quad V_{ac, eq, high} = \frac{2 \times 235 V_{pk}}{\sqrt{2}} = \mathbf{332.3 V_{ac}}$$

Step 3 — the turns ratio. The wound ratio is the input here, because 15 : 2 is what can be built; the plain calculation would give $n = 6.046$. The equivalent-model ratio follows from the wound one and the realised λ (step 8):

Equation 21, repeated

$$n = k n_T, \quad k = \sqrt{\frac{L_m}{L_m + L_r}} = \frac{1}{\sqrt{1 + \lambda}}$$

$$n = \frac{n_T}{\sqrt{1 + \lambda_{act}}} = \frac{7.500}{1.2450} = \mathbf{6.024}$$

Equation 19, repeated

$$V_{refl} = n V_{o, eff} = n(V_{out} + N_{rect} V_f)$$

$$V_{refl} = n V_{o, eff} = 6.024 \times 25.0 = \mathbf{150.6 \text{ V}}$$

Step 4 — the gain the tank is asked for, at the line peak ($\theta = 90^\circ$) of each corner:

Equation 12, repeated

$$M(\theta) = \frac{n V_{o, eff}}{V_{drive}(\theta)} = \frac{n V_{o, eff}}{\sqrt{2} V_{ac, eq} \sin \theta}$$

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$$M_{low} = \frac{2 \times 6.024 \times 25.0}{\sqrt{2} \times 173.24} = \mathbf{1.2294} \quad M_{high} = \frac{2 \times 6.024 \times 25.0}{\sqrt{2} \times 332.34} = \mathbf{0.6409}$$

Step 5 — the load the tank sees. The 4, not the textbook 8, because $P_{in,LLC}$ is the line-peak power (Section 6.4):

Equation 20, repeated

$$R_{ac} = \frac{4}{\pi^2} \frac{n^2 V_{o,eff}^2}{P_{in,LLC}}$$

$$R_{ac} = \frac{4}{\pi^2} \frac{6.024^2 \times 25.0^2}{670.9} = \mathbf{13.70 \, \Omega}$$

Step 6 — the cap on the quality factor and the design impedance. Two limits are evaluated at the low corner, the Q at which the tank still reaches M_{low} and the Q at which ZVS still completes inside t_D , and the smaller is kept. With Q defined by Equation 1:

$$Q_{ZVS} = \mathbf{1.7019} \quad Z_{0,design} = R_{ac} Q_{ZVS} = 13.70 \times 1.7019 = \mathbf{23.32 \, \Omega}$$

Step 7 — the resonant capacitor and inductor. The design impedance and the target f_r give C_r ; it is taken **up** on purpose, to gain Q margin. L_r then pairs with the selected C_r , which is why it is not $Z_0/2\pi f_r$:

Equation 2, repeated

$$f_r = \frac{1}{2\pi\sqrt{L_r C_r}}$$

$$C_r = \frac{1}{2\pi f_r Z_{0,design}} = \frac{1}{2\pi \times 150 \, \text{kHz} \times 23.32 \, \Omega} = 45.50 \, \text{nF} \rightarrow \mathbf{100 \, \text{nF}}$$

$$L_r = \frac{1}{(2\pi f_r)^2 C_r} = \frac{1}{(2\pi \times 150 \, \text{kHz})^2 \times 100 \, \text{nF}} = 11.26 \, \mu\text{H} \rightarrow \mathbf{11 \, \mu\text{H}}$$

Step 8 — λ and the magnetising inductance. Four candidates for λ are computed and the largest is binding; two of them carry $f_{sw,max}$ in a denominator:

Equation 16, repeated

$$\lambda_2 = \frac{\lambda_1}{1 - \left(\frac{f_r}{f_{sw,max}}\right)^2} \quad \lambda_{TD} = \frac{\lambda_1}{1 - \frac{\pi^2}{8} \left(\frac{f_r}{f_{sw,max}}\right)^2}$$

$$\lambda_1 = 0.560, \quad \lambda_2 = 1.009, \quad \lambda_{TD} = \mathbf{1.241} \quad \Rightarrow \quad L_m = \frac{L_r}{\lambda_{TD}} = \frac{11}{1.241} = 8.87 \, \mu\text{H}$$

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That 8.9 μH is the *no-load* condition at the high corner, which this design knowingly does not meet (Section 4.10), so it is not rounded to. L_m is chosen with n so that n_T lands on a windable ratio, and the λ that results is:

$$L_m = 20 \mu\text{H} \quad \lambda_{act} = \frac{L_r}{L_m} = \frac{11}{20} = 0.550$$

Step 9 — what the selected parts actually make. These, not the targets, are what every later check is measured against:

Equation 3, repeated

$$f_o = \frac{1}{2\pi\sqrt{(L_r + L_m) C_r}}$$

$$f_r = \frac{1}{2\pi\sqrt{11 \mu\text{H} \times 100 \text{ nF}}} = 151.7 \text{ kHz} \quad f_o = \frac{1}{2\pi\sqrt{31 \mu\text{H} \times 100 \text{ nF}}} = 90.4 \text{ kHz}$$

$$Z_0 = \sqrt{\frac{11 \mu\text{H}}{100 \text{ nF}}} = 10.49 \Omega \quad Q_{pk} = \frac{Z_0}{R_{ac}} = \frac{10.49}{13.70} = 0.766$$

$$n_T = n\sqrt{1 + \lambda_{act}} = 6.024 \times 1.2450 = 7.500 = 15 : 2$$

Step	Quantity	Eq.	Result
1	$P_{in}, P_{in,LLC}$	—	684.8 W, 670.9 W
2	$V_{ac,eq}$, low and high corner	18	173.2, 332.3 Vac
3	n, V_{refl}	21, 19	6.024, 150.6 V
4	M_{low}, M_{high}	12	1.2294, 0.6409
5	R_{ac}	20	13.70 Ω
6	$Q_{ZVS}, Z_{0,design}$	1	1.7019, 23.32 Ω
7	C_r, L_r	2	45.5 $\rightarrow$ 100 nF, 11.26 $\rightarrow$ 11 μH
8	$\lambda_{req}, L_m, \lambda_{act}$	16	1.241, 20 μH , 0.550
9	f_r, f_o, Q_{pk}, n_T	2, 3, 1	151.7 kHz, 90.4 kHz, 0.766, 7.500

Table 8: The design, step by step: the results of the nine steps.

Steps 7 and 8 are the judgement calls. C_r was taken to 100 nF against a calculated 45.5 nF and L_m to 20 μH against 8.9 μH ; both lower Q_{pk} and hold λ , which is where the ZVS margin comes from. Everything else is arithmetic any two engineers would reproduce identically.

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No-load solution: none. The no-load gain of this tank bottoms out at $M_{\infty} = 0.6452$, and the high corner asks for 0.6409, below it. No frequency reaches that gain without load, and the margin is under one per cent, so no full-load check would show it. Burst mode owns that region (Section 4.10).

7.5 ZVS over the whole operating space

The closed form of Section 6.5 reads 292 ns on this tank against a swept 460 ns, so it understates the margin by 36 %; on another tank in the same family it overstates by about 23 %. Read with $\lambda_{\text{act}} = 0.550$ instead of the design $\lambda = 1.241$ it returns -471 ns, a capacitive answer for a tank well inside the inductive region. So this design is judged by the sweep.

The capacitive edge of the full-load curve is where $\arg Z_{\text{in}} = 0$, at 110.1 kHz; the gain peak sits lower, at 105.4 kHz, and there the bridge still sees 8.4° of capacitive phase (Section 2.9). Neither is used for the verdict: the sweep measures T_{ZC} directly.

The worst point is not the corner the tank was designed at. Full load at the low equivalent input is where the gain requirement is set, and it gives $T_{\text{ZC}} = 460$ ns, a margin of 2.09. The smallest number in the table is **401 ns at 25 % load and the 332.3 Vac equivalent corner**, a margin of 1.82 against $t_D = 220$ ns. That is the number this design is held to.

Why light load at high line can be the ZVS corner. The charge that swings the bridge node is carried by the magnetising current, and its peak goes as $V_{\text{in}} / (4f_{\text{sw}} L_m)$. Shed load and the controller raises f_{sw} to cut the gain, so that current falls while the node still has the same capacitance to move. Sweeping only the design corner therefore reports a margin that is not there.

7.6 The gain chart of this design

Section 4.5 said how this chart is read. Figure 38 is it, on this tank, at the two equivalent inputs that decide the design. Everything on it comes from $\lambda_{\text{act}} = 0.550$ and $Q_{\text{pk}} = 0.766$, and the marked crossings are the same operating points the sweep in the rest of this chapter is built from.

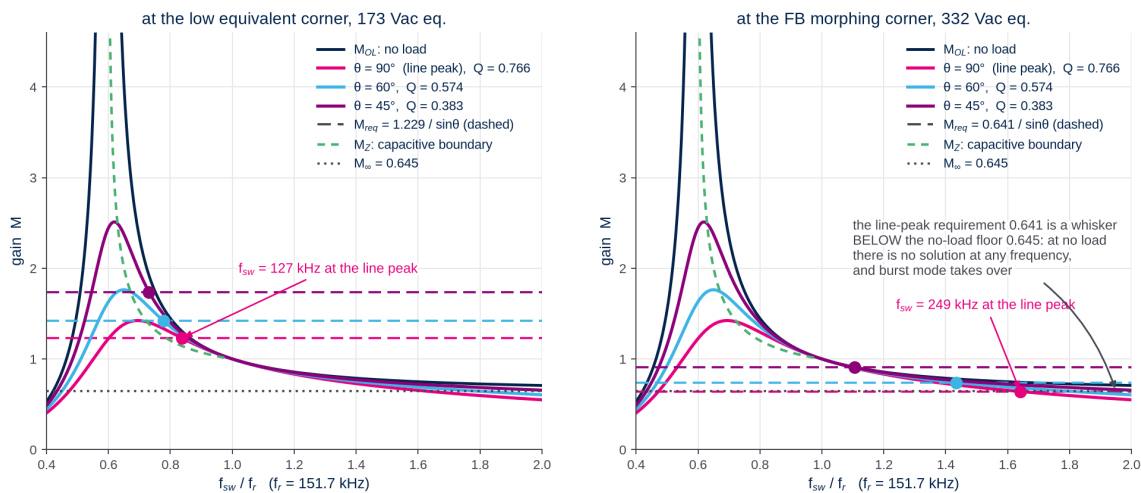


Figure 38: The gain chart of this design. Left: at the low equivalent corner, 173 Vac eq., where the required gain is highest and the crossings sit below f_r . Right: at the FB morphing corner, 332 Vac eq., where the required gain is lowest and the crossings are above f_r . Each phase is compared with the dashed line of its own colour and with no other. Every other line condition lies between these two panels.

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Corner	θ	$Q(\theta)$	$M_{\text{req}}(\theta)$	f_{sw}/f_r	f_{sw}
173 Vac eq.	90°	0.766	1.229	0.838	127.1 kHz
	60°	0.574	1.420	0.778	118.1 kHz
	45°	0.383	1.739	0.731	110.9 kHz
332 Vac eq.	90°	0.766	0.641	1.641	249.0 kHz
	60°	0.574	0.740	1.435	217.8 kHz
	45°	0.383	0.906	1.106	167.8 kHz

Table 9: The six marked crossings, read off Figure 38. $Q = Q_{\text{pk}} \sin^2 \theta$ and $M_{\text{req}} = M_{\text{pk}} / \sin \theta$, with $M_{\text{pk}} = 1.229$ at the low corner and 0.641 at the FB corner.

Three readings matter, and none of them is “the curve is above the line”.

- **The low corner sets the tank.** The 173 Vac equivalent is the lowest voltage the tank ever sees (Section 6.2), so its line-peak requirement $M_{\text{pk}} = 1.229$ is the largest gain the tank is ever asked for. It is met at $f_{\text{sw}}/f_r = 0.838$, comfortably to the right of M_{Z} : inductive, so the bridge switches softly there.
- **The FB corner sets the frequency.** The same chart drawn at 332 Vac equivalent puts the line-peak crossing at 249 kHz, and that number — not the mains maximum — is what the oscillator ceiling has to clear (Section 7.16).
- **The right-hand panel is also where λ is judged.** The line-peak requirement there is 0.641 and the no-load floor is $M_{\infty} = 1/(1+\lambda) = 0.645$. The requirement is **below** the floor, so at no load and high line there is no frequency that satisfies it and burst mode carries the converter. That is the whole content of the λ bound of Section 4.10, visible as two nearly coincident lines on the right of the picture.

Read the two panels as the two ends of one line cycle, not as two designs. The converter is at the left-hand chart when the mains is low and at the right-hand one when it is high, and at every ac input in between it is on a chart that lies between them with the curves unchanged and only the dashed lines moved. The curves do not depend on the input voltage at all: $M(f_n, Q)$ is fixed by λ_{act} and by Q_{pk} , which is a full-load number. **The input voltage enters only through M_{req} .**

7.7 Which side of resonance this design runs on

Section 6.6 says the turns ratio decides this. With $n_T = 7.50$, at all but the lowest line conditions the tank spends part of every line cycle above f_r , where the secondary loses zero-current turn-off, so the rectifier body diode and the SR dead time both need checking.

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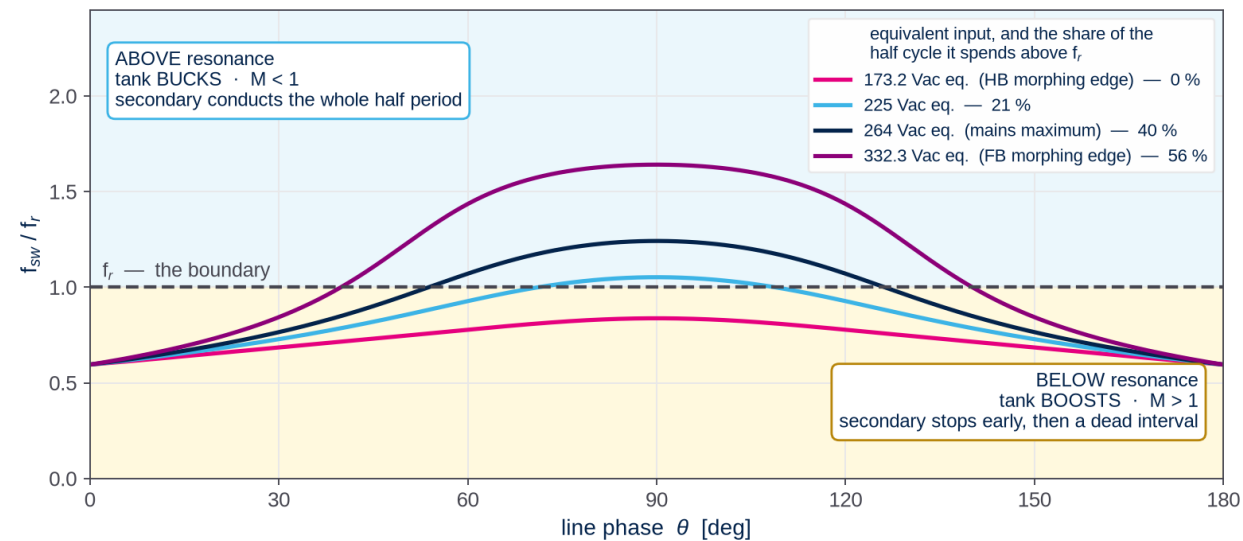


Figure 39: Which side of f_r the converter is on over a line half cycle, at four equivalent inputs. At the low corner it never leaves the boosting region; at the high corner it spends most of the cycle bucking. Every curve converges on f_o at the zero crossing. The percentage in each legend entry is the fraction of the 10 ms half cycle that curve spends above the dashed f_r line — the time the secondary spends without zero-current turn-off.

Line condition	V_{eq}	peak f_{sw}	side of f_r
minimum line	180 V	131.3 kHz	below
tank minimum	173 V	127.1 kHz	below
110 Vac	220 V	156.4 kHz	above
nominal	225 V	159.7 kHz	above
230 Vac	230 V	163.2 kHz	above
maximum line	264 V	188.5 kHz	above
FB threshold	332 V	249.0 kHz	above

Table 10: Peak f_{sw} over the half cycle against $f_r = 151.7$ kHz. 5 of the seven line conditions cross into above-resonance operation for part of the cycle.

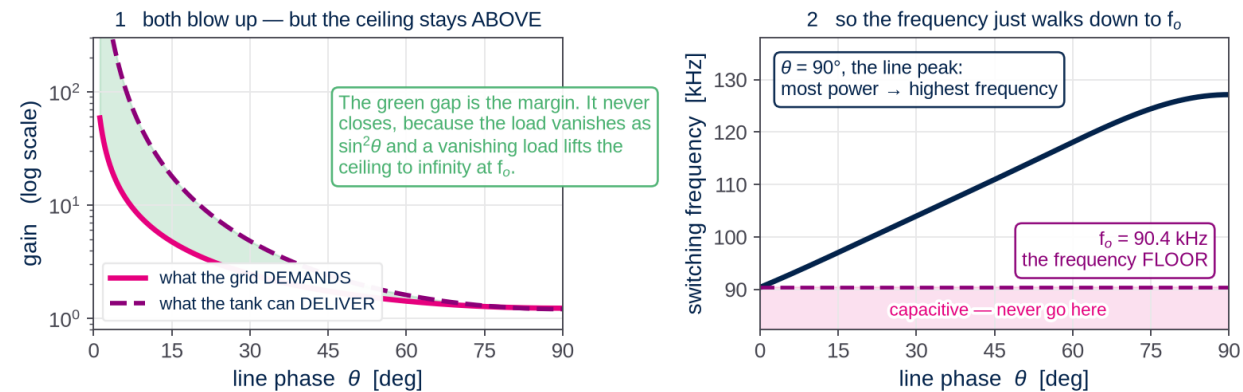


Figure 40: Near the zero crossing the required gain diverges and the load vanishes together, so the operating point converges on $f_o = 90.4$ kHz (Section 4.2).

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7.8 The currents this design has to carry

Ratings come from the worst switching cycle and losses from the line-cycle rms, so both are listed. The row to watch is the composite tank peak: it is what the primary devices and R_{CS} see, and it is neither the sum of the two component peaks nor the larger of them.

Where each kind of number comes from. The peaks are read off one switching cycle at the worst operating point — the low equivalent corner at the line peak, which is the row marked $\theta = 90^\circ$ in Table 9 — using the closed forms of Section 6.7. The composite peak is not read off either component: the two peak at different instants, so it is the peak of their sum,

$$I_{Lr,pk} = \max_t |i_{Lm}(t) + i_{trafo}(t)| = 18.32 \text{ A} < 12.41 + 16.37 = 28.78 \text{ A}$$

The rms values are not that cycle's. Each is the rms of one switching period, squared, averaged over the line half cycle and rooted — the integral evaluated by the five-point Simpson rule of Section 6.7:

$$I_{lc} = \sqrt{\frac{2}{\pi} \int_0^{\pi/2} I_{rms}^2(\theta) d\theta} \simeq \sqrt{\frac{1}{12} \sum_k w_k I_{rms}^2(\theta_k)}, \quad w = \{1, 4, 2, 4, 1\}$$

That is why the line-cycle primary rms, 10.93 A, is *lower* than the worst cycle's 13.76 A: the worst cycle is one instant of a half cycle that spends most of its time at less than full power.

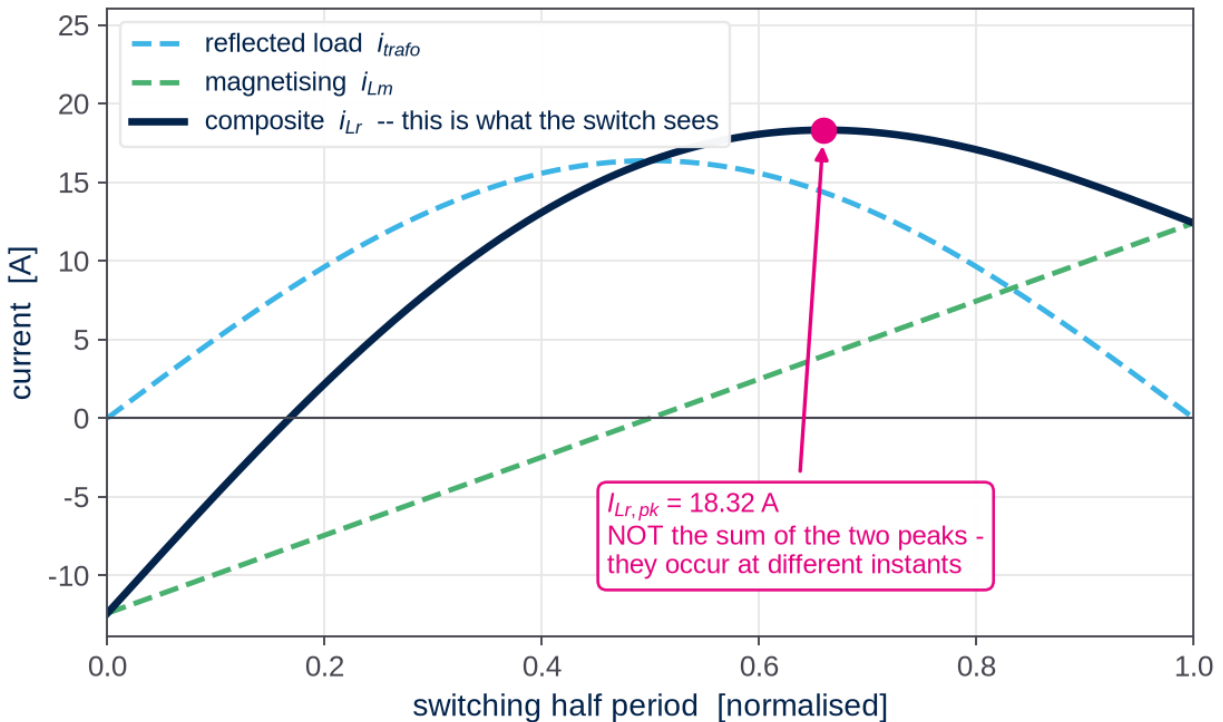


Figure 41: The composite tank current at the low equivalent corner, full load, and why its peak is not the sum of the two component peaks: they occur at different instants.

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Quantity	Value	Where it is used
f_{sw}	127.1 kHz	oscillator floor check
$I_{sec,pk}$	98.6 A	secondary rectifier peak
$I_{trafo,pk}$	16.37 A	reflected load component
$I_{Lm,pk}$	12.41 A	magnetising component
composite tank peak	18.32 A	primary device rating, OCP margin
$I_{pri,rms}$ (worst cycle)	13.76 A	primary rms rating
I_{pri} (line-cycle rms)	10.93 A	primary conduction loss
I_{rect} per leg (line-cycle rms)	28.34 A	secondary conduction loss
I_{Cout} (line-cycle rms)	30.24 A	output bank ripple current

Table 11: Currents at the low equivalent corner, full load, $\theta = 90^\circ$.

7.9 The transformer, as built

The tank asks for $L_r = 11 \mu H$, $L_m = 20 \mu H$ and $n_T = 7.50$. The transformer is one part: a 15-turn primary and two secondary windings of $N_s = 2$ turns, NS2 and NS3, joined at the centre tap, one conducting in each half period. Several units in series are the alternative when one window will not take the copper (Section 6.21); that is not needed here.

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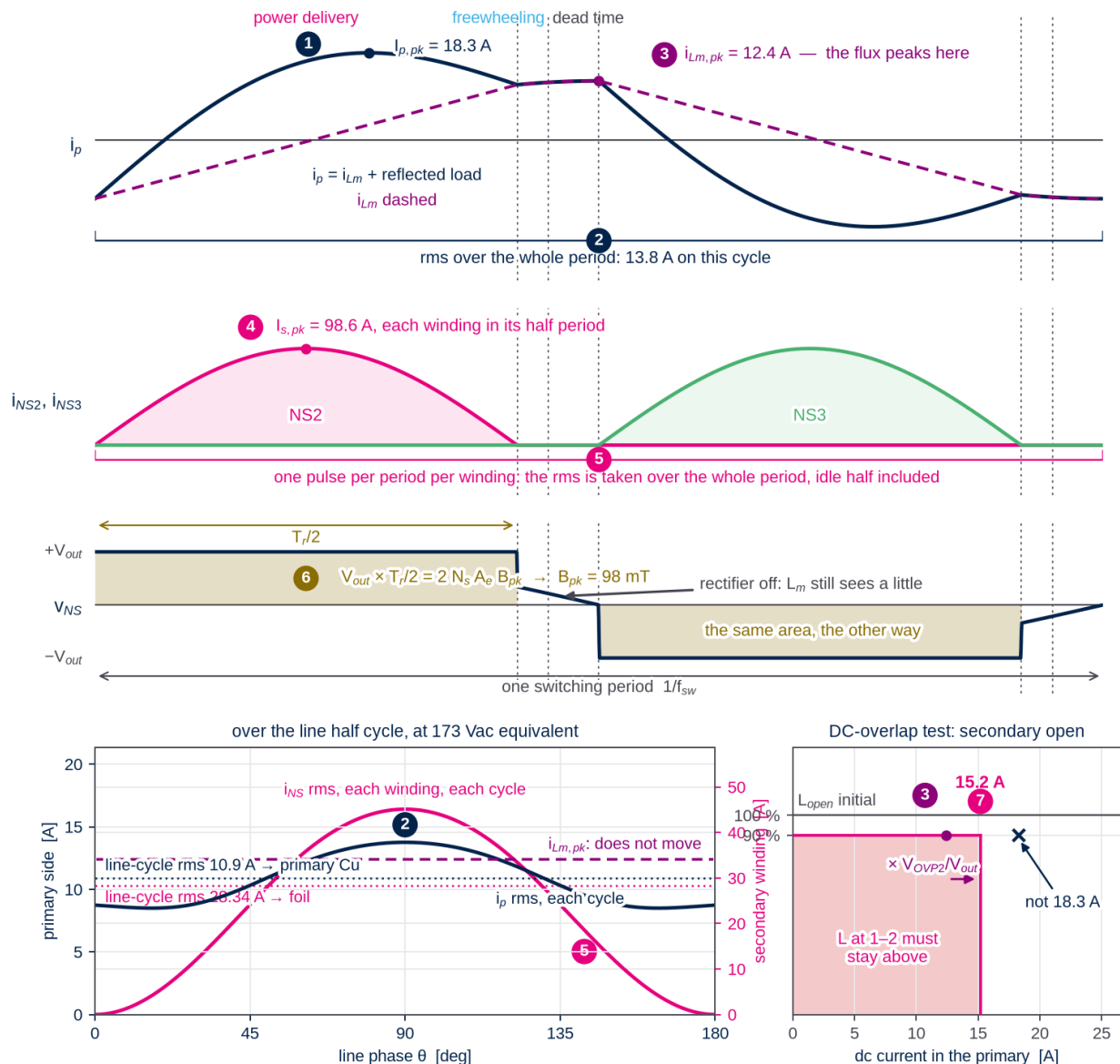


Figure 42: Where each number of the table below is read. Top: the primary winding current, the current in its two secondary windings and the secondary winding voltage over one switching period, at the worst cycle (line peak, 173 Vac equivalent, full load). Lower left: the rms of the two winding currents over the line half cycle, and the line-cycle values the copper is sized on. Lower right: the DC-overlap test, a bench condition. The circled marks are the rows of Table 3.

Quantity	Value	Note
Turns	N_p 15 T; NS2 2 T, NS3 2 T; NAUX 3 T	$n_T = 15 / 2 = 7.5$; the auxiliary is ZCD sense only
Open-circuit inductance	31.0 μ H	$L_r + L_m$, every other winding open
Short-circuit inductance	11.0 μ H	this is L_r : no separate resonant inductor
A_L	138 nH	open-circuit inductance / N_p^2 ; the core is gapped to it

Table 12: The transformer as wound and specified.

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Pins. The part is wound on the B66368 coil former of the ETD 49/25/16DG core, 20 pins in two rows of 10. NP1 and the ZCD auxiliary take one row, NS2 and NS3 the other. Each secondary terminal uses two pins, because one winding carries the whole secondary current. The centre tap is made on the board: the finish of NS2 and the start of NS3 come out on neighbouring pins.

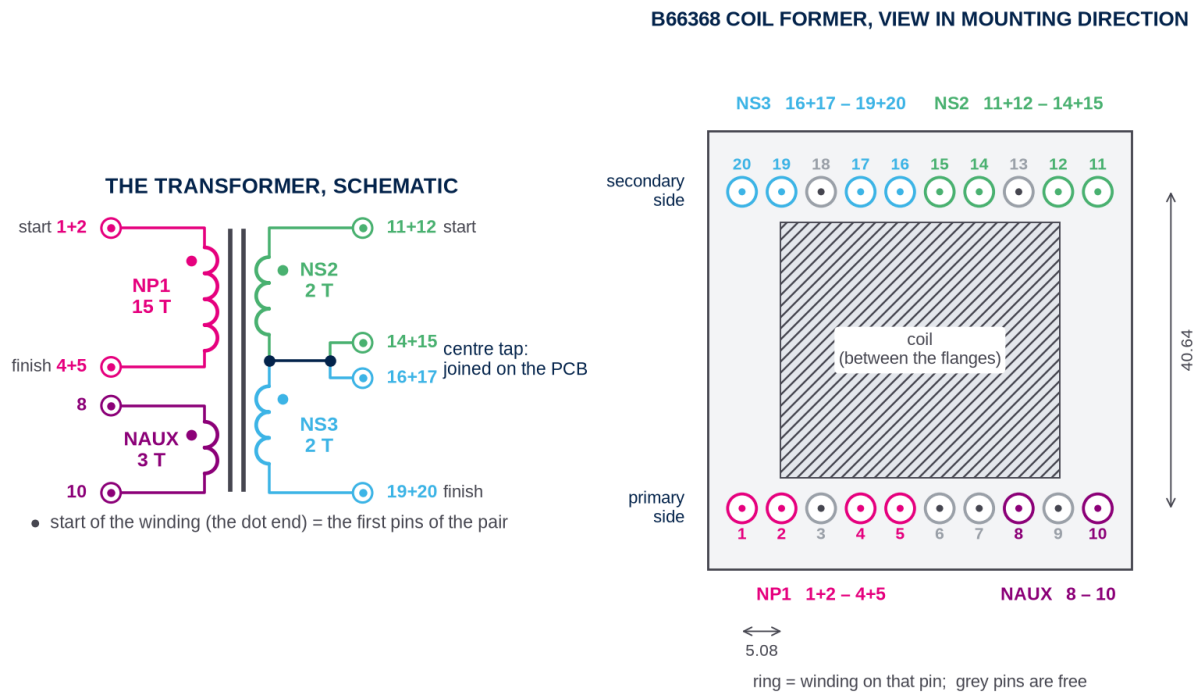


Figure 43: The transformer: the schematic symbol with its pin numbers, and the same pins on the B66368 coil former in the mounting-direction view of the TDK drawing (20 pins, pitch 5.08 mm, rows 40.64 mm apart). A ring in the colour of a winding marks the pins it uses; grey pins are free. The dot end of every winding is the first pins of its pair. The datasheet prints no pin numbers; the numbering shown counts along one row from pin 1 and back along the other, the usual convention. Confirm it against the bobbin drawing before the specification is released; the assignment does not depend on it, only the printed numbers do.

Winding	Pins (start – finish)	Turns	Conductor	Row
NP1 primary	1+2 – 4+5	15 T	Litz 310 × ø0.10 mm	front row 1-10
NAUX auxiliary (ZCD)	8 – 10	3 T	any wire, sense only	front row 1-10
NS2 secondary	11+12 – 14+15	2 T	foil 0.20 × 10.5 mm, 3 in parallel	back row 11-20
NS3 secondary	16+17 – 19+20	2 T	the same foil, on top of NS2	back row 11-20
Centre tap	14, 15, 16 and 17, joined on the PCB	—	—	back row 11-20
Free	3, 6, 7, 9, 13, 18	—	not connected	both rows

Table 13: Winding-to-pin assignment.

Polarity. With the dots as drawn, the ZCD pin is positive while the low-side switch of leg 1 is on, as the controller requires. NS2 and NS3 are wound in the same sense; the tap joins the finish of NS2 to the start of NS3.

Three more numbers follow from Table 12.

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Peak flux density, at f_r :

Equation 24, repeated

$$B_{pk} = \frac{V_{out, eff}}{4 f_r N_s A_e} \implies A_e \geq \frac{V_{out, eff}}{4 f_r N_s B_{max}}$$
$$B_{pk} = \frac{25.0 \text{ V}}{4 \times 151.75 \text{ kHz} \times 2 \times 211.0 \text{ mm}^2} = \mathbf{98 \text{ mT}}$$
$$A_e \geq \frac{25.0 \text{ V}}{4 \times 151.75 \text{ kHz} \times 2 \times 0.20 \text{ T}} = \mathbf{103 \text{ mm}^2}$$

Magnetising peak, over L_μ (it equals $i_{\mu, pk}$):

Equation 32, repeated

$$I_{eq} = \frac{B_{pk} N_p A_e}{L_\mu}$$
$$I_{eq} = \frac{0.0976 \text{ T} \times 15 \times 211.0 \text{ mm}^2}{24.90 \text{ } \mu\text{H}} = \mathbf{12.41 \text{ A}} = i_{\mu, pk}$$

Saturation test current, the magnetising peak at the OVP2 output voltage:

Equation 33, repeated

$$I_{sat} = I_{eq} \frac{V_{OVP2}}{V_{out, eff}}$$
$$I_{sat} = 12.41 \text{ A} \times \frac{30.61 \text{ V}}{25.0 \text{ V}} = 12.41 \times 1.2244 = \mathbf{15.2 \text{ A}}$$

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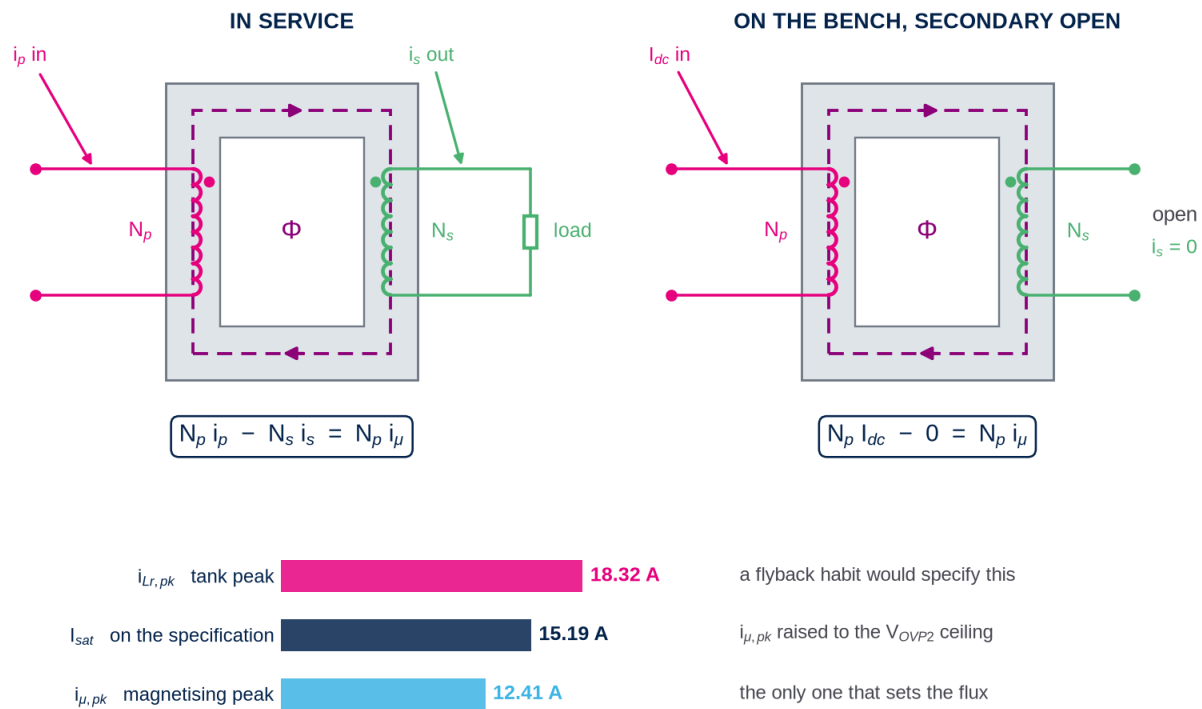


Figure 44: In service the secondary cancels most of the primary ampere-turns; with the secondary open, the whole test current magnetises the core. The design flux is therefore reached at 12.41 A on the bench, not at the 18.32 A tank peak.

7.10 The core and the winding

The core is TDK **ETD 49/25/16DG**, N95 (core B66367, coil former B66368). The centre leg is ground to A_L = 138 nH, about 1.1 mm of gap in total. Figure 45 shows the core in section and the winding in its window; Table 14 names each item and Table 15 gives the arithmetic behind the winding.

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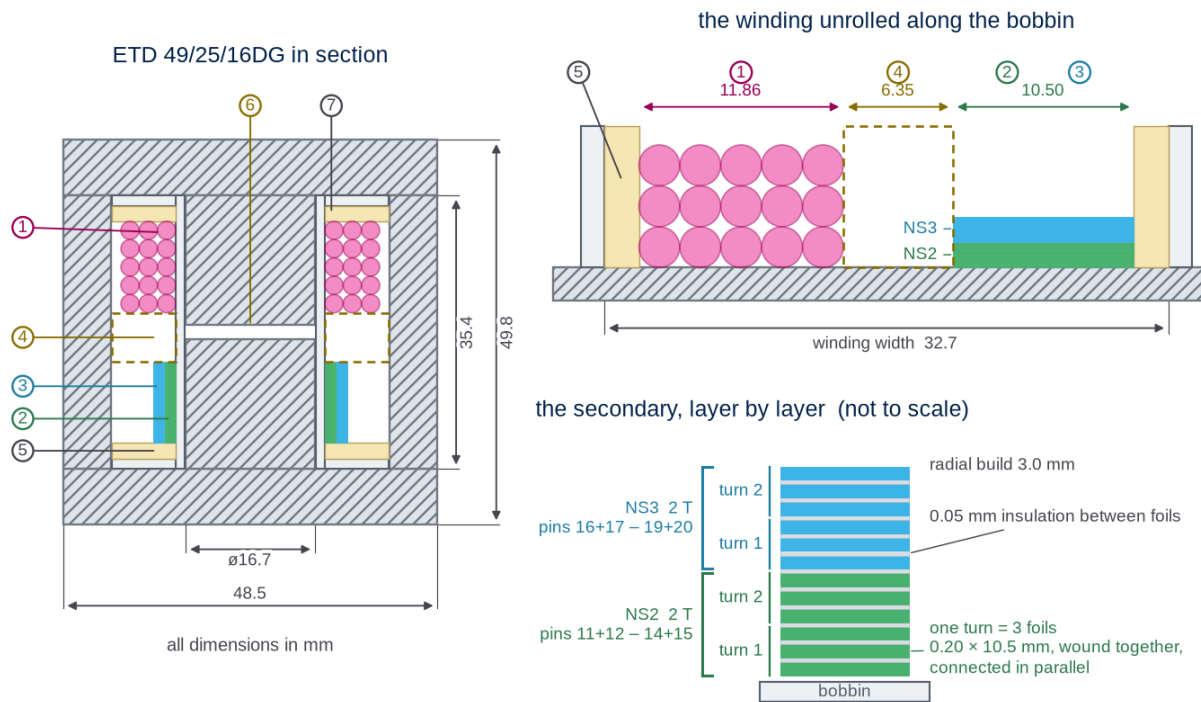


Figure 45: The core in section (left, to scale), the winding unrolled along the bobbin (top right, to scale) and the secondary layer by layer (bottom right, not to scale). Primary and secondary sit side by side with 6.35 mm between them. The circled numbers are the rows of Table 14.

Mark	Item	Turns	Conductor, placement
1	NP1 primary, pins 1+2 – 4+5	15 T	Litz 310 x Ø0.10 mm (bundle Ø2.37 mm), 3 layers of 5/5/5 turns
2	NS2 secondary, pins 11+12 – 14+15	2 T	foil 0.20 x 10.5 mm, 3 foils in parallel per turn; on the bobbin
3	NS3 secondary, pins 16+17 – 19+20	2 T	the same foil, on top of NS2
4	Separation	—	6.35 mm between primary and secondary; sets L_{short}
5	Margin tape	—	2.0 mm at each flange
6	Centre-leg gap	—	about 1.1 mm in total, ground to $A_L = 138$ nH
7	Window	—	$A_N = 269$ mm ² , 62 mm ² of copper in it
—	NAUX auxiliary, pins 8 – 10	3 T	any wire; ZCD sense only

Table 14: Items of the figure.

Table 15 is that arithmetic, and the three numbers in it that are not simply a division are worked here first.

How thick a conductor is worth using. Equation 28 again, with copper at 100 °C — $\rho = \rho_{20}(1 + 0.00393 \times 80) = 2.26 \times 10^{-8} \Omega\text{m}$ — at the series resonance:

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Equation 28, repeated

$$\delta = \sqrt{\frac{\rho}{\pi f \mu_0}}$$
$$\delta = \sqrt{\frac{2.26 \times 10^{-8}}{\pi \cdot 151.7 \times 10^3 \cdot 4\pi \times 10^{-7}}} = 0.194 \text{ mm}$$

A round conductor thicker than about $2\delta = 0.39 \text{ mm}$ carries no more current than one of 2δ (Section 6.16), so the primary is Litz. The standard **strand** — one of the individually insulated fine wires the bundle is made of — nearest and safely inside that is $d_s = \text{ø}0.10 \text{ mm}$, which is 3.9 times smaller than 2δ and therefore fully penetrated. Its copper cross-section a_s , and the number of strands the winding needs, are Equation 29:

Equation 29, repeated

$$a_s = \frac{\pi d_s^2}{4}, \quad n_s = \left\lceil \frac{A_{cu}}{a_s} \right\rceil$$
$$a_s = \frac{\pi \cdot (0.10)^2}{4} = 0.00785 \text{ mm}^2, \quad n_s = \left\lceil \frac{2.43}{0.00785} \right\rceil = 310$$

The served bundle is bigger than that bare copper, so its outside diameter comes from Equation 30 with a fill factor $k_{litz} = 0.55$:

Equation 30, repeated

$$d_{litz} = \sqrt{\frac{4A_{cu}}{\pi k_{litz}}}$$
$$d_{litz} = \sqrt{\frac{4 \cdot 2.43}{\pi \cdot 0.55}} = 2.37 \text{ mm}$$

The secondary is foil instead. Two turns of 6.30 mm^2 as round wire would be a bundle thicker than the whole build. At the standard gauge $t_f = 0.20 \text{ mm}$, Equation 31 gives

$$w_f = \frac{6.30}{0.20} = 31.5 \text{ mm}, \quad n_f = \left\lceil \frac{31.5}{12.0} \right\rceil = 3 \Rightarrow 3 \times 10.5 \text{ mm}$$

so one turn is **3 strips of 10.5 mm in parallel**, stacked radially rather than one strip laid axially.

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Step	Primary NP1	Secondary NS2
Rms current	10.93 A	28.34 A
Copper area at $J = 4.5 \text{ A/mm}^2$	$10.93 / 4.5 = \mathbf{2.43 \text{ mm}^2}$	$28.34 / 4.5 = \mathbf{6.30 \text{ mm}^2}$
Conductor ($\delta = 0.194 \text{ mm}$, Equation 28)	Litz of $\varnothing 0.10 \text{ mm}$ strands, each $a_s = 0.00785 \text{ mm}^2$ (Equation 29): $2.43 / 0.00785 = \mathbf{310 \text{ strands}}$, bundle $\varnothing 2.37 \text{ mm}$ (Equation 30)	foil 0.20 mm thick: $6.30 / 0.20 = 31.5 \text{ mm}$ wide, made as 3 strips of 10.5 mm in parallel
Turns	15, in 3 layers (5/5/5)	2
Width on the bobbin	$5 \times 2.37 = \mathbf{11.86 \text{ mm}}$	10.5 mm
Radial build	$3 \times 2.37 = 7.11 \text{ mm}$	NS2 + NS3: $12 \times (0.20 + 0.05) = 3.0 \text{ mm}$
Left between the windings	$32.7 - 2 \times 2.0 - 11.86 - 10.5 = \mathbf{6.35 \text{ mm}}$	
Copper in the window	$15 \times 2.43 + 4 \times 6.30 = 61.6 \text{ mm}^2$, 76 % of A_N at $k_u = 0.30$	

Table 15: The winding, from current to copper. NS3 is the same as NS2.

Reduced to what a supplier can measure at the terminals, this gives the specification sheet. L_μ is deliberately absent (Section 6.10).

Item	Value	Condition
Core	ETD 49/25/16DG, N95, A_L 138 nH	core B66367, coil former B66368; distributed gap
Turns	N_p 15 T; NS2 2 T and NS3 2 T; NAUX 3 T	two secondary windings, centre tap outside the part
Open-circuit inductance	$31.0 \mu\text{H}$, -4.1% at worst	primary, all other windings open
Leakage inductance	$11.0 \mu\text{H} \pm 10 \%$	primary, secondary shorted
DC overlap	$\geq 90 \%$ of initial inductance	test current 15 A (mark 7)
Primary current	10.9 A rms / 18.3 A pk	line-cycle rms (mark 2), composite peak (mark 1)
Secondary current, each winding	28.3 A rms / 99 A pk	line-cycle rms (mark 5), peak (mark 4)
Core area	$A_e \geq 103 \text{ mm}^2$	holds B_{pk} at or below 0.20 T (mark 6)
Switching frequency	127 to 249 kHz	at full load

Table 16: Transformer specification for the worked design.

7.11 The output bank, as sized

Both conditions of Section 6.23, on this specification; the larger wins.

The ripple condition, with Δv as a fraction and V_{out} in volts:

Equation 35, repeated

$$C_{out} \geq \frac{P_{out}}{2\pi f_{l,min} \Delta v V_{out}^2}$$

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$$C_{out} \geq \frac{657.5 \text{ W}}{2\pi \times 47 \text{ Hz} \times 0.05 \times 625 \text{ V}^2} = \mathbf{71.25 \text{ mF}}$$

The hold-up condition, starting half a ripple below V_{out} because the mains may disappear at the ripple trough:

Equation 36, repeated

$$C_{out} \geq \frac{2P_{out}T_{hold}}{\left(V_{out} - \frac{1}{2}\Delta v_{pp}\right)^2 - V_{o,min}^2}$$

$$C_{out} \geq \frac{2 \times 657.5 \text{ W} \times 0.012 \text{ s}}{24.41^2 - 19^2} = \mathbf{67.22 \text{ mF}} \quad (\text{started at } V_{out} : 59.77 \text{ mF})$$

Which one wins is settled by the screening inequality, at $k = V_{o,min}/V_{out} = 0.760$:

Equation 37, repeated

$$\left(1 - \frac{\Delta v}{2}\right)^2 - k^2 > 4\pi f_l \Delta v T_{hold}$$

$$0.373 > 0.354 \implies \text{ripple decides}$$

The two are only 1.060 apart; relaxed to 10 % ripple the two sides would swap. The bank fitted is the next assembly up from 71.2 mF: $\mathbf{470 \mu F \times 160 = 75.2 \text{ mF}}$.

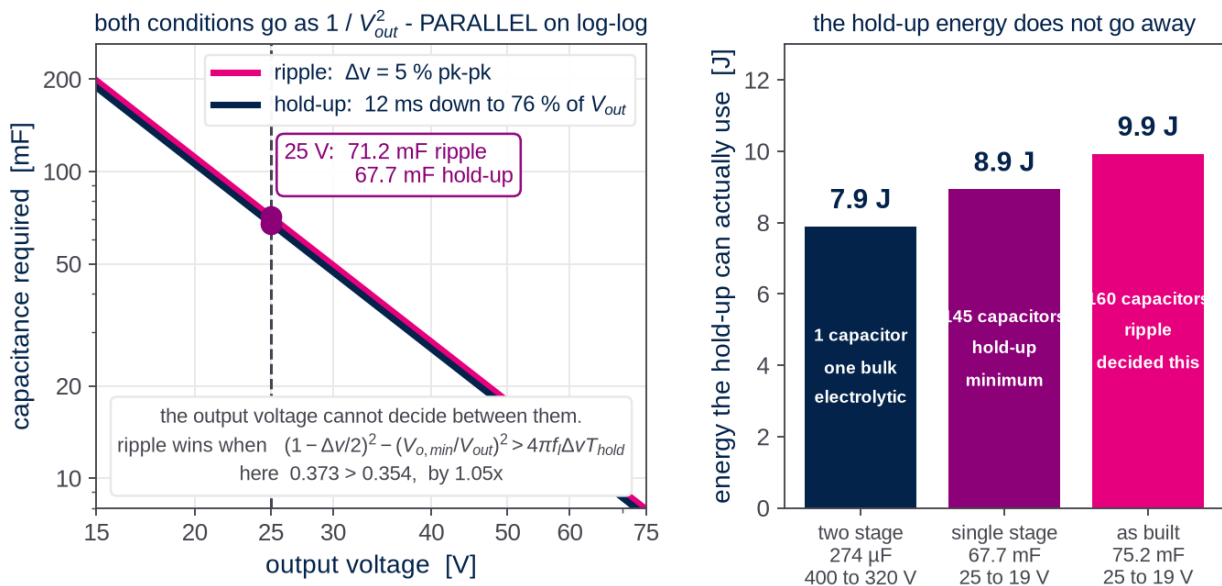


Figure 46: Left: both sizing conditions fall as $1/V_{out}^2$, so only the specification separates them; at 25 V ripple asks for 71.2 mF and hold-up for 67.2 mF. Right: the hold-up energy is the same 7.9 J either side of the boost stage; on a 400 V bus falling to 320 V it is a 274 μF part, on this output it is 67.2 mF, 245 times more, because the usable voltage window is so much smaller. That ratio is the price of the architecture.

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Quantity	Value	Against
Achieved ripple	1.18 V (4.74 %)	5 % allowed
Achieved hold-up	13.42 ms	12 ms required — 15.10 ms if started at V_{out} instead of at the trough
Ripple current, switching component	23.8 A rms	—
Ripple current, $2f_l$ component	18.6 A rms	orthogonal to the above
Ripple current, total	30.24 A rms	0.19 A in each of 160 parts
Bank ESR	0.1875 mΩ	30 mΩ each, 160 in parallel
Ripple and noise	17.4 mV	with the 10 μF ceramic bypass

Table 17: What the selected bank then delivers, and what it has to survive.

7.12 Where the power goes

The specification allowed 27.3 W of loss between $P_{in} = 684.8$ W and $P_{out} = 657.5$ W, split three ways before any component existed. The device losses computed afterwards do not have to agree with that split, and here they do not.

Item	Value	Note
Budget — input bridge	5.52 W	synchronous bridge assumed
Budget — EMI filter	8.34 W	—
Budget — LLC stage	13.42 W	$\eta_{HB} = 98\%$ of $P_{in,LLC}$
Primary conduction, standing device	9.65 W	the half-bridge leg that never switches — the single largest item
Primary switching, three positions	4.83 W	not a problem
Secondary rectifiers	5.65 W	1.411 W per device, 2 in parallel per leg
Current-sense resistors	2.86 W	4.54 W at the worst switching cycle
Output capacitor ESR	0.162 W	—
Itemised total	23.15 W	and the magnetics are not in it

Table 18: The budget as assumed, and the device losses as computed. The second block is not a breakdown of the first.

Two rows carry most of that total, and both are one product. **The standing primary device** is the low-side switch of the idle leg in half-bridge morphing: it is on continuously and carries the whole line-cycle rms, so its loss is I^2R with $R_{DS(on)}$ taken **hot**, at $k_T = 1.8$ times the 25 °C headline of 45 mΩ (Section 6.24):

$$P_{cond} = I_{pri}^2 \cdot k_T R_{DS(on), 25} = (10.93)^2 \cdot 1.8 \cdot 45 \times 10^{-3} = 9.65 \text{ W}$$

Each secondary rectifier carries the leg current divided by the 2 devices in parallel, with its own $k_T = 1.9$ on 3.7 mΩ:

$$P_{SR} = \left(\frac{28.34}{2} \right)^2 \cdot 1.9 \cdot 3.7 \times 10^{-3} = 1.411 \text{ W per device}$$

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The efficiency assumption is optimistic. The itemised device losses already come to 23.2 W against the 13.4 W that $\eta_{HB} = 98\%$ allows the LLC stage, before the transformer. Nothing electrical depends on it (95 % moves R_{ac} and R_{CS} by about 3 %); the thermal design does, and that is settled by measurement.

7.13 What the semiconductors have to be

Requirements, not part numbers; the four rules of Section 6.24 decide whether they are stated correctly.

Item	Requirement
Primary drain-source voltage	≥ 485 V, so a 600 V class part
Primary body diode	fast recovery: in full bridge the body diode conducts before turn-on
Primary $R_{DS(on)}$ per device	≤ 25.2 m Ω hot to meet the loss budget
Primary $C_{o(tr)}$	small enough that the tank can swing the node inside 220 ns
Secondary drain-source voltage	≥ 80 V including the centre-tap doubling
Secondary rectifier current	28.34 A rms per leg, 98.6 A peak
Secondary package	check the lead and clip rating, not only the die

Table 19: Semiconductor requirements for the worked design.

7.14 The voltage loop, as built

Sections 6.26 to 6.33 give the method. These are its numbers on this converter, in the order they are found. Every value comes from the design sheet.

Step 1 — the output divider. R_I is chosen as 100 k Ω and R_O follows from Equation 43 solved for it:

Equation 43, repeated

$$V_{out} = V_R \left(1 + \frac{R_I}{R_O} \right)$$

$$R_O = \frac{V_R R_I}{V_{out} - V_R} = \frac{2.495 \text{ V} \times 100 \text{ k}\Omega}{25.0 \text{ V} - 2.495 \text{ V}} = 11.09 \text{ k}\Omega \rightarrow \mathbf{11 \text{ k}\Omega}$$

$$V_{out} = 2.495 \text{ V} \times \left(1 + \frac{100}{11} \right) = \mathbf{25.18 \text{ V}}$$

Step 2 — the plant. The feedback voltage above its offset at rated power is $V_{FB} = K_{pwr} R_{CS} P_{in,LLC} = 0.167 \times 24 \text{ m}\Omega \times 670.9 \text{ W} = 2.689 \text{ V}$ (Equation 15). Then

Equation 39, repeated

$$G_{plant}(s) = \frac{v_{out}(s)}{v_{FB}(s)} = \frac{G_o}{s}, \quad G_o = \frac{P_{out}}{V_{out} V_{FB} C_{out}} \quad [\text{rad/s}]$$

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$$G_o = \frac{657.5 \text{ W}}{25.0 \text{ V} \times 2.689 \text{ V} \times 75.2 \text{ mF}} = \mathbf{130.1 \text{ rad/s}}, \quad f_{cto} = \frac{130.1}{2\pi} = 20.7 \text{ Hz}$$

Step 3 — the gain the loop may have at $2f_i$. The output ripple at the lowest line frequency, and the compensator gain that keeps the third harmonic at 5 %:

Equation 40, repeated

$$\Delta V_{loop} = \frac{P_{out}}{V_{out}} \frac{1}{2\pi f_i C_{out}}$$

$$\Delta V_{loop} = \frac{657.5 \text{ W}}{25.0 \text{ V}} \cdot \frac{1}{2\pi \times 47 \text{ Hz} \times 75.2 \text{ mF}} = \mathbf{1.184 \text{ V}}$$

Equation 42, repeated

$$G_{EA}(2f_i) \leq \frac{4 V_{FB} D_3}{\Delta V_{loop}}$$

$$G_{EA}(2f_i) \leq \frac{4 \times 2.689 \text{ V} \times 0.05}{1.184 \text{ V}} = \mathbf{0.4541}$$

Step 4 — the K factor, for a target phase margin of 50° ($\tan 50^\circ = 1.1918$) and $\alpha_v = 1.2$:

Equation 48, repeated

$$K_v = \frac{1}{2\alpha_v} \left[(1 + \alpha_v^2) \tan \Phi_M + \sqrt{(1 + \alpha_v^2 \tan \Phi_M)^2 + 4\alpha_v^2} \right]$$

$$K_v = \frac{1}{2 \times 1.2} \left[(1 + 1.2^2) \times 1.1918 + \sqrt{(1 + 1.2^2 \times 1.1918)^2 + 4 \times 1.2^2} \right] = \mathbf{2.7218}$$

Step 5 — the compensator gain constant:

Equation 49, repeated

$$EA_o = \frac{2\pi (2f_i) G_{EA}(2f_i)}{K_v^2}$$

$$EA_o = \frac{2\pi \times 94 \text{ Hz} \times 0.4541}{2.7218^2} = \mathbf{36.20 \text{ rad/s}}$$

Step 6 — where the zero and pole go. With $\Gamma_v = 0.744 \times 264.0 / 173.24 = 1.1338$:

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Equation 50, repeated

$$f_{MB} = \frac{1}{2\pi} \sqrt{\frac{G_o K_v E A_o}{\Gamma_v}}, \quad f_p = K_v f_{MB}, \quad f_z = \frac{f_{MB}}{K_v}$$

$$f_{MB} = \frac{1}{2\pi} \sqrt{\frac{130.06 \times 2.7218 \times 36.20}{1.1338}} = \mathbf{16.92 \text{ Hz}}$$

$$f_p = 2.7218 \times 16.92 = \mathbf{46.06 \text{ Hz}}, \quad f_z = \frac{16.92}{2.7218} = \mathbf{6.217 \text{ Hz}}$$

Step 7 — the bias parts. R_p from Equation 44 with the TL431 minimum current 0.8 mA, and the R_B window from Equation 47 with $V_Z = 12 \text{ V}$, $V_{Fo} = 1.15 \text{ V}$, $I_{FB,steady} = 150 \mu\text{A}$, $I_{FB,max} = 300 \mu\text{A}$, $CTR_s = 0.37$ and $CTR_m = 0.53$:

Equation 44, repeated

$$R_p \leq \frac{V_{Fo}}{I_{min}}$$

$$R_p \leq \frac{1.15 \text{ V}}{0.8 \text{ mA}} = 1.437 \text{ k}\Omega \rightarrow \mathbf{1.3 \text{ k}\Omega} \text{ (rounded down)}$$

Equation 47, repeated

$$R_{B,max} = \frac{V_Z - (V_R + V_{Fo})}{V_{Fo}/R_p + I_{FB,steady}/CTR_s}$$

$$R_{B,min} = \frac{V_Z - (V_R + V_{Fo})}{V_{Fo}/R_p + I_{FB,max}/CTR_m}$$

$$R_{B,max} = \frac{12 - (2.495 + 1.15) \text{ V}}{1.15/1.3 \text{ mA} + 150 \mu\text{A}/0.37} = \frac{8.355 \text{ V}}{0.885 \text{ mA} + 0.405 \text{ mA}} = \mathbf{6.48 \text{ k}\Omega}$$

$$R_{B,min} = \frac{8.355 \text{ V}}{0.885 \text{ mA} + 300 \mu\text{A}/0.53} = \mathbf{5.76 \text{ k}\Omega} \rightarrow R_B = \mathbf{6.2 \text{ k}\Omega}$$

Step 8 — the compensation parts, with $R_{FB} = 35 \text{ k}\Omega$, $C_{opto} = 3 \text{ nF}$ and the high-frequency pole placed at 1000 Hz:

Equation 51, repeated

$$C_{Fo} = \frac{f_z}{f_p} \cdot \frac{CTR_s R_{FB}}{R_I R_B E A_o}, \quad C_F = C_{Fo} \left(\frac{f_p}{f_z} - 1 \right)$$

$$R_F = \frac{1}{2\pi f_z C_F}, \quad C_{fx} = \frac{1}{2\pi f_{pHF} R_{FB}} - C_{opto}$$

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$$C_{Fo} = \frac{6.217}{46.06} \cdot \frac{0.37 \times 35 \text{ k}\Omega}{100 \text{ k}\Omega \times 6.2 \text{ k}\Omega \times 36.20} = 77.9 \text{ nF} \rightarrow \mathbf{82 \text{ nF}}$$

$$C_F = 82 \text{ nF} \times \left(\frac{46.06}{6.217} - 1 \right) = 526 \text{ nF} \rightarrow \mathbf{560 \text{ nF}}$$

$$R_F = \frac{1}{2\pi \times 6.217 \text{ Hz} \times 560 \text{ nF}} = 45.7 \text{ k}\Omega \rightarrow \mathbf{47 \text{ k}\Omega}$$

$$C_{fx} = \frac{1}{2\pi \times 1000 \text{ Hz} \times 35 \text{ k}\Omega} - 3 \text{ nF} = 1.547 \text{ nF} \rightarrow \mathbf{1.55 \text{ nF}}$$

Step 9 — what the standard parts actually give. Equation 46 with the rounded values:

Equation 46, repeated

$$EA_o = \frac{CTR R_{FB}}{(C_F + C_{Fo}) R_I R_B} \quad [\text{rad/s}], \quad f_z = \frac{1}{2\pi R_F C_F}$$

$$f_p = \frac{1}{2\pi R_F C_{ser}}, \quad C_{ser} = \frac{C_F C_{Fo}}{C_F + C_{Fo}}, \quad f_{px} = \frac{1}{2\pi R_{FB} (C_{opto} + C_{fx})}$$

$$EA_o = \frac{0.37 \times 35 \text{ k}\Omega}{(560 + 82) \text{ nF} \times 100 \text{ k}\Omega \times 6.2 \text{ k}\Omega} = \mathbf{32.53 \text{ rad/s}}$$

$$f_z = \frac{1}{2\pi \times 47 \text{ k}\Omega \times 560 \text{ nF}} = \mathbf{6.047 \text{ Hz}}$$

$$C_{ser} = \frac{560 \times 82}{560 + 82} \text{ nF} = 71.5 \text{ nF}, \quad f_p = \frac{1}{2\pi \times 47 \text{ k}\Omega \times 71.5 \text{ nF}} = \mathbf{47.34 \text{ Hz}}$$

$$f_{px} = \frac{1}{2\pi \times 35 \text{ k}\Omega \times (3 + 1.55) \text{ nF}} = \mathbf{999 \text{ Hz}}$$

Step 10 — the crossover and the phase margin. $G_o EA_o = 130.06 \times 32.53 = 4231$, so the iteration of Equation 53 starts at $\omega_c = \sqrt{4231} = 65.0 \text{ rad/s}$ (10.4 Hz) and settles after a few steps at

Equation 53, repeated

$$\omega_c \leftarrow \sqrt{G_o EA_o A(\omega_c)}, \quad \text{starting from } \omega_c = \sqrt{G_o EA_o}$$

$$\omega_c = \mathbf{110.4 \text{ rad/s}}, \quad f_c = \frac{110.4}{2\pi} = \mathbf{17.57 \text{ Hz}}, \quad |T(\omega_c)| = 1.0003 \quad (\text{check : } 1)$$

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Equation 54, repeated

$$\Phi_M = \arctan \frac{\omega_c}{\omega_z} - \arctan \frac{\omega_c}{\omega_p} - \arctan \frac{\omega_c}{\omega_{px}}$$

$$\Phi_M = \arctan \frac{110.4}{38.0} - \arctan \frac{110.4}{297.5} - \arctan \frac{110.4}{6279} = 71.0^\circ - 20.4^\circ - 1.0^\circ = \mathbf{49.6^\circ}$$

Step 11 — the gain margin:

Equation 55, repeated

$$f_{180} = \sqrt{f_p f_{px} - f_z(f_p + f_{px})}, \quad GM = -20 \log_{10} |T(f_{180})|$$

$$f_{180} = \sqrt{47.34 \times 999 - 6.047 \times (47.34 + 999)} \text{ Hz} = \mathbf{202 \text{ Hz}}$$

$$|T(f_{180})| = 0.0196, \quad GM = -20 \log_{10} 0.0196 = \mathbf{34.2 \text{ dB}}$$

Step 12 — the third harmonic the built loop causes. The compensator gain at $2f_l = 94 \text{ Hz}$ ($\omega = 590.6 \text{ rad/s}$), from Equation 45 without the plant, and Equation 41:

$$G_{EA}(2f_l) = \frac{32.53}{590.6} \cdot \frac{\sqrt{1 + (590.6/38.0)^2}}{\sqrt{1 + (590.6/297.5)^2} \sqrt{1 + (590.6/6279)^2}} = \mathbf{0.3843} \quad (\leq 0.4541)$$

Equation 41, repeated

$$D_3 = \frac{G_{EA}(2f_l) \Delta V_{loop}}{4 V_{FB}}$$

$$D_3 = \frac{0.3843 \times 1.184 \text{ V}}{4 \times 2.689 \text{ V}} = \mathbf{4.23 \%} \quad (\text{budget } 5 \%)$$

Step 13 — the feedback ripple at the burst point, with burst entry at 75 W:

Equation 56, repeated

$$\Delta V_{FB} = \frac{P_{in, BM}}{V_{out}} \frac{1}{2\pi f_l C_{out}} G_{EA}(2f_l)$$

$$\Delta V_{FB} = \frac{75 \text{ W}}{25.0 \text{ V}} \cdot \frac{1}{2\pi \times 47 \text{ Hz} \times 75.2 \text{ mF}} \times 0.3843 = \mathbf{51.9 \text{ mV}}$$

Equation 57, repeated

$$\Delta R_{BM} = \frac{\Delta V_{FB}}{2 \times 0.01 \text{ V/k}\Omega}$$

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$$\Delta R_{BM} = \frac{51.9 \text{ mV}}{2 \times 0.01 \text{ V/k}\Omega} = 2.60 \text{ k}\Omega, \quad R_{BM} = 30 - 2.60 = 27.4 \text{ k}\Omega$$

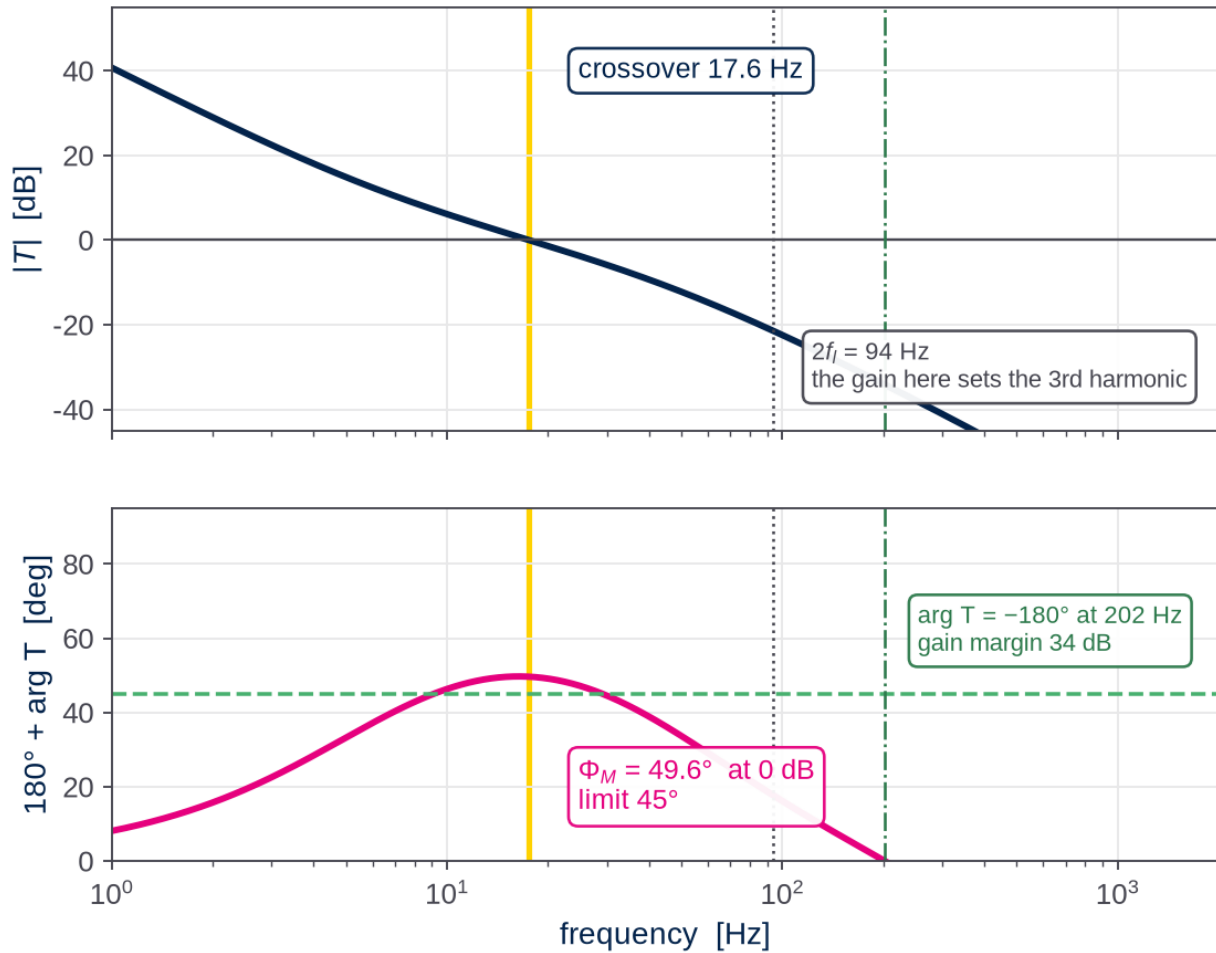


Figure 47: Open-loop gain $|T|$ and $180^\circ + \arg T$ of the loop as built (Steps 9 to 11). The distance from -180° is the phase margin only at the crossover, where $|T| = 0$ dB. The gain margin is read at f_{180° , where that distance reaches zero.

Quantity	Value	Against
Divider R_I / R_O	100 / 11 k Ω	regulates to 25.18 V
Bias R_P, R_B	1.3 k Ω , 6.2 k Ω	$R_P \leq 1.44$ k Ω ; R_B in 5.76 to 6.48 k Ω
Compensation C_{F0}, C_F, R_F, C_{fx}	82 nF, 560 nF, 47 k Ω , 1.55 nF	zero 6.05 Hz, pole 47.3 Hz, pole 999 Hz
Crossover f_c	17.57 Hz	the 15 to 20 Hz this topology lands in
Phase margin	49.6°	target 50°, floor 45°
Gain margin	34.2 dB at 202 Hz	6 dB floor, 10 dB comfortable
Third harmonic on the input current	4.23 %	budget 5 %
Feedback ripple at the burst point	52 mV	R_{BM} 30 $\rightarrow$ 27.4 k Ω

Table 20: The voltage loop, as built.

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7.15 The parts around the controller

These are the results. The sections after this one derive each value in the order it has to be done, because every one of them is sized against something already fixed.

Part	Value	Result
C_T / R_T	470 pF / 11 kΩ	f_{Min} 92.2 kHz, f_{Max} 319.6 kHz
R_{CS}	24.0 mΩ (120 mΩ × 5 in parallel)	OCP1 at 1.251 × the composite peak
R_{CFG}	30 kΩ	V_{BO} 84.8 V, morphing enabled
R_{BM}	30 kΩ	burst-mode entry point
ZCD divider	330 kΩ / 19 kΩ	OVP1 28.16 V, OVP2 30.61 V
C_{in}	2200 nF film	about 3.2 nF/W — there is no bulk capacitor
Compensation	82 nF / 560 nF / 47 kΩ / 1.55 nF	f_{cross} 17.57 Hz, Φ_M 49.64°

Table 21: Controller network for the worked design.

7.16 The oscillator: C_T first, then R_T

The VCO charges C_T from the sum of a fixed current V_{ref}/R_T and the error-amplifier current I_{EA} , up to $V_{\text{ref}} = 1.5$ V, then adds a fixed idle time:

Equation 59

$$\frac{T_{\text{sw}}}{2} = \frac{C_T V_{\text{ref}}}{\frac{V_{\text{ref}}}{R_T} + I_{\text{EA}}} + T_{\text{idle}}$$

I_{EA} is the only control input in the converter. More current means a higher frequency, which means less power. Everything the control loop does, it does through this one term.

C_T is chosen first because it sets the *span*, and R_T after it because it sets the *floor*:

Equation 60

$$C_{T, \text{max}} = \frac{I_{\text{EA}, \text{max}}}{2V_{\text{ref}}} \cdot \frac{(1 - 2T_{\text{idle}}f_{\text{sw}, \text{max}})(1 - 2T_{\text{idle}}f_{\text{sw}, \text{min}})}{f_{\text{sw}, \text{max}} - f_{\text{sw}, \text{min}}}, \quad C_{T, \text{min}} = \frac{1}{R_{T, \text{max}}} \left(\frac{1}{2f_{\text{sw}, \text{min}}} - T_{\text{idle}} \right)$$

Equation 61

$$R_{T, \text{ceil}} = \frac{1}{C_T} \left(\frac{1}{2f_{\text{sw}, \text{min}}} - T_{\text{idle}} \right)$$

with $f_{\text{sw}, \text{min}}$ set to $f_0 = 90.4$ kHz — the frequency floor of the whole design, for the reason in Section 4.2 — and $f_{\text{sw}, \text{max}} = 225$ kHz, $I_{\text{EA}, \text{max}} = 400$ μA, $V_{\text{ref}} = 1.5$ V and $T_{\text{idle}} = 250$ ns:

$$C_{T, \text{max}} = \frac{400 \times 10^{-6}}{2 \cdot 1.5} \cdot \frac{(1 - 2 \cdot 250 \times 10^{-9} \cdot 225 \times 10^3)(1 - 2 \cdot 250 \times 10^{-9} \cdot 90.4 \times 10^3)}{(225 - 90.4) \times 10^3} = 703 \text{ pF}$$

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and the floor condition, with the 30 kΩ datasheet maximum on R_T , gives $C_{T,min} = 176$ pF. The datasheet allows 270 to 1000 pF, so the usable window is 270 to 703 pF and the part is taken from the middle of it rather than from either edge: **$C_T = 470$ pF, C0G**. Then

$$R_{T,ceil} = \frac{1}{470 \times 10^{-12} \left(\frac{1}{2 \cdot 90.4 \times 10^3} - 250 \times 10^{-9} \right)} = 11.24 \text{ k}\Omega \rightarrow 11 \text{ k}\Omega$$

$R_{T,ceil}$ is a maximum, not a minimum, whatever the name suggests. $f_{Min} = 1/[2(C_T R_T + T_{idle})]$ falls as R_T grows. Round this number up and the VCO clamp drops below f_o , leaving only the anti-capacitive protection between the converter and the capacitive region. Round it **down**.

The three frequencies the selected pair actually produce are

Equation 62

$$f_{Min} = \frac{1}{2(C_T R_T + T_{idle})}, \quad f_{Max} = \frac{1}{2 \left(\frac{C_T}{\frac{I_{EA,max}}{V_{ref}} + \frac{1}{R_T}} + T_{idle} \right)}$$

Putting the selected pair into the first of those, with $T_{idle} = 250$ ns: $f_{Min} = 1/[2(470 \text{ pF} \times 11 \text{ k}\Omega + 0.25 \text{ }\mu\text{s})] = 1/(2 \times 5.420 \text{ }\mu\text{s}) = \mathbf{92.25 \text{ kHz}}$, which is what has to clear f_o .

Quantity	Value	Limit	Margin
f_{Min}	92.2 kHz	above $f_o = 90.4$ kHz	1.021
f_{Max}	319.6 kHz	above $f_{sw,max} = 249.0$ kHz	1.283
f_{SU}	503.5 kHz	below the 675 kHz silicon ceiling	1.341
$R_T C_T$	5.17 μs	2.5 to 12 μs	2.068 / 2.321
R_T	11 kΩ	5 to 30 kΩ	2.200 / 2.727
C_T	470 pF	270 to 1000 pF	1.741 / 2.128

Table 22: Oscillator: what the two parts produce, and the limits each result has to clear.

T_{idle} is the weakest number in this chapter. The draft datasheet gives 700 ns in the text and a table that back-solves to 250 ns; this design is worked at 250 ns. At 700 ns the same R_T and C_T put f_{Min} essentially on top of f_o and the floor margin disappears. **It is not a precision question, it is whether the two clamps still bracket the operating range**, and it is the first thing to measure on hardware.

7.17 The current-sense resistor, which sets three things at once

R_{CS} is asked for by two conditions and the smaller one wins:

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Equation 63

$$R_{CS, max1} = \frac{16.8 \Omega \cdot W}{P_{in}} \quad R_{CS, max2} = \frac{0.55 V}{I_{Lr, pk}}$$

The first is the maximum-power law of Section 4.6; the second is the OCP1 trip point. Both are one division:

$$R_{CS, max1} = \frac{16.8}{684.8} = 0.02453 \Omega = 24.53 \text{ m}\Omega, \quad R_{CS, max2} = \frac{0.55}{18.32} = 0.03003 \Omega = 30.03 \text{ m}\Omega$$

so the power law is the limiting one. Dissipation then fixes how many resistors that has to be split over: at the worst switching cycle the sense element burns $I_{pri, rms}^2 R_{CS}$, and a 1 W part cannot take it alone.

$$P_{CS} = I_{pri, rms}^2 R_{CS} = (13.76)^2 \cdot 24.0 \times 10^{-3} = 4.54 \text{ W} \Rightarrow N = 5 \text{ in parallel}$$

5 resistors of 120 mΩ in parallel give **$R_{CS} = 24.0 \text{ m}\Omega$** , which is what the rest of this section is read at.

The denominator of $R_{CS, max2}$ is the composite tank peak, not the reflected load current. R_{CS} sits in the bridge return, so it carries the magnetising component too. Using $I_{trafo, pk} = 16.37 \text{ A}$ instead of $I_{Lr, pk} = 18.32 \text{ A}$ loosens the limit by about 12 % and the over-current protection with it.

Result	Value	Against
Maximum input power	700.0 W	$P_{in} = 684.8 \text{ W}$
OCP1 trip	22.92 A	composite peak 18.32 A, margin 1.251
OCP2 trip	31.25 A	immediate stop, 50 μs, restart at f_{SU}
Sense dissipation	4.54 W total, 0.909 W each	1 W parts, margin 1.100

Table 23: What the selected R_{CS} then fixes.

7.18 Burst mode: one resistor, read once at power-up

Equation 64

$$R_{BM} = 16.7 \frac{\text{k}\Omega}{\text{V}^2} R_{CS} P_{in, BM}, \quad V_{BM, eq} = \frac{R_{BM}}{100 \text{ k}\Omega} + 0.5 \text{ V}$$

Burst is asked to start at 11 % of rated input power, that is $P_{in, BM} = 11 \% \times 684.8 \text{ W} = 75.3 \text{ W}$:

$$R_{BM} = 16.7 \cdot 24.0 \times 10^{-3} \cdot 75.3 = 30.1 \text{ k}\Omega \rightarrow 30 \text{ k}\Omega, \quad V_{BM, eq} = \frac{30}{100} + 0.5 = 0.800 \text{ V}$$

The valid range is 15 to 140 kΩ. Tie the pin to ground and burst mode is off altogether; deep burst stays active either way.

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This threshold and the feedback ripple have to be checked against each other, or the converter chatters in and out of burst once per half line cycle — Section 6.33.

7.19 Brown-out and bridge configuration: the CFG pin

One resistor does two unrelated jobs, and both are read once at power-up:

Equation 65

$$V_{BO,pk} = R_{CFG} \times 4 \frac{V}{k\Omega}, \quad V_{BO,rms} = \frac{V_{BO,pk}}{\sqrt{2}}$$

The brown-out threshold is compared against the *peak* of the mains, so it has to be converted before it is compared with an rms specification — a $\sqrt{2}$ that is easy to drop. Asking for brown-out just below the minimum line inverts Equation 65:

$$R_{CFG,max} = \frac{\sqrt{2} \cdot 90}{4} = 31.82 \text{ k}\Omega \rightarrow 30 \text{ k}\Omega$$

That is a maximum, so it rounds down, and the threshold the selected part actually sets is

$$V_{BO,rms} = \frac{30 \cdot 4}{\sqrt{2}} = 84.85 \text{ V}_{rms}$$

against a 90 Vac minimum — margin 1.061. Rounding up instead would stop the converter starting at low line.

R _{CFG}	LOUT2	Configuration
15 kΩ	open	morphing, fixed brown-out (60 V / 70 V peak)
15 to 47 kΩ	open	morphing, adjustable brown-out — this design
47 to 100 kΩ	open	fixed full bridge
15 kΩ	to GND	fixed half bridge, split C _r , fixed brown-out
15 to 100 kΩ	to GND	fixed half bridge, split C _r , adjustable brown-out

Table 24: What R_{CFG} and LOUT2 select together. The 235 and 245 V thresholds are fixed inside the IC and cannot be moved.

For a universal-input design there is only one window. The datasheet floor is 15 kΩ and brown-out at the minimum line is the ceiling at 31.8 kΩ, so the 47 kΩ morphing limit never comes into play. And hang nothing on LOUT2. The pin-strap drives 300 μA into it and needs 2.5 V, so any pull-down under about 8 kΩ reads as a request for a fixed half bridge.

7.20 Output sensing and over-voltage: the ZCD divider

The auxiliary winding is divided down to the ZCD pin, and the divider ratio alone sets both over-voltage thresholds — the absolute resistor values only set the bias current:

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Equation 66

$$R_{ZCD,L} = \frac{2.3 \text{ V}}{I_{bias}}, \quad R_{ZCD,H} = R_{ZCD,L} \left(\frac{n_{aux}}{n_{sec}} \frac{V_{OVP1,out}}{2.3 \text{ V}} - 1 \right)$$

Equation 67

$$V_{OVP1} = \frac{2.3 \text{ V}}{n_{aux}/n_{sec}} \left(\frac{R_{ZCD,H}}{R_{ZCD,L}} + 1 \right), \quad V_{OVP2} = \frac{2.5}{2.3} V_{OVP1}$$

Aiming OVP1 10 % above the output makes the target 27.50 V. The lower resistor comes from the bias current the pin is to be driven with, 120 μ A, and the upper one from the ratio the target asks for, with $n_{aux}/n_{sec} = 1.5$:

$$R_{ZCD,L} = \frac{2.3}{120 \times 10^{-6}} = 19.17 \text{ k}\Omega \rightarrow 19 \text{ k}\Omega$$

$$R_{ZCD,H} = 19 \left(\frac{1.5 \cdot 27.5}{2.3} - 1 \right) = 321.8 \text{ k}\Omega \rightarrow 330 \text{ k}\Omega$$

The upper one is deliberately the *next E24 value above* the calculation, so OVP1 lands clear of the ripple crest rather than on it. Putting the fitted pair back into Equation 67 gives what the divider really does:

$$V_{OVP1} = \frac{2.3}{1.5} \left(\frac{330}{19} + 1 \right) = 28.16 \text{ V}, \quad V_{OVP2} = \frac{2.5}{2.3} \cdot 28.16 = 30.61 \text{ V}$$

Start-up hands over to the loop at 16.53 V of output.

Two ways to get this wrong, and both stop the converter completely. Swapping the two resistors inverts the ratio and OVP1 trips at well under a volt of output, so the converter never starts. And the winding polarity must make ZCD *positive* when the low side of leg 1 is on; reversed, the bridge hard switches from the first pulse.

The turns-ratio ceiling in the datasheet does not apply here. It asks that the rectified auxiliary voltage stay under the V_{CC} rating at OVP2, which is a question only when that winding supplies V_{CC} . In this design V_{CC} comes from a housekeeping rail and the winding senses only, so its voltage is free — 37.5 V nominal, 45.9 V at OVP2 — and the only remaining constraint is that the turns come out whole. Carrying that check over from a self-supplied design reports a failure that is not one.

7.21 HVSU and V_{CC}

- **HVSU connects ahead of the input bridge**, on the ac side, through one 1000 V diode per line. Behind the bridge both the X-capacitor discharge and the brown-out detection stop working, because neither can see the mains disappear.
- The V_{CC} capacitor has to hold the IC from $V_{CCon} = 17 \text{ V}$ down to $V_{CCoff} = 8 \text{ V}$ for as long as it takes the supply to take over. The start-up unit stops 120 ms after V_{CCon} is reached, whether or not the supply has taken over by then.

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- Where a linear regulator feeds V_{CC} , put a diode between its output and the pin so the start-up charging current cannot be pushed back into it.
- A 100 nF bypass sits at the pin itself.

7.22 Controller pin rules

These follow from the datasheet and each one causes immediate misbehaviour if broken.

Pin	Rule
ISEN	No filter and no series resistor. The maximum-power law and the over-current threshold share this pin.
CFG, BM	No capacitor. Both are strapped at power-up. An out-of-range CFG resistor latches the controller off.
DRV_EN, RT	No capacitor. Both are sampled briefly every few milliseconds and a capacitor is read as a fault.
LOUT2	No pull-down if morphing is used. A few kilohms to ground reads as a fixed half bridge.
HVSU	Connect ahead of the input bridge, on the ac side. Behind the bridge, both X-capacitor discharge and brown-out detection stop working.
ZCD	Auxiliary winding polarity must make ZCD positive when the low side of leg 1 is on. Reversed, the bridge hard switches.
HOUTx, LOUTx	Logic-level outputs, not gate drivers. External half-bridge drivers are required.

Table 25: Pin rules that must not be broken.

7.23 Protections, and where each threshold is set

Every one of these is a design constant fixed by a resistor this note has already chosen, so a change made for one reason moves a threshold set for another. The table is here because the thresholds are otherwise spread over five sections.

Protection	Set by	This design
Brown-out, on the rectified mains	R_{CFG} , read at power-up	V_{BO} 84.8 V, clear of the 90 Vac minimum by 1.061
Cycle-by-cycle over-current, OCP1	R_{CS} — the same resistor as the maximum-power law	22.92 A against a composite peak of 18.32 A, a margin of 1.251
Second-level over-current, OCP2	R_{CS} , fixed ratio to OCP1	31.25 A
Output over-voltage, OVP1	the ZCD divider on the auxiliary winding	28.16 V, above the ripple crest and below OVP2
Second-level over-voltage, OVP2	the same divider	30.61 V
Capacitive-mode protection	internal; it watches the ZCD edge against the gate	the last line of defence only — the frequency floor f_{Min} is the first
Burst mode at light load	R_{BM} , read at power-up	30 k Ω ; the $2f_1$ feedback ripple is 52 mV, so see Section 6.33
Loss of ZVS at the worst corner	not a protection — a design margin	swept, not estimated; Section 6.5

Table 26: Protections and the component that sets each one.

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Two of these share one resistor. R_{CS} sets the over-current thresholds *and* the maximum-power law, because the controller reads both from the same pin. Changing it to fix a power limit moves OCP1, and the other way round. That is also why the pin tolerates no filter.

7.24 System design rules

The same material as a checklist, in the order a design meets it. Every line is argued for somewhere above; collected here so a reviewer can run down the list without re-reading the argument.

- **Design the tank at the real ends of the equivalent range.** With morphing that is 173.2 and 332.3 Vac, not 2×90 and 264. Looking only at the mains range misses both worst corners.
- **Expect λ near 0.5.** The 5-to-10 inductance ratio of a two-stage LLC will not start at low line.
- **Keep the oscillator floor above f_o ,** and treat the anti-capacitive protection as the last line, not the first.
- **Size the output bank from both conditions** — ripple and hold-up — take the larger, and then check the rms ripple current separately. The capacitance and the current are different questions.
- **Put the voltage-loop crossover in the tens of hertz,** then deal with the feedback ripple that survives by moving R_{BM} , not by speeding the loop up.
- **Measure both morphing edges,** at full load and at light load. The half-bridge edge is the worst case for gain and ZVS, the full-bridge edge for switching frequency.
- **Check the turns ratio as a reflected voltage,** $nV_{o,eff}$, never as a bare ratio — that is what keeps n and n_T from being interchanged.
- **Judge the over-current margin on the composite tank peak,** not on the reflected load current.
- **Include the $2f_l$ component** in the output capacitor ripple current, and judge a centre tap at the output node rather than per winding.
- **Carry the conduction ratio $\sqrt{d}$ into the secondary rms.** Dropping it overstates the rms by about 12 % here, and the loss — which goes as the square — by about 26 %.
- **Decide how L_r is built before specifying the transformer,** and put open-circuit and short-circuit inductance on the drawing alongside the turns.
- **Use capacitor ESR at the switching frequency,** not the 120 Hz $\tan\delta$.
- **Never round L_m up.** It lowers λ and takes the ZVS margin with it.
- **Verify ZVS by sweeping the selected tank,** not with the closed form — and if the closed form is used anyway, feed it λ_{act} and the Q of the moment, and know that its error has no fixed sign.
- **Record calculated and selected values separately,** and make every downstream check read the selected one. A sheet that quietly reads the calculated value reports margins the hardware does not have.

7.25 What to measure first on hardware

In order. The first four decide whether the design is sound at all; the rest confirm margins that are computed but never yet seen.

- **$f_{sw}(\theta)$ over a line half cycle.** This settles T_{idle} , and with it both oscillator clamps — the thinnest margin in the design. Everything else in the controller network hangs off it.
- **Cold start into the full output bank.** A 75.2 mF bank is a far heavier start-up load than a conventional design presents, and the start-up window is finite. This risk is new to this architecture, so there is no earlier design to compare with.

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- **ZVS at the half-bridge morphing edge** ($245 V_{pk}$, full load), on the gate and mid-point waveforms. The calculation says 460 ns against 220 ns; confirm it is not hard switching, and that the adaptive dead time actually settles where it is assumed to. **Then measure the worst point of the sweep as well** — 25 % load at the 332.3 Vac equivalent corner, 401 ns — because that, not the design corner, is the figure this design is held to.
- **Switching frequency at the full-bridge morphing edge** ($235 V_{pk}$, full load). Calculated 249.0 kHz; check it does not run into the VCO ceiling and get power-limited.
- **Input current and THD at 90 Vac, full load**, to put a number on the zero-crossing dead zone.
- **Output ripple against what the load actually tolerates**. The bank is sized from a specification; this is where the specification is tested.
- **Ripple current and temperature rise in the output bank** — 30.24 A rms calculated, 0.19 A per capacitor.
- **Device temperatures, primary and secondary**. This is where the accepted loss-budget shortfall is decided: measure it in **half-bridge morphing**, above $245 V_{pk}$, because in full bridge the standing device is switching and dissipates far less.
- **Feedback ripple against the burst threshold** at light load.
- **An ac dropout at the worst line phase** — cut the mains at the trough of the ripple and confirm the output is still above 19 V after 12 ms.
- **A step across the morphing band**, both directions, on a programmable source. A ramp gives the loop time to follow and hides the transition.

8 List of symbols

The symbols used in this note. The controller's own electrical parameters are in its datasheet.

Symbol	Meaning
The tank and its gain	
C_r, L_r, L_m	resonant capacitor, series inductance and magnetising inductance of the tank model
L_p	$L_r + L_m$, the inductance the tank shows at no load
$Z_0, Z_{0,design}$	characteristic impedance $\sqrt{L_r/C_r}$; the value the tank is designed to, $R_{ac}Q_{ZVS}$
Z_{in}	input impedance the bridge drives; $\arg Z_{in} = 0$ is the capacitive boundary
f_r, f_o	series resonance, and the lower resonance with L_m included
f_{sw}, f_n	switching frequency, and the same normalised to f_r
$f_{sw,max}, f_{sw,min}$	the specified maximum and minimum switching frequency; $f_{sw,max}$ is an input to λ
T_{sw}, T_r	switching period $1/f_{sw}$, and the resonant period $1/f_r$
λ, m	L_r/L_m , and $(L_r+L_m)/L_r = 1 + 1/\lambda$
$\lambda_{act}, \lambda_{req}$	the λ of the selected parts, and the λ the design asked for
$\lambda_1, \lambda_2, \lambda_{TD}, \lambda_3$	the four candidate λ : minimum gain, the same with the frequency ceiling, the dead-time condition, and the practical floor; the smallest wins
Q, Q_{pk}, Q_{ZVS}	quality factor Z_0/R_{ac} ; its value at the line peak; the cap the ZVS condition puts on it
R_{ac}	the rectifier and load as one resistance at the fundamental
M	tank gain, $n V_{o,eff}$ over the drive
M_{req}, M_{pk}	the gain the operating point demands, and its value at the line peak

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Symbol	Meaning
$M_{\min}, M_{\max}$	the gain range the tank must cover, from the two equivalent-input corners
$M_{\text{low}}, M_{\text{high}}$	the required gain at the low and the high equivalent corner
$M_{\text{no load}}, M_{\infty}$	the gain curve at $Q = 0$, and the floor $1/(1+\lambda)$ it flattens onto
M_Z	the locus of the gain peaks, which is the capacitive/inductive boundary
x, u, q	the substitutions that turn the gain equation into a cubic: $1/f_n^2$, $\sin^2\theta$, and $Q_{pk}^2 u^2$
d	conduction ratio f_{sw}/f_r of the secondary, 1 above resonance
n, n_T	equivalent-model turns ratio, and the physical (wound) turns ratio
N_{rect}	devices in the secondary conduction path: 1 centre tap, 2 full bridge
V_{drive}	the amplitude the bridge applies to the tank: the rectified mains in full bridge, half of it in half bridge
V_{refl}	reflected output voltage $n V_{o,\text{eff}}$
$V_{o,\text{eff}}$	$V_{\text{out}} + N_{\text{rect}} V_f$
V_f	forward drop of one secondary rectifier
The mains, and power factor	
θ	line phase angle
$f_l, f_{l,\min}, \omega_l$	line frequency, its lowest specified value, and $2\pi f_l$
v_{ac}, V_{ac}, I_{ac}	the instantaneous mains voltage, and the rms voltage and current drawn
$V_{\text{rms}}, I_{\text{rms}}, I_{1,\text{rms}}$	rms input voltage and current, and the rms of the current's fundamental alone
$p_{\text{in}}(t), P$	instantaneous and average input power
PF, THD	power factor, and total harmonic distortion of the input current
ϕ_1, D	displacement angle of the fundamental, and the distortion factor $I_{1,\text{rms}}/I_{\text{rms}}$
D_3	third harmonic on the input current, as a fraction of the fundamental
$V_{\text{in}}, V_{\text{pk}}$	the rail the bridge works from, and the peak of the mains
$V_{ac,\text{eq}}, V_{\text{eq}}, V_{ac,\text{eq,low}}, V_{ac,\text{eq,high}}$	equivalent input voltage the tank sees after morphing, and its value at the two morphing thresholds
$V_{\text{eq,min}}, V_{\text{eq,max}}$	the two ends of that equivalent range, at the two morphing thresholds
Power and loss	
$P_{\text{in}}, P_{\text{out}}$	input power, output power
$P_{\text{LLC}}, P_{\text{EMI}}, P_{\text{BR}}$	the three budgeted losses: the LLC stage, the EMI filter, the input bridge
$P_{\text{in,LLC}}$	power into the tank, $P_{\text{out}}/\eta_{\text{HB}}$
η_{HB}	assumed efficiency of the LLC stage
$P_{\text{cond}}, P_{\text{mos,dc}}$	conduction loss of the primary device that stands on in half-bridge morphing
$P_{\text{SR}}, P_{\text{SR,leg}}$	secondary rectifier loss, per device and per rectifier leg
P_{CS}	dissipation in the current-sense resistors
P_{budget}	the loss a device position is allowed, before parts are chosen

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Symbol	Meaning
P_v	core loss per unit volume, from the material curve
$E, C, V, V_{\min}$	the energy a capacitance C gives up between two voltages, in the hold-up expression
Currents and times of one switching cycle	
$i_{Lr}, I_{Lr,pk}$	tank current, and its peak: reflected load plus magnetising
$i_{trafo}, I_{trafo,pk}$	the reflected load current alone, and its peak
$i_{Lm}, I_{Lm,pk}$	magnetising current of the tank model, and its peak
$i_{\mu}, i_{\mu,pk}$	magnetising current of the physical transformer, and its peak
i_p, i_s	the instantaneous primary and secondary winding currents
$I_{sec,pk}$	peak secondary current per rectifier leg
$I_{pri,rms}$	primary rms over the worst switching cycle
$I_{lc}, I_{pri}, I_{rect}$	a line-cycle rms: the rms of each switching period, averaged in I^2 over the line half cycle; for the primary and for one rectifier leg
I_{Cout}, I_{out}	ripple current in the output bank, and the load current
$t_D, T_{ZC}, T_{ZC,min}$	bridge dead time; the time the tank current takes to reach zero after the transition, and its smallest value over the operating space
V_d, V_{ds}	bridge mid-point voltage, and a device's drain-source voltage
$S_1, S_2, S_3, S_4, D_1, D_2$	the four bridge switches, and the two secondary rectifiers
The transformer and its core	
$B(t), B_{pk}, B_{max}$	flux density, its peak, and the ceiling the design keeps under
N	turns of the winding a voltage is applied to, in Faraday's law
A_e, A_{min}	core effective area, and its narrowest section
A_N	winding window area of the coil former
A_L, g, μ_0	inductance factor of the gapped core, the total centre-leg gap, and the permeability of free space
N_p, N_s, N_x	primary turns, secondary turns per winding, number of units in the assembly
$N_{aux}, n_{aux}/n_{sec}$	auxiliary winding turns, and its turns ratio to one secondary
L_{open}, L_{short}	primary inductance with the secondaries open, and shorted
L_{μ}, L_{L1}, L_{L2}	physical magnetising inductance and the two leakage inductances
L_1, L_2	the open-circuit inductance measured at the primary and at one secondary
$L, \Delta L$	an inductance and the change of it a tolerance allows
I_{eq}, I_{sat}	open-circuit current that reproduces the operating flux; the DC-overlap test current
l_N	mean length of one turn on the coil former
The copper	
A_{cu}, J	copper cross-section one winding needs, and the current density it is sized to
$N_w, A_{cu,w}$	turns and copper area of winding w , in the sum over all the windings

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Symbol	Meaning
k_u	window utilisation: the share of A_N that ends up copper
$\delta, \rho, \rho_{20}, f$	skin depth, the resistivity of copper at the working temperature and at 20 °C, and the frequency the depth is evaluated at
d_s, a_s, n_s	Litz strand diameter, the copper cross-section of one strand, and the number of strands a winding needs
d_{litz}, k_{litz}	outside diameter of the served bundle, and the share of that area which is copper
$t_f, w_f, w_{f,max}, n_f$	foil thickness, the width one turn needs, the widest strip the bobbin takes, and the number of strips in parallel
Semiconductors	
$R_{DS(on)}, R_{DS(on),25}, k_T$	on-resistance, its 25 °C headline value, and the multiplier from 25 °C to $T_{j,max}$
$T_j, T_{j,max}, T_a$	junction temperature, its rating, and ambient temperature
$C_{oss}, C_{o(er)}, C_{o(tr)}, Q_{oss}$	the small-signal, energy-related and time-related output capacitance of a MOSFET, and the charge that time-related one stands for
The controller and its network	
V_{FB}, v_{FB}	feedback voltage above its 0.5 V offset at rated power, and its small-signal part
$V_{os}, K_{HV}, K_M, K_{FF}$	the offset and the three internal gains of the feedback law, quoted from the datasheet block diagram
K_{pwr}	slope of the power command, P_{out} per volt of V_{FB}
$R_{CS}, R_{CS,max1}, R_{CS,max2}$	current-sense resistor, and the two conditions that bound it: the maximum-power law and the OCP1 trip
I_{OCP1}, I_{OCP2}	over-current thresholds set by R_{CS}
C_T, R_T	oscillator timing capacitor and resistor
$C_{T,min}, C_{T,max}, R_{T,ceil}, R_{T,max}$	the window those two have to be chosen inside
$V_{ref}, I_{EA}, I_{EA,max}$	the oscillator reference, the error-amplifier current that is the only control input, and its maximum
f_{Min}, f_{Max}, f_{SU}	oscillator floor, ceiling and start-up frequency set by R_T, C_T and T_{idle}
T_{idle}	oscillator idle time
$R_{BM}, r_{BM}, V_{BM,eq}, P_{in,BM}$	burst-mode resistor, the fraction of rated power burst starts at, the equivalent FB threshold, and the input power that corresponds to
$R_{CFG}, R_{CFG,max}$	configuration resistor, and the largest value the minimum line allows
$V_{BO}, V_{BO,pk}, V_{BO,rms}$	brown-out threshold, as a mains peak and as an rms voltage
$R_{ZCD,H}, R_{ZCD,L}, I_{bias}$	the ZCD divider, and the current it is designed to draw
$V_{OVP1}, V_{OVP2}, V_{OVP1,out}$	the two over-voltage thresholds, and the output voltage OVP1 is aimed at
$V_{CC}, V_{CCon}, V_{CCoff}$	the IC supply, and the two thresholds of its under-voltage lockout
The voltage loop	
$G_{plant}(s), G_{EA}(s), T(s)$	plant, compensator and loop gain; $T = G_{plant} G_{EA}$
G_o, EA_o	gain constant of the plant integrator and of the compensator, in rad/s
$\omega_z, \omega_p, \omega_{px}, \omega_c$	compensator zero, pole, high-frequency pole, and the crossover, in rad/s

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Symbol	Meaning
$f_z, f_p, f_{px}, f_{180}$	the same zero and poles in hertz, and the frequency where $\arg T = -180^\circ$
$f_c, f_{\text{cross}}, \Phi_M, \text{GM}$	loop crossover frequency, phase margin, gain margin
f_{cto}	where the bare plant integrator would cross 0 dB, $G_o/2\pi$
$f_{\text{PHF}}, f_{\text{MB}}$	the geometric mean of the two poles, and the frequency the K-factor method balances the loop about
K_v, Γ_v, α_v	K factor of the Type II compensator, and the input-voltage margin factor and weighting constant it is built from
Z_f	feedback impedance of the compensator, C_{F0} in parallel with $R_F + C_F$
R_I, R_O	output divider of the compensator
C_{F0}, C_F, R_F, C_{fx}	compensator parts: the two capacitors and the resistor of the TL431 network, and the FB-pin capacitor
$C_{\text{opto}}, C_{\text{ser}}$	the optocoupler's own output capacitance, and $C_F + C_{F0}$ in series
$R_B, R_{B,\text{max}}, R_{B,\text{min}}, R_P, R_{\text{FB}}$	optocoupler LED resistor and the window it has to sit in, the TL431 bias resistor, and the pull-up inside the FB pin
$\text{CTR}, \text{CTR}_s, \text{CTR}_m, i_{\text{LED}}$	optocoupler current transfer ratio, its value at the steady-state and at the maximum LED current, and the LED current itself
$I_{\text{FB,steady}}, I_{\text{FB,max}}, I_{\text{min}}$	FB-pin current at steady state and at maximum, and the least current the TL431 needs to stay in regulation
V_R, V_Z, V_{F0}	TL431 reference, the regulated rail that feeds the LED, and the LED forward drop
v_{out}, v_C	small-signal output voltage, and the voltage on a capacitor in a waveform
$\Delta V_{\text{loop}}, \Delta V_{\text{FB}}$	$2f_i$ output ripple seen by the loop, and the ripple it leaves on the FB pin
The output, and hold-up	
$C_{\text{out}}, C_{\text{in}}$	output capacitor bank, input film capacitor
$V_{\text{out}}, V_o, V_{\text{out,eff}}, I_{\text{out}}$	output voltage and load current; $V_{\text{out,eff}}$ is $V_{o,\text{eff}}$ spelled out
$\Delta v, \Delta v_{\text{pp}}$	allowed and achieved $2f_i$ output ripple
$T_{\text{hold}}, t_{\text{hold}}, V_{o,\text{min}}$	required and achieved hold-up time, and the lowest output allowed at its end
ESR	equivalent series resistance of the bank, taken at the switching frequency
Verification margins	
$k, X, X_{\text{act}}, X_{\text{req}}$	a verification margin $X_{\text{act}}/X_{\text{req}}$, where X is whatever quantity the row is about; every k in this note is ≥ 1 when the design passes
$k_{\text{floor}}, k_{\text{ceil}}$	how far f_{Min} clears f_o , and how far f_{Max} clears the highest operating frequency
k_{Ploss}	the loss budget of a device position over the loss computed for it
w_k, θ_k	the weights and the nodes of the five-point Simpson rule the line-cycle rms values are integrated with

Table 27: Symbols.

9 References

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10 Errata and open items

Everything in this chapter is something the note works around rather than something it solves. It is last because none of it is needed to follow the design, and first to be re-read before a production release — most of it is a question that only the released datasheet or a prototype can close.

10.1 Datasheet: the draft is not self-consistent

Item	What the draft says	What is used here, and why
Oscillator idle time T_{idle}	Section 5.3.2 says 700 ns; Table 5 back-solves to about 250 ns from the frequency expressions	250 ns. The two clamps bracket the operating range at 250 ns and stop doing so at 700 ns, so this is a measurement, not a rounding
Brown-out expression	$V_{BO} = \min(60\text{ V}, R_{CFG} \cdot 4\text{ V/k}\Omega)$ — read literally this is 60 V for every resistor, which contradicts the configuration table	Read as $R_{CFG} \cdot 4\text{ V/k}\Omega$ with 15 k Ω as the floor. That is the only reading consistent with the table
Brown-out units	the threshold is a <i>peak</i> voltage; board notes and design tools quote rms	Converted explicitly wherever it is compared ($\sqrt{2} = 1.414$)
Thermal resistance	TBD	No junction temperature can be predicted from the datasheet alone; the loss budget is checked against measured rise instead
ZCD absolute maximum	lower limit given as TBD	The divider is sized from the OVP thresholds only
Driver naming in Section 5.3.1	LOUT1 and LOUT2 are transposed in one sentence against the block diagram and the configuration table	The block diagram and the table are taken as correct: LOUT2 is the pin that is strapped and held high in half bridge

Table 28: Points where the draft datasheet contradicts itself or leaves a value open. Every one of them must be re-checked against the released document.

10.2 Traps in the obvious way of building a design sheet

These are not datasheet problems. They are places where the obvious way to set a design sheet up gives an answer that is wrong in a way nothing flags, so they are worth stating as traps rather than as corrections to any particular tool.

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Where	The trap	Consequence if missed
Equivalent input range	taking the maximum from the ac maximum instead of the full-bridge morphing edge	The frequency corner is never evaluated. Here it is the difference between 264 and 332.3 Vac equivalent, and it also sets λ
λ from the minimum-gain condition	the same corner again, one step earlier	The required λ comes out far too small — a factor of over a hundred in one case — and L_m with it
Line-cycle rms currents	evaluating one phase, usually $\theta = \pi/4$, and calling it the line-cycle value	The output bank ripple current and the rectifier loss are both understated. A Simpson rule over the half cycle costs nothing and is exact enough
Secondary rms above resonance	squaring a full half sine when the current is truncated	Up to 8 % at the high corner. Below resonance the factor is exactly 1, which is why it can sit unnoticed for a long time
Primary device rating	rating on the reflected load current $I_{trafo,pk}$	About 12 % under-rated — the switch carries the <i>composite</i> tank current
Loss and thermal resistance	computing loss from the 25 °C $R_{DS(on)}$ and then asking for a heatsink that holds 125 °C	The required thermal resistance comes out only about half as demanding as it really is
ZVS check	the closed-form shortcut, at the design Q and with the design λ	Error of either sign; on this tank 36 % conservative, on another about 23 % optimistic. Optimistic is the dangerous direction
Selected versus calculated	a check that reads the calculated value while the board carries the selected one	Every downstream margin is reported for a design that was not built

Table 29: Traps that produce a plausible wrong number.

10.3 Constants used here without a derivation

- The **$\pi^2/8$** in the dead-time form of the λ condition. It behaves like a correction for the fundamental content of a square wave, but no source states it. It is indicative; the binding check is the ZVS sweep.
- The **exponent 5** in the closed-form ZVS shortcut. This is a fitted number, and it is the main reason the shortcut errs unpredictably near the boundary.
- The **0.744** in the input-voltage margin factor of the compensator design. It scales the crossover target and nothing downstream is sensitive to it at the percent level.
- The **16.8 $\Omega \cdot W$** maximum-power constant is consistent with the feedback span and the multiplier gains quoted in the datasheet — $2.8 \text{ V} / 0.167 = 16.8$ — so it is derived rather than assumed, and is listed here only because the gains themselves are draft values.

10.4 Open items in this design

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Item	State	What closes it
T_{idle}	two candidate values, 250 and 700 ns	Measure $f_{sw}(\theta)$ on the first board
Primary conduction loss	0.311 against the budget — not met, and accepted	No single 600 V device meets a 3 W budget in the standing position. Whether it needs a heatsink is a thermal measurement, not a calculation
Secondary loss budget	1.063 — essentially exhausted	One more device in parallel per leg recovers it; the decision waits on the measured temperature
Transformer inductance tolerance	the tank tolerates a 4.1 % fall in open-circuit inductance and the usual specification asks for ± 10 %	Either tighten the low side with the supplier, or lower R_T to lift f_{Min} first. Before the transformer is ordered
Output bank volume and height	75.2 mF, 160 parts	A mechanical question, not an electrical one, and it can send the whole architecture back to the output voltage
Standby power	burst entry assumed at 75 W	Measure, then trade R_{BM} against the feedback ripple
Efficiency assumption	$\eta_{HB} = 98$ % assumed	Optimistic. Taking 95 % instead moves R_{CS} and R_{ac} by about 3 %, so nothing downstream is sensitive — but it should be replaced by a measurement

Table 30: What is not settled, and what would settle it.

Every cross-check here is a consistency check, not a correctness check. The numbers agree across three independent implementations, which means the three implement the same equations — not that the equations describe the hardware. That is what the measurement list in Section 7.25 is for.